Math 2117: Final Exam Boss Notes

Compiled for Exam Preparation

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1 A01: Vector Basics and Direction Cosines

Exam Weight: Tier 3 | **Appeared in**: 2019, 2020, 2021, 2023 | **Typical Marks**: 3–4

This section covers the absolute basics of vectors: definitions, unit vectors, and direction cosines. These concepts form the foundation for all subsequent vector calculus operations.

1.1 Definitions

- **Scalar Quantity**: A physical quantity that has only magnitude but no direction (e.g., Mass, Temperature, Time).
- **Vector Quantity**: A physical quantity that has both magnitude and direction, and obeys vector addition rules (e.g., Velocity, Force, Acceleration).

A vector field

$$\vec{V}(x,y) = P(x,y)\hat{i} + Q(x,y)\hat{j}$$

assigns a vector to every point in space. - A **source** field (e.g.,

$$\vec{V} = x\hat{i} + y\hat{j}$$

) has all vectors pointing radially outward. - A **sink** field (e.g.,

$$\vec{V} = -x\hat{i} - y\hat{j}$$

) has all vectors pointing radially inward.

1.2 Unit Vectors and Resultants

The **resultant** of two vectors $\vec{r}_1$ and $\vec{r}_2$ is simply their vector sum:

$$\vec{R} = \vec{r}_1 + \vec{r}_2.$$

The **unit vector** parallel to any vector $\vec{R}$ is the vector divided by its magnitude:

$$\hat{u} = \frac{\vec{R}}{|\vec{R}|}$$

1.2.1 Worked Example (PYQ 2020)

Problem: Find a unit vector parallel to the resultant of

$$\vec{r}_1 = 2\hat{\imath} + 4\hat{\jmath} - 5\hat{k}$$

and

$$\vec{r}_2 = \hat{\imath} + 2\hat{\jmath} + \hat{k}.$$

Solution:

$$\vec{R} = \vec{r}_1 + \vec{r}_2 = 3\hat{\imath} + 6\hat{\jmath} - 4\hat{k}$$

$$|\vec{R}| = \sqrt{3^2 + 6^2 + (-4)^2} = \sqrt{9 + 36 + 16} = \sqrt{61}$$

Unit vector

$$\hat{u} = \frac{3\hat{\imath} + 6\hat{\jmath} - 4\hat{k}}{\sqrt{61}}$$

1.3 Direction Cosines

Let a position vector be

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$

with magnitude

$$r = \sqrt{x^2 + y^2 + z^2}.$$

The angles α, β, γ that the vector makes with the positive x, y, z axes respectively are defined by their **direction cosines**:

$$\cos\alpha = \frac{x}{r}, \quad \cos\beta = \frac{y}{r}, \quad \cos\gamma = \frac{z}{r}$$

A fundamental identity is that the sum of their squares is always 1:

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{x^2}{r^2} + \frac{y^2}{r^2} + \frac{z^2}{r^2} = \frac{x^2+y^2+z^2}{r^2} = 1$$

1.3.1 Worked Example (PYQ 2019)

Problem: Find the acute angles which the line joining $P(1,-3,2)$ and $Q(3,-5,1)$ makes with the coordinate axes. **Solution**:

$$\vec{PQ} = (3-1)\hat{i} + (-5-(-3))\hat{j} + (1-2)\hat{k} = 2\hat{i} - 2\hat{j} - \hat{k}$$

Magnitude

$$|\vec{PQ}| = \sqrt{4+4+1} = 3$$

Direction cosines:

$$\cos\alpha = \frac{2}{3} \implies \alpha = \cos^{-1}\left(\frac{2}{3}\right)$$

$$\cos\beta = -\frac{2}{3}$$

. For the acute angle, take the absolute value:

$$\beta = \cos^{-1}\left(\frac{2}{3}\right)$$

$$\cos\gamma = -\frac{1}{3}$$

. For the acute angle, take the absolute value:

$$\gamma = \cos^{-1}\left(\frac{1}{3}\right)$$

1.4 Exam Patterns

- Direction cosine questions often ask for the acute angle. If your cosine is negative, drop the minus sign before taking the inverse cosine to find the acute angle relative to that axis line.
- Basic definitions (scalar vs vector) sometimes appear as 1-2 mark theoretical add-ons to larger questions.

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2 A02: Dot Product and Cross Product

Exam Weight: Tier 2 | **Appeared in**: 2020, 2021, 2023, CT | **Typical Marks**: 3–4

The dot product and cross product are the two fundamental ways to multiply vectors. They have distinct physical interpretations and mathematical properties.

2.1 Dot Product (Scalar Product)

Definition:

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$$

- **Output**: A scalar (a number).
- **Physical Significance**: Represents the projection of one vector onto another. It is used to calculate work done by a force (

$$W = \vec{F} \cdot \vec{d}$$

). - **Orthogonality**: If two non-zero vectors are perpendicular (orthogonal), their dot product is 0.

2.1.1 Worked Example: Mutually Orthogonal Vectors (CT)

Problem: Show that

$$\vec{A} = \frac{1}{3}(2\hat{\imath} - 2\hat{\jmath} + \hat{k}),$$

$$\vec{B} = \frac{1}{3}(\hat{\imath} + 2\hat{\jmath} + 2\hat{k}),$$

$$\vec{C} = \frac{1}{3}(2\hat{\imath} + \hat{\jmath} - 2\hat{k})$$

are mutually orthogonal unit vectors. **Solution**: Magnitude check for unit vectors:

$$|\vec{A}| = \sqrt{(2/3)^2 + (-2/3)^2 + (1/3)^2} = \sqrt{9/9} = 1$$

. (Same applies for $\vec{B}$ and $\vec{C}$). Dot product check for orthogonality:

$$\vec{A} \cdot \vec{B} = \frac{1}{9}(2(1) + (-2)(2) + 1(2)) = \frac{1}{9}(2 - 4 + 2) = 0.$$

$$\vec{B} \cdot \vec{C} = \frac{1}{9}(1(2) + 2(1) + 2(-2)) = \frac{1}{9}(2 + 2 - 4) = 0.$$

$$\vec{C} \cdot \vec{A} = \frac{1}{9}(2(2) + 1(-2) + (-2)(1)) = \frac{1}{9}(4 - 2 - 2) = 0.$$

Since all magnitudes are 1 and all dot products are 0, they are mutually orthogonal unit vectors.

2.2 Cross Product (Vector Product)

Definition:

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \sin\theta \, \hat{n}$$

- **Output**: A new vector, perpendicular to the plane containing $\vec{A}$ and $\vec{B}$.
- **Physical Significance**: Represents a rotational effect. It is used to calculate torque (

$$\vec{\tau} = \vec{r} \times \vec{F}$$

). The magnitude equals the area of the parallelogram formed by the two vectors. - **Parallelism**: If two non-zero vectors are parallel, their cross product is $\vec{0}$.

2.2.1 Worked Example: Finding Perpendicular Vectors (PYQ 2020)

Problem: Find a vector of magnitude 5 perpendicular to both

$$\vec{A} = 2\hat{\imath} + \hat{\jmath} - 3\hat{k}$$

and

$$\vec{B} = \hat{\imath} - 2\hat{\jmath} + \hat{k}.$$

Solution: The cross product yields a perpendicular vector:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 1 & -3 \\ 1 & -2 & 1 \end{vmatrix} = \hat{\imath}(1-6) - \hat{\jmath}(2-(-3)) + \hat{k}(-4-1) = -5\hat{\imath} - 5\hat{\jmath} - 5\hat{k}$$

Simplify to a base direction vector

$$\vec{v} = \hat{\imath} + \hat{\jmath} + \hat{k}.$$

Find the unit normal vector $\hat{n}$:

$$\hat{n} = \pm \frac{\hat{\imath} + \hat{\jmath} + \hat{k}}{\sqrt{1^2 + 1^2 + 1^2}} = \pm \frac{\hat{\imath} + \hat{\jmath} + \hat{k}}{\sqrt{3}}$$

Multiply by the desired magnitude 5:

$$\vec{P} = \pm \frac{5}{\sqrt{3}} (\hat{\imath} + \hat{\jmath} + \hat{k})$$

2.3 Exam Patterns

- When asked to find a vector perpendicular to two other vectors, always use the cross product, normalize it to a unit vector, and then multiply by the requested magnitude. Include the $\pm$ sign to show both possible directions.
- "Show vectors are orthogonal" simply means proving their dot product is zero.

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3 A03: Area, Volume, and Triple Products

Exam Weight: Tier 2 | **Appeared in**: 2019, 2020, 2021, 2023 | **Typical Marks**: 3–4

Triple products allow you to find volumes of 3D shapes or determine if three vectors lie in the same flat plane (coplanar). Cross products natively provide the area of 2D shapes.

3.1 Area of a Triangle

The magnitude of the cross product of two vectors is the area of the parallelogram they form. Therefore, the area of a triangle formed by two edge vectors is half of that:

$$\text{Area} = \frac{1}{2}|\vec{A} \times \vec{B}|$$

3.1.1 Worked Example (PYQ 2021)

Problem: Find the area of a triangle with vertices at $P(3, -1, 2)$, $Q(1, -1, -3)$ and $R(4, -3, 1)$. **Solution**: Construct two edge vectors originating from the same point P:

$$\vec{PQ} = (1-3)\hat{\imath} + (-1-(-1))\hat{\jmath} + (-3-2)\hat{k} = -2\hat{\imath} - 5\hat{k}$$

$$\vec{PR} = (4-3)\hat{\imath} + (-3-(-1))\hat{\jmath} + (1-2)\hat{k} = \hat{\imath} - 2\hat{\jmath} - \hat{k}$$

Calculate the cross product:

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ -2 & 0 & -5 \\ 1 & -2 & -1 \end{vmatrix} = \hat{\imath}(0-10) - \hat{\jmath}(2-(-5)) + \hat{k}(4-0) = -10\hat{\imath} - 7\hat{\jmath} + 4\hat{k}$$

Find magnitude and area:

$$|\vec{PQ} \times \vec{PR}| = \sqrt{(-10)^2 + (-7)^2 + 4^2} = \sqrt{100 + 49 + 16} = \sqrt{165}$$

Area

$$= \frac{\sqrt{165}}{2}$$

3.2 Scalar Triple Product and Volume

The scalar triple product

$$\vec{A} \cdot (\vec{B} \times \vec{C})$$

calculates the volume of a parallelepiped (a 3D box with sheared sides) whose adjacent edges are $\vec{A}, \vec{B}, \vec{C}$.

It is evaluated as a 3×3 determinant:

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

Geometric Proof: The cross product $\vec{B} \times \vec{C}$ gives a normal vector whose magnitude is the area of the base. The dot product

$$\vec{A} \cdot (\vec{B} \times \vec{C})$$

projects the third edge $\vec{A}$ onto this normal vector, effectively multiplying the base area by the perpendicular height. Thus:

$$\text{Volume} = |\vec{A} \cdot (\vec{B} \times \vec{C})|$$

3.2.1 Worked Example: Volume (PYQ 2019)

Problem: Find the volume of the parallelepiped whose edges are

$$\vec{A} = 2\hat{\imath} - 3\hat{\jmath} + 4\hat{k},$$

$$\vec{B} = \hat{\imath} + 2\hat{\jmath} - \hat{k},$$

$$\vec{C} = 3\hat{\imath} - \hat{\jmath} + 2\hat{k}.$$

Solution:

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} 2 & -3 & 4 \\ 1 & 2 & -1 \\ 3 & -1 & 2 \end{vmatrix}$$

$$= 2(4-1) - (-3)(2-(-3)) + 4(-1-6) = 2(3) + 3(5) + 4(-7) = 6 + 15 - 28 = -7$$

Volume is the absolute value:

$$|-7| = 7$$

cubic units.

3.3 Coplanar Vectors

If three vectors lie in the exact same plane, the 3D parallelepiped they form has a height of zero. Therefore, its volume is zero.

Condition for Coplanarity: Three vectors are coplanar if and only if their scalar triple product is zero.

3.3.1 Worked Example: Unknown Constant (PYQ 2023)

Problem: Find the constant a such that

$$2\hat{\imath} - \hat{\jmath} + \hat{k},$$

$$\hat{\imath} + 2\hat{\jmath} - 3\hat{k},$$

$$3\hat{\imath} + a\hat{\jmath} + 5\hat{k}$$

are coplanar. **Solution**: Set the scalar triple product determinant to 0:

$$\begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -3 \\ 3 & a & 5 \end{vmatrix} = 0$$

$$2(10 - (-3a)) - (-1)(5 - (-9)) + 1(a - 6) = 0$$

$$2(10 + 3a) + 14 + a - 6 = 0 \implies 20 + 6a + 14 + a - 6 = 0 \implies 7a + 28 = 0 \implies a = -4$$

3.4 Exam Patterns

- When given three points and asked for an area, always form two vectors originating from the **same point** (e.g., $\vec{PQ}$ and $\vec{PR}$). Do not cross arbitrary position vectors.
- Scalar triple product calculations are very prone to arithmetic sign errors. Always write out the 2×2 determinant expansions carefully.

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4 A04: Vector Differentiation

Exam Weight: Tier 3 | **Appeared in**: 2019, 2024 | **Typical Marks**: 3–4

Differentiating a vector function works exactly like scalar differentiation, component by component. The product rule still applies to both dot and cross products.

4.1 The Basics

If a position vector is given as a function of time t:

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

The derivative with respect to t is:

$$\frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

4.1.1 Product Rules

When differentiating the product of two vector functions $\vec{A}(t)$ and $\vec{B}(t)$:

Dot Product Rule:

$$\frac{d}{dt}(\vec{A} \cdot \vec{B}) = \vec{A} \cdot \frac{d\vec{B}}{dt} + \frac{d\vec{A}}{dt} \cdot \vec{B}$$

Cross Product Rule (Order matters!):

$$\frac{d}{dt}(\vec{A} \times \vec{B}) = \vec{A} \times \frac{d\vec{B}}{dt} + \frac{d\vec{A}}{dt} \times \vec{B}$$

4.2 Must-Know Proof: Constant Magnitude (PYQ 2019)

Problem: If $\vec{A}$ has a constant magnitude, show that $\vec{A}$ and $\frac{d\vec{A}}{dt}$ are perpendicular, provided

$$\left|\frac{d\vec{A}}{dt}\right| \neq 0.$$

Proof: Let the constant magnitude of $\vec{A}$ be

$$|\vec{A}| = c$$

(where c is a constant). We know that the dot product of a vector with itself is its magnitude squared:

$$\vec{A} \cdot \vec{A} = |\vec{A}|^2 = c^2$$

Differentiate both sides with respect to t:

$$\frac{d}{dt}(\vec{A} \cdot \vec{A}) = \frac{d}{dt}(c^2)$$

Apply the dot product rule to the left side, and note that the derivative of a constant is 0:

$$\vec{A} \cdot \frac{d\vec{A}}{dt} + \frac{d\vec{A}}{dt} \cdot \vec{A} = 0$$

Since the dot product is commutative (

$$\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$$

):

$$2\left(\vec{A} \cdot \frac{d\vec{A}}{dt}\right) = 0 \implies \vec{A} \cdot \frac{d\vec{A}}{dt} = 0$$

Because their dot product is zero and neither vector is the zero vector, $\vec{A}$ and $\frac{d\vec{A}}{dt}$ must be mutually perpendicular. ■

4.3 Exam Patterns

- The "constant magnitude implies perpendicular derivative" proof is a classic theoretical question. Memorize the 4-step proof above.
- The fundamental concept of taking a derivative component-by-component is heavily used in the next topic: finding velocity and acceleration.

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5 A05: Velocity, Acceleration, and Particle Motion

Exam Weight: Tier 2 | **Appeared in**: 2019, 2020, 2023, 2024 | **Typical Marks**: 4

A particle's motion in space is described by its position vector $\vec{r}(t)$. Taking derivatives with respect to time t gives its velocity and acceleration.

5.1 The Core Formulas

Given a position vector $\vec{r}(t)$: 1. **Velocity**:

$$\vec{v}(t) = \frac{d\vec{r}}{dt}$$

2. **Acceleration**:

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

If asked for a component of velocity or acceleration in a specific direction, take the dot product of your result with the **unit vector** $\hat{u}$ of that direction.

5.2 Worked Example 1: Directional Components (PYQ 2019, 2023)

Problem: A particle moves along

$$x = 2t^2, y = t^2 - 4t, z = 3t - 5$$

. Find the components of velocity and acceleration at $t = 1$ in the direction of

$$\vec{d} = \hat{\imath} - 3\hat{\jmath} + 2\hat{k}.$$

Solution: **Step 1: Setup** Position:

$$\vec{r}(t) = 2t^2\hat{\imath} + (t^2 - 4t)\hat{\jmath} + (3t - 5)\hat{k}$$

Direction Unit Vector:

$$\hat{u} = \frac{\hat{\imath} - 3\hat{\jmath} + 2\hat{k}}{\sqrt{1^2 + (-3)^2 + 2^2}} = \frac{\hat{\imath} - 3\hat{\jmath} + 2\hat{k}}{\sqrt{14}}$$

Step 2: Velocity

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = 4t\hat{\imath} + (2t - 4)\hat{\jmath} + 3\hat{k}$$

At $t = 1$:

$$\vec{v}(1) = 4\hat{i} - 2\hat{j} + 3\hat{k}$$

Component in direction $\hat{u}$:

$$v_d = \vec{v}(1) \cdot \hat{u} = (4\hat{i} - 2\hat{j} + 3\hat{k}) \cdot \left(\frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}\right) = \frac{4(1) - 2(-3) + 3(2)}{\sqrt{14}} = \frac{4 + 6 + 6}{\sqrt{14}} = \frac{16}{\sqrt{14}}$$

Step 3: Acceleration

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = 4\hat{i} + 2\hat{j}$$

At $t = 1$:

$$\vec{a}(1) = 4\hat{i} + 2\hat{j}$$

(Acceleration is constant) Component in direction $\hat{u}$:

$$a_d = \vec{a}(1) \cdot \hat{u} = (4\hat{i} + 2\hat{j}) \cdot \left(\frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}\right) = \frac{4(1) + 2(-3) + 0}{\sqrt{14}} = \frac{4 - 6}{\sqrt{14}} = -\frac{2}{\sqrt{14}}$$

5.3 Worked Example 2: Trigonometric Motion (PYQ 2024)

Problem: A particle moves so that its position vector is

$$\vec{r} = \cos\omega t\hat{i} + \sin\omega t\hat{j}$$

where ω is constant. Show that velocity $\vec{v}$ is perpendicular to $\vec{r}$, and that $\vec{r} \times \vec{v}$ is a constant vector.

Solution: **Step 1: Velocity**

$$\vec{v} = \frac{d\vec{r}}{dt} = -\omega\sin\omega t\hat{i} + \omega\cos\omega t\hat{j}$$

Step 2: Show Perpendicular

$$\vec{r} \cdot \vec{v} = (\cos\omega t)(-\omega\sin\omega t) + (\sin\omega t)(\omega\cos\omega t) = -\omega\sin\omega t\cos\omega t + \omega\sin\omega t\cos\omega t = 0$$

Since the dot product is zero, $\vec{v}$ is perpendicular to $\vec{r}$.

Step 3: Show Constant Cross Product

$$\vec{r} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\omega t & \sin\omega t & 0 \\ -\omega\sin\omega t & \omega\cos\omega t & 0 \end{vmatrix}$$

$$= \hat{k}\left[(\cos\omega t)(\omega\cos\omega t) - (\sin\omega t)(-\omega\sin\omega t)\right] = \hat{k}(\omega\cos^2\omega t + \omega\sin^2\omega t)$$

$$= \omega(\cos^2\omega t + \sin^2\omega t)\hat{k} = \omega\hat{k}$$

Since ω is a constant, $\omega\hat{k}$ is a constant vector.

5.4 Exam Patterns

- When asked for a "component in the direction of", do not forget to divide the direction vector by its magnitude to make it a unit vector before dotting.
- Simple derivatives involving e^t, sin, and cos are common. Remember the chain rule for arguments like ωt or $3t$.

6 A06: Curvature, Torsion, and Frenet-Serret

Exam Weight: Tier 3 | **Appeared in**: 2020, 2023, CT | **Typical Marks**: 4–5

This section analyzes the geometric properties of a 3D curve in space. It relies heavily on taking multiple derivatives and cross products of the position vector $\vec{r}(t)$.

6.1 Formulas for Curvature and Torsion

Let $\vec{r}(t)$ be the position vector of a curve parameterized by t. You need the first three derivatives: 1. $\vec{r}'$ (Velocity) 2. $\vec{r}''$ (Acceleration) 3. $\vec{r}'''$ (Jerk)

Curvature (K): Measures how sharply a curve bends.

$$K = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$$

Torsion (T): Measures how sharply a curve twists out of its plane.

$$T = \frac{(\vec{r}' \times \vec{r}'') \cdot \vec{r}'''}{|\vec{r}' \times \vec{r}''|^2}$$

(Note: The numerator is a scalar triple product).

6.2 Frenet-Serret Formulas

The Frenet-Serret formulas describe the kinematic properties of a particle moving along a continuous, differentiable curve in 3D space.

Let s be the arc length along a space curve. - $\vec{T}$ = unit tangent vector - $\vec{N}$ = unit principal normal vector - $\vec{B}_b$ = unit binormal vector - κ = curvature - τ = torsion

The three formulas are: 1.

$$\frac{d\vec{T}}{ds} = \kappa\vec{N}$$

2.

$$\frac{d\vec{N}}{ds} = -\kappa\vec{T} + \tau\vec{B}_b$$

3.

$$\frac{d\vec{B}_b}{ds} = -\tau\vec{N}$$

6.3 Worked Example: Calculating and (PYQ 2020, 2023)

Problem: For the space curve

$$x = t, y = t^2, z = \frac{2}{3}t^3$$

, find the curvature K and the torsion T.

Solution: **Step 1: Find Derivatives**

$$\vec{r}(t) = t\hat{i} + t^2\hat{j} + \frac{2}{3}t^3\hat{k}$$

$$\vec{r'} = \hat{i} + 2t\hat{j} + 2t^2\hat{k}$$

$$\vec{r''} = 2\hat{j} + 4t\hat{k}$$

$$\vec{r'''} = 4\hat{k}$$

Step 2: Cross Product and Magnitudes

$$\vec{r'} \times \vec{r''} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2t & 2t^2 \\ 0 & 2 & 4t \end{vmatrix} = \hat{i}(8t^2 - 4t^2) - \hat{j}(4t - 0) + \hat{k}(2 - 0) = 4t^2\hat{i} - 4t\hat{j} + 2\hat{k}$$

Magnitude $|\vec{r'}|$:

$$|\vec{r'}| = \sqrt{1^2 + (2t)^2 + (2t^2)^2} = \sqrt{1 + 4t^2 + 4t^4} = \sqrt{(2t^2 + 1)^2} = 2t^2 + 1$$

Magnitude

$$|\vec{r'} \times \vec{r''}| :$$

$$|\vec{r'} \times \vec{r''}| = \sqrt{(4t^2)^2 + (-4t)^2 + 2^2} = \sqrt{16t^4 + 16t^2 + 4} = \sqrt{4(4t^4 + 4t^2 + 1)} = 2(2t^2 + 1)$$

Step 3: Calculate Curvature (K)

$$K = \frac{|\vec{r'} \times \vec{r''}|}{|\vec{r'}|^3} = \frac{2(2t^2 + 1)}{(2t^2 + 1)^3} = \frac{2}{(2t^2 + 1)^2}$$

Step 4: Calculate Torsion (T) Calculate the scalar triple product

$$(\vec{r'} \times \vec{r''}) \cdot \vec{r'''} :$$

$$(4t^2\hat{i} - 4t\hat{j} + 2\hat{k}) \cdot (4\hat{k}) = 8$$

$$T = \frac{(\vec{r}' \times \vec{r}'') \cdot \vec{r}'''}{|\vec{r}' \times \vec{r}''|^2} = \frac{8}{[2(2t^2+1)]^2} = \frac{8}{4(2t^2+1)^2} = \frac{2}{(2t^2+1)^2}$$

In this specific curve, curvature equals torsion!

6.4 Exam Patterns

- This specific problem (

$$x = t, y = t^2, z = \frac{2}{3}t^3$$

) is practically the only curvature/torsion problem asked. The trick is recognizing that $1 + 4t^2 + 4t^4$ is a perfect square $(2t^2+1)^2$. - Be able to state the Frenet-Serret formulas verbatim.

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7 A07: Gradient and Normal to a Surface

Exam Weight: Tier 1 | **Appeared in**: 2017, 2018, 2019, 2020, 2021, 2024 (6 papers) | **Typical Marks**: 2–5

The gradient is the foundation of vector calculus. It appears directly (find the gradient) and indirectly (in directional derivatives, surface normals, angle between surfaces).

7.1 The Del Operator

The del operator (nabla) is the vector differential operator:

$$\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

It acts on scalar functions (gradient), dot-products with vectors (divergence), and cross-products with vectors (curl).

7.2 The Gradient

7.2.1 Definition

The gradient of a scalar function $\phi(x, y, z)$ is:

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

Input: a scalar function. Output: a vector field.

7.2.2 Geometric Meaning

The gradient vector $\nabla\phi$ at any point is **perpendicular to the surface** $\phi(x, y, z) = c$ at that point.

The magnitude $|\nabla\phi|$ gives the maximum rate of change of ϕ at that point.

The direction of $\nabla\phi$ points toward the direction of greatest increase.

7.2.3 Properties

1. $$\nabla(f \pm g) = \nabla f \pm \nabla g$$

2. $$\nabla(fg) = f\nabla g + g\nabla f$$

3. $$\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}$$

4. ∇r^n:

$$\nabla r^n = nr^{n-2}\vec{r} \quad \text{(where } r = |\vec{r}|\text{)}$$

7.3 Proof: Gradient is Normal to a Surface (PYQ 2020)

7.3.1 Statement

$\nabla\phi$ is perpendicular to the surface $\phi(x, y, z) = c$.

7.3.2 Proof

Let

$$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}$$

be any curve lying on the surface $\phi = c$.

Since all points of this curve satisfy $\phi(x(t), y(t), z(t)) = c$, differentiate with respect to t:

$$\frac{\partial\phi}{\partial x}\frac{dx}{dt} + \frac{\partial\phi}{\partial y}\frac{dy}{dt} + \frac{\partial\phi}{\partial z}\frac{dz}{dt} = 0$$

This is a dot product:

$$\nabla\phi \cdot \frac{d\vec{r}}{dt} = 0$$

Since $\frac{d\vec{r}}{dt}$ is tangent to the curve, and the dot product is zero, $\nabla\phi$ is perpendicular to every tangent vector on the surface. So $\nabla\phi$ is normal to the surface.

$$\blacksquare$$

7.4 Unit Normal Vector

The unit normal to a surface $\phi = c$ at a point is:

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$$

7.4.1 Worked Example: Unit Normal (CT)

Problem: Find the unit outward normal to

$$(x - 1)^2 + y^2 + (z + 2)^2 = 9$$

at $(3, 1, -4)$.

Solution:

Let

$$f = (x-1)^2 + y^2 + (z+2)^2 - 9.$$

$$\nabla f = 2(x-1)\hat{i} + 2y\hat{j} + 2(z+2)\hat{k}$$

At $(3, 1, -4)$:

$$\nabla f = 4\hat{i} + 2\hat{j} - 4\hat{k}.$$

Magnitude:

$$|\nabla f| = \sqrt{16+4+16} = 6.$$

$$\hat{n} = \frac{4\hat{i} + 2\hat{j} - 4\hat{k}}{6} = \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$$

7.5 Angle Between Two Surfaces (PYQ 2017, 2019, 2021 — 3 Times)

7.5.1 The Method

The angle between two surfaces

$$\phi_1 = c_1$$

and

$$\phi_2 = c_2$$

at a point is the angle between their normal vectors:

$$\cos\theta = \frac{\nabla\phi_1 \cdot \nabla\phi_2}{|\nabla\phi_1||\nabla\phi_2|}$$

7.5.2 Must-Know Problem (Appeared 3 Times)

Problem: Find the angle between

$$x^2 + y^2 + z^2 = 9$$

and

$$z = x^2 + y^2 - 3$$

at $(2, -1, 2)$.

Solution:

Surface 1:

$$f_1 = x^2 + y^2 + z^2 - 9$$

. Gradient:

$$\nabla f_1 = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}.$$

At $(2, -1, 2)$:

$$\vec{n}_1 = 4\hat{i} - 2\hat{j} + 4\hat{k}.$$

Surface 2:

$$f_2 = x^2 + y^2 - z - 3$$

. Gradient:

$$\nabla f_2 = 2x\hat{i} + 2y\hat{j} - \hat{k}.$$

At $(2, -1, 2)$:

$$\vec{n}_2 = 4\hat{i} - 2\hat{j} - \hat{k}.$$

Dot product:

$$\vec{n}_1 \cdot \vec{n}_2 = 16 + 4 - 4 = 16.$$

Magnitudes:

$$|\vec{n}_1| = 6,$$

$$|\vec{n}_2| = \sqrt{21}.$$

$$\cos\theta = \frac{16}{6\sqrt{21}} = \frac{8}{3\sqrt{21}}$$

$$\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$$

7.6 Finding the Gradient at a Point (PYQ 2024)

Problem: If

$$Q = 3x^2y - y^3z^2$$

, find ∇Q at $(1, -2, -1)$.

Solution:

$$\frac{\partial Q}{\partial x} = 6xy, \quad \frac{\partial Q}{\partial y} = 3x^2 - 3y^2z^2, \quad \frac{\partial Q}{\partial z} = -2y^3z$$

At $(1, -2, -1)$:

$$\frac{\partial Q}{\partial x} = 6(1)(-2) = -12$$

$$\frac{\partial Q}{\partial y} = 3(1) - 3(4)(1) = -9$$

$$\frac{\partial Q}{\partial z} = -2(-8)(-1) = -16$$

$$\nabla Q = -12\hat{\imath} - 9\hat{\jmath} - 16\hat{k}$$

7.7 Exam Patterns

- "Find the angle between two surfaces" has appeared 3 times (2017, 2019, 2021). The exact same surfaces. Memorize it.
- "Find the gradient at a point" is a 2-3 mark easy question.
- "Show that gradient is normal to the surface" appeared in 2020 for 3 marks.
- The gradient formula is used in directional derivative (A08) and tangent plane (A09) problems. It is foundational.

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8 A08: Directional Derivative

Exam Weight: Tier 1 | **Appeared in**: 2017, 2018, 2019, 2021 | **Typical Marks**: 3–4

The directional derivative measures how fast a scalar function changes in a specific direction. It builds directly on the gradient.

8.1 Definition

The directional derivative of ϕ at point P in the direction of unit vector $\hat{u}$ is:

$$D_{\hat{u}}\phi = \nabla\phi \cdot \hat{u}$$

It gives the rate of change of ϕ at P in the direction $\hat{u}$.

8.1.1 Key Facts

- Maximum value of $D_{\hat{u}}\phi$ is $|\nabla\phi|$, achieved in the direction of $\nabla\phi$ itself.
- Minimum value is $-|\nabla\phi|$, in the opposite direction.
-

$$D_{\hat{u}}\phi = 0$$

when $\hat{u}$ is perpendicular to $\nabla\phi$ (moving along the level surface).

8.2 The Three-Step Method

1. **Find the gradient** $\nabla\phi$ and evaluate at the given point.
2. **Find the unit direction vector**

$$\hat{u} = \frac{\vec{d}}{|\vec{d}|}.$$

3. **Dot product**:

$$D_{\hat{u}}\phi = \nabla\phi \cdot \hat{u}.$$

8.3 Worked Example 1 (PYQ 2018)

Problem: Find the directional derivative of

$$\phi = x^2yz + 4xz^2$$

at $(1, -2, -1)$ in the direction

$$2\hat{i} - \hat{j} - 2\hat{k}.$$

Solution:

Step 1: Gradient.

$$\nabla\phi = (2xyz + 4z^2)\hat{i} + x^2z\hat{j} + (x^2y + 8xz)\hat{k}$$

At $(1, -2, -1)$:

$$\nabla\phi = (4 + 4)\hat{i} + (1)(-1)\hat{j} + (-2 - 8)\hat{k} = 8\hat{i} - \hat{j} - 10\hat{k}$$

Step 2: Unit vector.

$$\hat{u} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{4 + 1 + 4}} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$$

Step 3: Dot product.

$$D_{\hat{u}}\phi = \frac{8(2) + (-1)(-1) + (-10)(-2)}{3} = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

8.4 Worked Example 2 (PYQ 2017)

Problem: Find the directional derivative of

$$\phi = 4e^{2x-y+z}$$

at $(1, 1, -1)$ toward $(-3, 5, 6)$.

Solution:

Step 1: Direction from $P(1, 1, -1)$ to $Q(-3, 5, 6)$:

$$\vec{d} = -4\hat{i} + 4\hat{j} + 7\hat{k}, \quad |\vec{d}| = \sqrt{16 + 16 + 49} = 9$$

$$\hat{u} = \frac{-4\hat{i} + 4\hat{j} + 7\hat{k}}{9}$$

Step 2: Gradient of

$$\phi = 4e^{2x-y+z} :$$

$$\nabla\phi = 4e^{2x-y+z}(2\hat{i} - \hat{j} + \hat{k})$$

At $(1, 1, -1)$: exponent $= 2 - 1 - 1 = 0$, so

$$e^0 = 1 :$$

$$\nabla\phi = 8\hat{i} - 4\hat{j} + 4\hat{k}$$

Step 3:

$$D_{\hat{u}}\phi = \frac{8(-4) + (-4)(4) + 4(7)}{9} = \frac{-32 - 16 + 28}{9} = -\frac{20}{9}$$

8.5 Worked Example 3 (PYQ 2021)

Problem: Find the directional derivative of

$$f = x^2 + xy + z^2$$

at $A(1, -1, -1)$ in the direction of AB where $B(3, 2, 1)$.

Solution:

Direction:

$$\vec{AB} = 2\hat{i} + 3\hat{j} + 2\hat{k}.$$

Unit vector:

$$\hat{u} = \frac{2\hat{i} + 3\hat{j} + 2\hat{k}}{\sqrt{17}}.$$

Gradient:

$$\nabla f = (2x + y)\hat{i} + x\hat{j} + 2z\hat{k}.$$

At $(1, -1, -1)$:

$$\nabla f = \hat{i} + \hat{j} - 2\hat{k}.$$

$$D_{\hat{u}} f = \frac{2 + 3 - 4}{\sqrt{17}} = \frac{1}{\sqrt{17}}$$

8.6 Worked Example 4 (PYQ 2019)

Problem: Find directional derivative of

$$\phi = x^2yz + 4xz^2$$

at $(1, -2, 1)$ in direction

$$2\hat{i} - \hat{j} - 2\hat{k}.$$

Solution:

At $(1, -2, 1)$:

$$\nabla\phi = 0\hat{i} + \hat{j} + 6\hat{k}.$$

$$D_{\hat{u}}\phi = \frac{0(2) + 1(-1) + 6(-2)}{3} = \frac{-13}{3}$$

8.7 Exam Patterns

- Directional derivative appears in about 4 out of 7 papers.
- The same function

$$\phi = x^2yz + 4xz^2$$

has appeared in 2018 and 2019 with different points. - Always show the three steps clearly: gradient, unit vector, dot product. - The direction is sometimes given as a vector (just normalize it) and sometimes as “toward point Q” (compute $\vec{PQ}$ first, then normalize).

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9 A09: Tangent Plane and Normal Line

Exam Weight: Tier 2 | **Appeared in**: 2018 | **Typical Marks**: 3

The gradient vector ∇F is perpendicular to the surface $F(x, y, z) = C$. We use this fact to find the equations of the tangent plane and the normal line at a given point.

9.1 Formulas

Let a surface be given by $F(x, y, z) = C$, and let $P(x_0, y_0, z_0)$ be a point on this surface.

The normal vector to the surface at P is the gradient evaluated at P:

$$\vec{n} = \nabla F|_P = \frac{\partial F}{\partial x}\hat{i} + \frac{\partial F}{\partial y}\hat{j} + \frac{\partial F}{\partial z}\hat{k}$$

Let the components of this normal vector be

$$\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}.$$

9.1.1 Equation of the Tangent Plane

The tangent plane passes through P and is perpendicular to $\vec{n}$. Its equation is:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Or, in terms of partial derivatives:

$$\frac{\partial F}{\partial x}(x - x_0) + \frac{\partial F}{\partial y}(y - y_0) + \frac{\partial F}{\partial z}(z - z_0) = 0$$

9.1.2 Equation of the Normal Line

The normal line passes through P and is parallel to $\vec{n}$. Its symmetric equations are:

$$\frac{x - x_0}{A} = \frac{y - y_0}{B} = \frac{z - z_0}{C}$$

9.2 Worked Example 1 (PYQ 2018)

Problem: Find an equation for the tangent plane to the surface

$$2xz^2 - 3xy - 4x = 7$$

at the point $(1, -1, 2)$.

Solution:

Step 1: Define the surface function $F(x, y, z)$.

$$F(x, y, z) = 2xz^2 - 3xy - 4x - 7 = 0$$

Step 2: Find the partial derivatives (the components of the gradient).

$$\frac{\partial F}{\partial x} = 2z^2 - 3y - 4$$

$$\frac{\partial F}{\partial y} = -3x$$

$$\frac{\partial F}{\partial z} = 4xz$$

Step 3: Evaluate the partial derivatives at the point $(1, -1, 2)$.

$$A = \frac{\partial F}{\partial x}(1, -1, 2) = 2(2)^2 - 3(-1) - 4 = 8 + 3 - 4 = 7$$

$$B = \frac{\partial F}{\partial y}(1, -1, 2) = -3(1) = -3$$

$$C = \frac{\partial F}{\partial z}(1, -1, 2) = 4(1)(2) = 8$$

The normal vector is

$$\vec{n} = 7\hat{\imath} - 3\hat{\jmath} + 8\hat{k}.$$

Step 4: Write the equation of the tangent plane.

Using the formula

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0 :$$

$$7(x - 1) - 3(y - (-1)) + 8(z - 2) = 0$$

$$7(x - 1) - 3(y + 1) + 8(z - 2) = 0$$

$$7x - 7 - 3y - 3 + 8z - 16 = 0$$

$$7x - 3y + 8z = 26$$

The equation of the tangent plane is $7x - 3y + 8z = 26$.

*(Note: If the question also asked for the normal line, it would be:

$$\frac{x - 1}{7} = \frac{y + 1}{-3} = \frac{z - 2}{8}$$

)*

9.3 Exam Patterns

- "Find the equation of the tangent plane" is a straightforward 3-mark application of the gradient.
- Always bring all terms to one side to define $F(x, y, z) = 0$ before taking derivatives.
- The normal vector components (A, B, C) become the coefficients of x, y, z in the final plane equation.

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10 A10: Divergence and Curl

> **Exam Weight**: Tier 1 | **Appeared in**: Core concept used in almost every vector calculus question.

While gradient operates on scalars, divergence and curl operate on vector fields. They describe the physical behavior of the field (expansion/compression and rotation).

10.1 Divergence (Dot Product)

10.1.1 Definition

The divergence of a vector field

$$\vec{F} = F_1\hat{\imath} + F_2\hat{\jmath} + F_3\hat{k}$$

is the dot product of the Del operator and $\vec{F}$:

$$\text{div } \vec{F} = \nabla \cdot \vec{F} = \left(\hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right) \cdot (F_1\hat{\imath} + F_2\hat{\jmath} + F_3\hat{k})$$

$$\text{div } \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Input: Vector field. Output: **Scalar quantity**.

10.1.2 Physical Significance

Divergence represents the net outward flux (flow) per unit volume from a point. - **Positive divergence**: The point is a source (fluid expanding outward). - **Negative divergence**: The point is a sink (fluid compressing inward).

10.1.3 Solenoidal Vector

If

$$\nabla \cdot \vec{F} = 0$$

everywhere, the vector field is called **solenoidal**. It behaves like an incompressible fluid with no sources or sinks.

10.2 Curl (Cross Product)

10.2.1 Definition

The curl of a vector field $\vec{F}$ is the cross product of the Del operator and $\vec{F}$:

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Expanding this gives:

$$\nabla \times \vec{F} = \hat{i}\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right) - \hat{j}\left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}\right) + \hat{k}\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)$$

Input: Vector field. Output: **Vector quantity**.

10.2.2 Physical Significance

Curl measures the rotational tendency or "spin" of the vector field at a point (like a vortex or whirlpool). The direction of the curl vector is the axis of rotation, given by the right-hand rule.

10.2.3 Irrotational Vector

If

$$\nabla \times \vec{F} = \vec{0}$$

everywhere, the vector field is called **irrotational**. This means the field has absolutely no rotational flow or twisting motion.

10.3 Properties of Position Vector

Let the position vector be

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}.$$

Then

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}.$$

Two absolute must-know facts:

1. **Divergence of $\vec{r}$ is 3**:

$$\nabla \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3$$

2. **Curl of $\vec{r}$ is zero**:

$$\nabla \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \hat{i}(0) - \hat{j}(0) + \hat{k}(0) = \vec{0}$$

10.4 Worked Example 1: Solenoidal Condition

Problem: Determine a such that the vector

$$\vec{F} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x-az)\hat{k}$$

is solenoidal.

Solution:

For $\vec{F}$ to be solenoidal,

$$\nabla \cdot \vec{F} = 0.$$

$$\frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x-az) = 0$$

$$1 + 1 - a = 0 \implies a = 2$$

10.5 Worked Example 2: Evaluating Divergence

Problem: Prove that

$$\text{div}(r^n\vec{r}) = (n+3)r^n$$

. For what value of n is it solenoidal?

Solution:

We know

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

, and similarly for y and z.

$$\nabla \cdot (r^n x\hat{i} + r^n y\hat{j} + r^n z\hat{k}) = \frac{\partial}{\partial x}(r^n x) + \frac{\partial}{\partial y}(r^n y) + \frac{\partial}{\partial z}(r^n z)$$

Apply the product rule to the x term:

$$\frac{\partial}{\partial x}(r^n x) = r^n(1) + x\left(nr^{n-1}\frac{\partial r}{\partial x}\right) = r^n + nxr^{n-1}\left(\frac{x}{r}\right) = r^n + nx^2r^{n-2}$$

Summing the terms for x, y, z:

$$\text{div}(r^n\vec{r}) = 3r^n + nr^{n-2}(x^2+y^2+z^2)$$

Since

$$x^2 + y^2 + z^2 = r^2 :$$

$$\mathrm{div}(r^n\vec{r}) = 3r^n + nr^{n-2}(r^2) = 3r^n + nr^n = (n+3)r^n$$

For the vector to be solenoidal, the divergence must be zero:

$$(n+3)r^n = 0 \implies n = -3$$

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11 A11: Irrotational and Conservative Fields

Exam Weight: Tier 1 | **Appeared in**: 2018 (twice), 2019, 2020, 2021, CT | **Typical Marks**: 3–9

This is one of the most frequently tested topics in the entire course. You must know how to prove a field is irrotational (curl = 0) and how to find its scalar potential.

11.1 Conservative Force Fields

11.1.1 Definition

A force field $\vec{F}$ is **conservative** if the work done in moving a particle between two points is independent of the path taken.

Mathematical equivalent: A vector field $\vec{F}$ is conservative (or irrotational) if and only if its curl is zero:

$$\nabla \times \vec{F} = \vec{0}$$

11.1.2 Scalar Potential

If $\vec{F}$ is a conservative field, it can be written as the gradient of a scalar function ϕ:

$$\vec{F} = \nabla\phi$$

This function $\phi(x, y, z)$ is called the **scalar potential** of $\vec{F}$.

*(Note:

$$\nabla \times (\nabla\phi) = \vec{0}$$

is a fundamental vector identity. The curl of a gradient is always zero).*

11.2 The Method: Finding Scalar Potential

To find ϕ such that

$$\vec{F} = \nabla\phi$$

, where

$$\vec{F} = F_1\hat{\imath} + F_2\hat{\jmath} + F_3\hat{k} ::$$

1. Set up three equations:

$$\frac{\partial\phi}{\partial x} = F_1 \quad (1)$$

$$\frac{\partial \phi}{\partial y} = F_2 \quad (2)$$

$$\frac{\partial \phi}{\partial z} = F_3 \quad (3)$$

2. Integrate (1) with respect to x. Add a function $g(y, z)$ instead of a constant C.
3. Differentiate your result with respect to y, and equate it to (2) to find

$$\frac{\partial g}{\partial y}.$$

4. Integrate to find $g(y, z)$, adding a function $h(z)$.
5. Substitute $g(y, z)$ back into your ϕ equation.
6. Differentiate with respect to z, equate to (3) to find $h'(z)$.
7. Integrate to find $h(z)$, adding a final constant C.

11.3 Must-Know Problem 1 (PYQ 2020)

Problem: Show that

$$\vec{A} = (6xy + z^3)\hat{\imath} + (3x^2 - z)\hat{\jmath} + (3xz^2 - y)\hat{k}$$

is irrotational. Find ϕ such that

$$\vec{A} = \nabla \phi.$$

Solution:

Step 1: Show Irrotational.

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy + z^3 & 3x^2 - z & 3xz^2 - y \end{vmatrix}$$

- $\hat{\imath}$-component:

$$\frac{\partial}{\partial y}(3xz^2 - y) - \frac{\partial}{\partial z}(3x^2 - z) = -1 - (-1) = 0$$

- $\hat{\jmath}$-component:

$$\frac{\partial}{\partial z}(6xy + z^3) - \frac{\partial}{\partial x}(3xz^2 - y) = 3z^2 - 3z^2 = 0$$

- $\hat{k}$-component:

$$\frac{\partial}{\partial x}(3x^2 - z) - \frac{\partial}{\partial y}(6xy + z^3) = 6x - 6x = 0$$

Curl is zero. It is irrotational.

Step 2: Find Potential.

$$\frac{\partial \phi}{\partial x} = 6xy + z^3 \implies \phi = 3x^2y + xz^3 + g(y,z) \quad \text{(Eq A)}$$

Differentiate (Eq A) w.r.t y:

$$\frac{\partial \phi}{\partial y} = 3x^2 + \frac{\partial g}{\partial y} = 3x^2 - z \implies \frac{\partial g}{\partial y} = -z$$

Integrate w.r.t y: $g(y,z) = -yz + h(z)$. Substitute back into (Eq A):

$$\phi = 3x^2y + xz^3 - yz + h(z) \quad \text{(Eq B)}$$

Differentiate (Eq B) w.r.t z:

$$\frac{\partial \phi}{\partial z} = 3xz^2 - y + h'(z) = 3xz^2 - y \implies h'(z) = 0 \implies h(z) = C$$

Final Potential:

$$\phi = 3x^2y + xz^3 - yz + C$$

11.4 Must-Know Problem 2 (PYQ 2021)

Problem: Show that

$$\vec{F} = (y^2 + 2xz^2)\hat{i} + (2xy - z)\hat{j} + (2x^2z - y + 2z)\hat{k}$$

is irrotational and find scalar potential.

Solution:

Step 1: Show Irrotational.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 + 2xz^2 & 2xy - z & 2x^2z - y + 2z \end{vmatrix}$$

- $\hat{i}$: $(-1) - (-1) = 0$
- $\hat{j}$: $(4xz) - (4xz) = 0$
- $\hat{k}$: $(2y) - (2y) = 0$

Step 2: Find Potential.

$$\frac{\partial \phi}{\partial x} = y^2 + 2xz^2 \implies \phi = xy^2 + x^2z^2 + g(y,z)$$

$$\frac{\partial \phi}{\partial y} = 2xy + \frac{\partial g}{\partial y} = 2xy - z \implies \frac{\partial g}{\partial y} = -z \implies g(y,z) = -yz + h(z)$$

$$\phi = xy^2 + x^2z^2 - yz + h(z)$$

$$\frac{\partial\phi}{\partial z} = 2x^2z - y + h'(z) = 2x^2z - y + 2z \implies h'(z) = 2z \implies h(z) = z^2 + C$$

Final Potential:

$$\phi = xy^2 + x^2z^2 - yz + z^2 + C$$

11.5 Must-Know Problem 3: Finding Unknown Constants (PYQ 2018)

Problem: Find constants a, b, c so that

$$\vec{V} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)\hat{k}$$

is irrotational.

Solution:

Set

$$\nabla \times \vec{V} = 0:$$

$$\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + 2y + az & bx - 3y - z & 4x + cy + 2z \end{vmatrix} = 0$$

- $\hat{\imath}$-component:

$$\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z) = c - (-1) = c + 1 = 0 \implies c = -1$$

- $\hat{\jmath}$-component:

$$\frac{\partial}{\partial z}(x + 2y + az) - \frac{\partial}{\partial x}(4x + cy + 2z) = a - 4 = 0 \implies a = 4$$

- $\hat{k}$-component:

$$\frac{\partial}{\partial x}(bx - 3y - z) - \frac{\partial}{\partial y}(x + 2y + az) = b - 2 = 0 \implies b = 2$$

Constants are $a = 4, b = 2, c = -1$.

11.6 Exam Patterns

- "Show $\vec{F}$ is conservative and find scalar potential" is a 6-mark staple.
- "Find a, b, c such that $\vec{F}$ is irrotational" is a 3-mark easy question.
- Always write the final potential function with the integration constant $+C$.
- The determinant calculation for curl must be shown explicitly to get full marks.

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12 A12: Vector Identities and Proofs

Exam Weight: Tier 2 | **Appeared in**: 2018, 2019, 2021, 2024 | **Typical Marks**: 3–5

Vector identity proofs test your ability to apply the Del operator systematically using the chain, product, and quotient rules.

12.1 The Four Fundamental Identities

You should memorize these results, as they often simplify complex problems:

1. **Curl of Gradient is Zero**:

$$\nabla \times (\nabla \phi) = \vec{0}$$

2. **Divergence of Curl is Zero**:

$$\nabla \cdot (\nabla \times \vec{A}) = 0$$

3. **Curl of Curl**:

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$$

4. **Divergence of Gradient is Laplacian**:

$$\nabla \cdot (\nabla \phi) = \nabla^2 \phi$$

12.2 Proof 1: (PYQ 2018)

Statement: If $\vec{A}$ is a constant vector, then

$$\nabla(\vec{r} \cdot \vec{A}) = \vec{A}.$$

Proof:

Let the constant vector be

$$\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}.$$

Let the position vector be

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}.$$

Calculate the dot product:

$$\vec{r} \cdot \vec{A} = A_1 x + A_2 y + A_3 z$$

Now calculate the gradient of this scalar:

$$\nabla(\vec{r}\cdot\vec{A}) = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)(A_1x + A_2y + A_3z)$$

$$\nabla(\vec{r}\cdot\vec{A}) = \hat{i}\frac{\partial}{\partial x}(A_1x + A_2y + A_3z) + \hat{j}\frac{\partial}{\partial y}(A_1x + A_2y + A_3z) + \hat{k}\frac{\partial}{\partial z}(A_1x + A_2y + A_3z)$$

$$\nabla(\vec{r}\cdot\vec{A}) = A_1\hat{i} + A_2\hat{j} + A_3\hat{k} = \vec{A} \qquad \blacksquare$$

12.3 Proof 2: Curl of Curl Identity (PYQ 2018, 2021)

Statement:

$$\nabla\times(\nabla\times\vec{A}) = \nabla(\nabla\cdot\vec{A}) - \nabla^2\vec{A}.$$

Proof:

Let

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}.$$

Let

$$\vec{V} = \nabla\times\vec{A}$$

. The components of $\vec{V}$ are:

$$V_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}, \quad V_y = \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}, \quad V_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}$$

We want to find:

$$\nabla\times\vec{V}$$

Evaluate its x-component:

$$(\nabla\times\vec{V})_x = \frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}$$

Substitute V_z and V_y:

$$(\nabla\times\vec{V})_x = \frac{\partial}{\partial y}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) - \frac{\partial}{\partial z}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)$$

$$(\nabla\times\vec{V})_x = \frac{\partial^2 A_y}{\partial y\partial x} - \frac{\partial^2 A_x}{\partial y^2} - \frac{\partial^2 A_x}{\partial z^2} + \frac{\partial^2 A_z}{\partial z\partial x}$$

Add and subtract

$$\frac{\partial^2 A_x}{\partial x^2}:$$

$$(\nabla \times \vec{V})_x = \left(\frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_y}{\partial y \partial x} + \frac{\partial^2 A_z}{\partial z \partial x}\right) - \left(\frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_x}{\partial y^2} + \frac{\partial^2 A_x}{\partial z^2}\right)$$

Factor out the operators:

$$(\nabla \times \vec{V})_x = \frac{\partial}{\partial x}\left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) A_x$$

$$(\nabla \times \vec{V})_x = \frac{\partial}{\partial x}(\nabla \cdot \vec{A}) - \nabla^2 A_x$$

By symmetry for y and z components, assembling the vector gives:

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} \qquad \blacksquare$$

12.4 Proof 3: (PYQ 2019)

Statement: Prove that the Laplacian of $1/r$ is zero (except at origin).

Proof:

Let

$$r = \sqrt{x^2 + y^2 + z^2}$$

. We know

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

, etc.

First, find the gradient of r^{-1}:

$$\frac{\partial}{\partial x}(r^{-1}) = -r^{-2}\frac{\partial r}{\partial x} = -r^{-2}\left(\frac{x}{r}\right) = -\frac{x}{r^3}$$

So,

$$\nabla(r^{-1}) = -\frac{x\hat{i} + y\hat{j} + z\hat{k}}{r^3} = -\frac{\vec{r}}{r^3}.$$

Now find the divergence of this gradient (which is ∇^2):

$$\nabla^2(r^{-1}) = \nabla \cdot \left(-\frac{\vec{r}}{r^3}\right) = -\left[\frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) + \frac{\partial}{\partial y}\left(\frac{y}{r^3}\right) + \frac{\partial}{\partial z}\left(\frac{z}{r^3}\right)\right]$$

Using the quotient rule on the x-component:

$$\frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) = \frac{r^3(1) - x(3r^2\frac{\partial r}{\partial x})}{r^6} = \frac{r^3 - 3x^2 r}{r^6} = \frac{1}{r^3} - \frac{3x^2}{r^5}$$

Summing the x, y, and z terms:

$$\nabla^2(r^{-1}) = -\left[\left(\frac{1}{r^3} - \frac{3x^2}{r^5}\right) + \left(\frac{1}{r^3} - \frac{3y^2}{r^5}\right) + \left(\frac{1}{r^3} - \frac{3z^2}{r^5}\right)\right]$$

$$\nabla^2(r^{-1}) = -\left[\frac{3}{r^3} - \frac{3(x^2+y^2+z^2)}{r^5}\right]$$

Since

$$x^2 + y^2 + z^2 = r^2 :$$

$$\nabla^2(r^{-1}) = -\left[\frac{3}{r^3} - \frac{3r^2}{r^5}\right] = -\left[\frac{3}{r^3} - \frac{3}{r^3}\right] = 0 \qquad \blacksquare$$

12.5 Exam Patterns

- The identity proof for

$$\nabla \times (\nabla \times \vec{A})$$

12.6 is tedious but important (appeared in 2018 and 2021). Write out the x-component clearly, then use symmetry.

$$\nabla^2(1/r) = 0$$

is a classic vector calculus result. The trick is using

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

correctly with the quotient rule.

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13 A13: Line Integrals and Work Done

> **Exam Weight**: Tier 1-2 | **Appeared in**: 2017, 2018, 2020, 2021, 2024, CT | **Typical Marks**: 4–6

A line integral evaluates a function along a curve. The most common application is calculating the work done by a force field along a path.

13.1 Formula for Work Done

The work done by a force field $\vec{F}$ in moving a particle along a curve C is:

$$W = \int_C \vec{F} \cdot d\vec{r}$$

If

$$\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$$

and

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} ::$$

$$W = \int_C (F_1 dx + F_2 dy + F_3 dz)$$

13.1.1 How to Evaluate

Convert everything to a single variable (a parameter like t, or x if the path is $y = f(x)$). Then integrate normally.

13.2 Worked Example 1: Parametric Path (PYQ 2024)

Problem: Find the work done in a force field

$$\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$$

along the curve

$$x = t^2 + 1, y = 2t^2, z = t^3$$

from $t = 1$ to $t = 2$.

Solution:

Step 1: Find differentials.

$$x = t^2 + 1 \implies dx = 2tdt$$

$$y = 2t^2 \implies dy = 4tdt$$

$$z = t^3 \implies dz = 3t^2dt$$

Step 2: Substitute into the dot product

$$\vec{F} \cdot d\vec{r} = 3xydx - 5zdy + 10xdz.$$

- $$3xydx = 3(t^2 + 1)(2t^2)(2tdt) = (12t^5 + 12t^3)dt$$

- $$-5zdy = -5(t^3)(4tdt) = -20t^4dt$$

- $$10xdz = 10(t^2 + 1)(3t^2dt) = (30t^4 + 30t^2)dt$$

Step 3: Sum and integrate.

$$W = \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2)dt$$

$$W = \left[2t^6 + 2t^5 + 3t^4 + 10t^3\right]_1^2$$

Upper limit ($t = 2$): $2(64) + 2(32) + 3(16) + 10(8) = 128 + 64 + 48 + 80 = 320$ Lower limit ($t = 1$): $2 + 2 + 3 + 10 = 17$

$$W = 320 - 17 = 303$$

13.3 Worked Example 2: Planar Path (PYQ 2017)

Problem: Evaluate

$$\int_C (3xy\hat{\imath} - y^2\hat{\jmath}) \cdot d\vec{r}$$

when C is

$$y = 2x^2$$

from $(0, 0)$ to $(1, 2)$.

Solution:

We have

$$\vec{F} \cdot d\vec{r} = 3xydx - y^2dy.$$

Path:

$$y = 2x^2 \implies dy = 4xdx.$$

Limits for x: 0 to 1.

Substitute y and dy:

$$\int_0^1 \left[3x(2x^2)dx - (2x^2)^2(4x)dx\right] = \int_0^1 (6x^3 - 16x^5)dx$$

$$= \left[\frac{3}{2}x^4 - \frac{8}{3}x^6\right]_0^1 = \frac{3}{2} - \frac{8}{3} = \frac{9-16}{6} = -\frac{7}{6}$$

13.4 Worked Example 3: Segmented Path (PYQ 2021)

Problem: Evaluate

$$\int_C \vec{A} \cdot d\vec{r}$$

where

$$\vec{A} = (3x^2 + 6y)\hat{\imath} - 14yz\hat{\jmath} + 20xz^2\hat{k}$$

along straight lines from $(0,0,0)$ to $(1,0,0)$ to $(1,1,0)$ to $(1,1,1)$.

Solution:

Break into three paths: C_1, C_2, C_3.

$$\vec{A} \cdot d\vec{r} = (3x^2 + 6y)dx - 14yzdy + 20xz^2dz.$$

Path C_1: $(0,0,0)$ to $(1,0,0)$. $y = 0, z = 0 \implies dy = 0, dz = 0$. x goes 0 to 1.

$$\int_{C_1} = \int_0^1 (3x^2 + 0)dx = [x^3]_0^1 = 1$$

Path C_2: $(1,0,0)$ to $(1,1,0)$. $x = 1, z = 0 \implies dx = 0, dz = 0$. y goes 0 to 1.

$$\int_{C_2} = \int_0^1 -14y(0)dy = 0$$

Path C_3: $(1,1,0)$ to $(1,1,1)$. $x = 1, y = 1 \implies dx = 0, dy = 0$. z goes 0 to 1.

$$\int_{C_3} = \int_0^1 20(1)z^2dz = \left[\frac{20}{3}z^3\right]_0^1 = \frac{20}{3}$$

Total Integral =

$$1 + 0 + \frac{20}{3} = \frac{23}{3}.$$

13.5 Exam Patterns

- Parametric path questions (like Example 1) are very common. Substitute x, y, z and dx, dy, dz perfectly. Watch your algebra.
- Segmented straight line paths (Example 3) are simple but test your logic. On each segment, two variables are constant (so their differentials are zero).
- Always explicitly write the dot product

$$\vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

first.

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14 A14: Surface Integrals

Exam Weight: Tier 3 | **Appeared in**: 2017, 2020 | **Typical Marks**: 6

A surface integral calculates the total flux of a vector field passing through a surface. It is the 2D equivalent of a line integral. Most surface integrals in exams are evaluated using the Divergence Theorem or Stokes' Theorem, but occasionally you must calculate them directly.

14.1 The Formula

The flux of a vector field $\vec{A}$ through a surface S is:

$$\text{Flux} = \iint_S \vec{A} \cdot \hat{n} dS$$

where $\hat{n}$ is the unit outward normal vector to the surface.

14.1.1 Evaluation by Projection

To evaluate this directly, we project the surface S onto one of the coordinate planes (e.g., the xy-plane).

Let R be the projected region in the xy-plane. The differential area dS on the surface relates to the area $dxdy$ on the plane by:

$$dS = \frac{dxdy}{|\hat{n} \cdot \hat{k}|}$$

If projecting onto the yz-plane:

$$dS = \frac{dydz}{|\hat{n} \cdot \hat{i}|}.$$

If projecting onto the zx-plane:

$$dS = \frac{dzdx}{|\hat{n} \cdot \hat{j}|}.$$

14.2 Must-Know Problem: Cylinder Surface (PYQ 2017, 2020)

Problem: Evaluate

$$\iint_S \vec{A} \cdot \hat{n} dS$$

where

$$\vec{A} = z\hat{i} + x\hat{j} - 3y^2 z\hat{k}$$

and S is the surface of the cylinder

$$x^2 + y^2 = 16$$

included in the first octant between $z = 0$ and $z = 5$.

Solution:

Step 1: Find the normal vector $\hat{n}$. The surface is

$$g(x, y, z) = x^2 + y^2 - 16 = 0.$$

Gradient:

$$\nabla g = 2x\hat{\imath} + 2y\hat{\jmath}.$$

Magnitude:

$$|\nabla g| = \sqrt{4x^2 + 4y^2} = 2\sqrt{16} = 8.$$

Unit normal:

$$\hat{n} = \frac{2x\hat{\imath} + 2y\hat{\jmath}}{8} = \frac{x\hat{\imath} + y\hat{\jmath}}{4}$$

Step 2: Calculate $\vec{A} \cdot \hat{n}$.

$$\vec{A} \cdot \hat{n} = (z\hat{\imath} + x\hat{\jmath} - 3y^2z\hat{k}) \cdot \left(\frac{x\hat{\imath} + y\hat{\jmath}}{4}\right) = \frac{xz + xy}{4}$$

Step 3: Choose a projection plane. We project the cylinder onto the yz-plane. The region R is bounded by $y \in [0, 4]$ and $z \in [0, 5]$. The area element is:

$$dS = \frac{dydz}{|\hat{n} \cdot \hat{\imath}|} = \frac{dydz}{|x/4|} = \frac{4}{x}dydz$$

Step 4: Set up and evaluate the integral.

$$\iint_S (\vec{A} \cdot \hat{n})dS = \iint_R \left(\frac{xz + xy}{4}\right)\left(\frac{4}{x}dydz\right)$$

The x and 4 cancel perfectly:

$$\iint_R (z + y)dydz$$

Integrate over the rectangular region R:

$$\int_0^4 \int_0^5 (y + z)dzdy$$

Integrate with respect to z:

$$\int_0^4 \left[yz + \frac{z^2}{2}\right]_0^5 dy = \int_0^4 \left(5y + \frac{25}{2}\right) dy$$

Integrate with respect to y:

$$\left[\frac{5y^2}{2} + \frac{25}{2}y\right]_0^4 = \frac{5(16)}{2} + \frac{25(4)}{2} = 40 + 50 = 90$$

The surface integral value is 90.

14.3 Exam Patterns

- Direct evaluation of surface integrals is rare. This single cylinder problem appeared identically in 2017 and 2020.
- The projection method is crucial. If the surface equation contains x and y but no z, projecting onto the yz or zx plane is usually easier than projecting onto the xy plane.
- Most surface integrals in exams will ask you to use Gauss's Divergence Theorem or Stokes' Theorem instead of direct evaluation.

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15 A15: Volume Integrals

Exam Weight: Tier 2-3 | **Appeared in**: 2018, 2019, 2021 | **Typical Marks**: 4–6

Volume integrals (triple integrals) are used to sum a quantity over a 3D region. You must carefully determine the limits of integration for x, y, and z.

15.1 The Formula

The volume integral of a scalar function $\phi(x, y, z)$ over a volume V is:

$$\iiint_V \phi \, dV = \iiint_V \phi \, dx \, dy \, dz$$

For a vector field

$$\vec{F} = F_1\hat{\imath} + F_2\hat{\jmath} + F_3\hat{k},$$

You integrate each component separately:

$$\iiint_V \vec{F} \, dV = \left(\iiint_V F_1 \, dV\right)\hat{\imath} + \left(\iiint_V F_2 \, dV\right)\hat{\jmath} + \left(\iiint_V F_3 \, dV\right)\hat{k}$$

15.2 Setting Up Integration Limits

The hardest part is finding the limits. For a region bounded by planes: 1. **Inner integral (z)**: Express z in terms of x and y using the top and bottom bounding surfaces. 2. **Middle integral (y)**: Set $z = 0$ (or analyze the projection on the xy-plane) to find y in terms of x. 3. **Outer integral (x)**: Find the absolute minimum and maximum constant values for x.

15.3 Worked Example 1: Scalar Volume Integral (PYQ 2018)

Problem: Let

$$\phi = 45x^2y$$

. Evaluate

$$\iiint_V \phi \, dV$$

where V is the closed region bounded by planes $4x + 2y + z = 8, x = 0, y = 0, z = 0$.

Solution:

Step 1: Find limits. - z goes from 0 to $8 - 4x - 2y$. - In the xy-plane ($z = 0$), the boundary is $4x + 2y = 8 \implies y = 4 - 2x$. So y goes from 0 to $4 - 2x$. - Set $y = 0$ to find x: $4x = 8 \implies x = 2$. So x goes from 0 to 2.

Step 2: Set up integral.

$$\int_0^2 \int_0^{4-2x} \int_0^{8-4x-2y} 45x^2y \, dz \, dy \, dx$$

Step 3: Integrate z.

$$45 \int_0^2 x^2 \int_0^{4-2x} y[z]_0^{8-4x-2y} dydx = 45 \int_0^2 x^2 \int_0^{4-2x} (8y - 4xy - 2y^2) dydx$$

Step 4: Integrate y.

$$45 \int_0^2 x^2 \left[4y^2 - 2xy^2 - \frac{2}{3}y^3 \right]_0^{4-2x} dx$$

Notice

$$4y^2 - 2xy^2 = y^2(4 - 2x).$$

Substitute upper limit $y = 4 - 2x$:

$$45 \int_0^2 x^2 \left[(4-2x)^2(4-2x) - \frac{2}{3}(4-2x)^3 \right] dx = 45 \int_0^2 x^2 \left[\frac{1}{3}(4-2x)^3 \right] dx$$

Step 5: Integrate x.

$$15 \int_0^2 x^2(4-2x)^3 dx = 120 \int_0^2 x^2(2-x)^3 dx$$

Expand

$$(2-x)^3 = 8 - 12x + 6x^2 - x^3 :$$

$$120 \int_0^2 (8x^2 - 12x^3 + 6x^4 - x^5) dx$$

$$120 \left[\frac{8}{3}x^3 - 3x^4 + \frac{6}{5}x^5 - \frac{x^6}{6} \right]_0^2 = 120 \left(\frac{64}{3} - 48 + \frac{192}{5} - \frac{64}{6} \right) = 128$$

15.4 Worked Example 2: Integral of Divergence (PYQ 2019)

Problem: Evaluate

$$\iiint_V \nabla \cdot \vec{F} \, dV$$

where

$$\vec{F} = (2x^2 - 3z)\hat{\imath} - 2xy\hat{\jmath} - 4x\hat{k}$$

and V is bounded by $x = 0, y = 0, z = 0, 2x + 2y + z = 4$.

Solution:

Step 1: Find divergence.

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(2x^2 - 3z) + \frac{\partial}{\partial y}(-2xy) + \frac{\partial}{\partial z}(-4x) = 4x - 2x + 0 = 2x$$

Step 2: Find limits. z from 0 to $4 - 2x - 2y$. y from 0 to $2 - x$. x from 0 to 2.

Step 3: Integrate.

$$\int_0^2 \int_0^{2-x} \int_0^{4-2x-2y} 2x\, dz\, dy\, dx = \int_0^2 \int_0^{2-x} 2x(4 - 2x - 2y) dy dx$$

$$= \int_0^2 2x \left[(4 - 2x)y - y^2\right]_0^{2-x} dx = \int_0^2 2x \left[2(2 - x)(2 - x) - (2 - x)^2\right] dx$$

$$= \int_0^2 2x(2 - x)^2 dx = \int_0^2 (8x - 8x^2 + 2x^3) dx = \left[4x^2 - \frac{8}{3}x^3 + \frac{1}{2}x^4\right]_0^2 = \frac{8}{3}$$

15.5 Exam Patterns

- Volume integrals in this course are mostly an exercise in polynomial integration and algebra. Make sure your expansions are careful.
- Setting up limits from an intersecting plane like $ax + by + cz = d$ is a recurring pattern. Always work from $z \to y \to x$.
- You will also compute volume integrals as the right-hand side of the Gauss Divergence Theorem.

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16 A16: Green's Theorem

> **Exam Weight**: Tier 1 | **Appeared in**: 2017 (twice), 2019, 2020, 2021, 2023, 2024 | **Typical Marks**: 5–6

Green's theorem relates a line integral around a closed boundary curve C to a double integral over the plane region R bounded by C. It is one of the most frequently tested topics.

16.1 Statement of the Theorem

Let R be a closed, flat region bounded by a simple, closed curve C in the xy-plane. If $P(x, y)$ and $Q(x, y)$ are continuous functions with continuous first-order partial derivatives in R, then:

$$\oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy$$

Here the curve C is traversed in the counterclockwise (positive) direction.

16.2 Must-Know Proof (PYQ 2017, 2021, 2024)

You must be able to prove Green's theorem. We split it into two parts: 1.

$$\oint_C Pdx = -\iint_R \frac{\partial P}{\partial y} dxdy$$

2.

$$\oint_C Qdy = \iint_R \frac{\partial Q}{\partial x} dxdy$$

Proof of Part 1: Let region R be bounded by curves

$$y = y_1(x)$$

(lower) and

$$y = y_2(x)$$

(upper) from $x = a$ to $x = b$.

Evaluate the double integral:

$$\iint_R \frac{\partial P}{\partial y} dxdy = \int_a^b \left[\int_{y_1(x)}^{y_2(x)} \frac{\partial P}{\partial y} dy \right] dx = \int_a^b [P(x, y_2(x)) - P(x, y_1(x))] \, dx \quad \text{— (Eq 1)}$$

Now evaluate the line integral $\oint_C Pdx$. The curve C has two paths: - C_1: Along $y_1(x)$ from a to b. - C_2: Along $y_2(x)$ from b to a.

$$\oint_C Pdx = \int_a^b P(x, y_1(x))dx + \int_b^a P(x, y_2(x))dx$$

Reverse limits on the second integral:

$$\oint_C Pdx = \int_a^b P(x, y_1(x))dx - \int_a^b P(x, y_2(x))dx = -\int_a^b \left[P(x, y_2(x)) - P(x, y_1(x))\right] dx \quad \text{— (Eq 2)}$$

Comparing (Eq 1) and (Eq 2) proves Part 1. By symmetry (bounding x with functions of y), Part 2 is proven. Adding them yields Green's theorem.

16.3 Worked Example 1: Evaluation (PYQ 2017)

Problem: Evaluate

$$\oint_C (2x + y^2)dx + (3y - 4x)dy$$

where C is the boundary of the region between

$$y = x^2$$

and

$$y^2 = x.$$

Solution: Identify P and Q:

$$P = 2x + y^2 \implies \frac{\partial P}{\partial y} = 2y$$

$$Q = 3y - 4x \implies \frac{\partial Q}{\partial x} = -4$$

Apply Green's theorem:

$$\iint_R (-4 - 2y)dydx$$

Limits: The curves intersect at $(0, 0)$ and $(1, 1)$. For a given x, y goes from the lower curve

$$y = x^2$$

to the upper curve $y = \sqrt{x}$.

$$\int_0^1 \int_{x^2}^{\sqrt{x}} (-4 - 2y)dydx = \int_0^1 \left[-4y - y^2\right]_{x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 \left[(-4\sqrt{x} - x) - (-4x^2 - x^4)\right] dx = \int_0^1 (-4x^{1/2} - x + 4x^2 + x^4) dx$$

$$= \left[-\frac{8}{3}x^{3/2} - \frac{1}{2}x^2 + \frac{4}{3}x^3 + \frac{1}{5}x^5\right]_0^1 = -\frac{8}{3} - \frac{1}{2} + \frac{4}{3} + \frac{1}{5} = -\frac{49}{30}$$

16.4 Worked Example 2: Verification (PYQ 2020, 2023)

Problem: Verify Green's theorem for $\oint_C (xy + y^2)dx + x^2 dy$ where C is bounded by $y = x$ and

$$y = x^2.$$

Solution:

Step 1: Evaluate via Double Integral (Green's Theorem)

$$P = xy + y^2 \implies \frac{\partial P}{\partial y} = x + 2y$$

$$Q = x^2 \implies \frac{\partial Q}{\partial x} = 2x$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - (x + 2y) = x - 2y$$

Limits: x from 0 to 1, y from x^2 to x.

$$\int_0^1 \int_{x^2}^x (x - 2y) dy dx = \int_0^1 \left[xy - y^2\right]_{x^2}^x dx = \int_0^1 (0 - (x^3 - x^4)) dx$$

$$= \int_0^1 (x^4 - x^3) dx = \left[\frac{x^5}{5} - \frac{x^4}{4}\right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}$$

Step 2: Evaluate via Line Integral Path 1 (

$$y = x^2, dy = 2x dx$$

from 0 to 1):

$$\int_0^1 \left[(x^3 + x^4)dx + x^2(2x dx)\right] = \int_0^1 (3x^3 + x^4) dx = \left[\frac{3}{4}x^4 + \frac{x^5}{5}\right]_0^1 = \frac{19}{20}$$

Path 2 ($y = x, dy = dx$ from 1 to 0):

$$\int_1^0 (x^2 + x^2)dx + x^2 dx = \int_1^0 3x^2 dx = [x^3]_1^0 = -1$$

Total line integral =

$$\frac{19}{20} - 1 = -\frac{1}{20}.$$

Since both equal $-\frac{1}{20}$, the theorem is verified.

16.5 Exam Patterns

- "Verify Green's theorem" means you must compute BOTH the line integral (summing along all paths) and the double integral independently to show they match. This takes time, be careful with signs.
- The proof of Green's theorem is a high-value question. Memorize the algebraic steps for Part 1.
- Sometimes evaluating the double integral just yields the Area of the region (e.g., if

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -1$$

). Don't overcomplicate it if it's a known shape like a triangle.

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17 A17: Stokes' Theorem

Exam Weight: Tier 2 | **Appeared in**: 2023, 2024 | **Typical Marks**: 5

Stokes' theorem is the 3D extension of Green's theorem. It relates a line integral around a closed curve in 3D space to a surface integral over any surface bounded by that curve.

17.1 Statement of the Theorem

Let S be an open, two-sided surface bounded by a closed, non-self-intersecting curve C. If $\vec{A}$ is a vector field with continuous first-order partial derivatives, then:

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot \hat{n} dS$$

Here the boundary curve C is traversed in the positive direction (determined by the right-hand rule with respect to the normal vector $\hat{n}$).

17.2 Must-Know Proof (PYQ 2023)

You must be able to prove Stokes' theorem by projecting the surface onto a coordinate plane and using Green's theorem.

Let the surface equation be $z = f(x, y)$ over a region R in the xy-plane. Let

$$\vec{A} = A_1\hat{\imath} + A_2\hat{\jmath} + A_3\hat{k}.$$

We prove it for A_1 first:

$$\oint_C A_1 dx = \iint_S [\nabla \times (A_1\hat{\imath})] \cdot \hat{n} dS.$$

Step 1: The Line Integral Project the curve C onto the xy-plane as C'.

$$\oint_C A_1(x, y, z)dx = \oint_{C'} A_1(x, y, f(x, y))dx$$

Apply Green's theorem to C':

$$= \iint_R -\frac{\partial}{\partial y}[A_1(x, y, f(x, y))]dA$$

Using the chain rule:

$$\oint_C A_1 dx = \iint_R \left(-\frac{\partial A_1}{\partial y} - \frac{\partial A_1}{\partial z}\frac{\partial z}{\partial y}\right) dxdy \quad \text{— (Eq 1)}$$

Step 2: The Surface Integral Calculate the curl of $A_1\hat{\imath}$:

$$\nabla \times (A_1\hat{i}) = \frac{\partial A_1}{\partial z}\hat{j} - \frac{\partial A_1}{\partial y}\hat{k}$$

For surface $z - f(x, y) = 0$, the normal vector gives:

$$\hat{n}dS = \left(-\frac{\partial z}{\partial x}\hat{i} - \frac{\partial z}{\partial y}\hat{j} + \hat{k}\right) dxdy$$

Calculate the dot product:

$$[\nabla \times (A_1\hat{i})] \cdot \hat{n}dS = \left(-\frac{\partial A_1}{\partial z}\frac{\partial z}{\partial y} - \frac{\partial A_1}{\partial y}\right) dxdy \quad \text{— (Eq 2)}$$

Comparing (Eq 1) and (Eq 2) proves the theorem for A_1. The process is identical for A_2 and A_3. Adding them together yields Stokes' theorem.

17.3 Worked Example: Verification (PYQ 2024)

Problem: Verify Stokes' theorem for

$$\vec{A} = (y - z + 2)\hat{i} + (yz + 4)\hat{j} - xz\hat{k}$$

where S is the surface of the cube $x = 0, y = 0, z = 0, x = 2, y = 2, z = 2$ above the xy plane (i.e., open at the bottom).

Solution:

Step 1: The Line Integral The boundary curve C is the square in the $z = 0$ plane from $x = 0$ to $x = 2$ and $y = 0$ to $y = 2$. On $z = 0, dz = 0$, the vector field is

$$\vec{A} = (y + 2)\hat{i} + 4\hat{j}.$$

$$\oint_C \vec{A} \cdot d\vec{r} = \oint_C (y + 2)dx + 4dy$$

Path 1 ($y = 0, dy = 0, x$ from 0 to 2):

$$\int_0^2 2dx = 4$$

Path 2 ($x = 2, dx = 0, y$ from 0 to 2):

$$\int_0^2 4dy = 8$$

Path 3 ($y = 2, dy = 0, x$ from 2 to 0):

$$\int_2^0 4dx = -8$$

Path 4 ($x = 0, dx = 0, y$ from 2 to 0):

$$\int_2^0 4dy = -8$$

Total Line Integral $= 4 + 8 - 8 - 8 = -4$.

Step 2: The Surface Integral Calculate Curl:

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y - z + 2 & yz + 4 & -xz \end{vmatrix} = -y\hat{i} + (z-1)\hat{j} - \hat{k}$$

Evaluate over the 5 faces of the cube above $z = 0$: 1. **Top (

$$z = 2, \hat{n} = \hat{k}$$

)**:

$$(\nabla \times \vec{A}) \cdot \hat{n} = -1$$

. Integral:

$$\iint (-1) dxdy = -4.$$

2. **Front (

$$x = 2, \hat{n} = \hat{i}$$

)**:

$$(\nabla \times \vec{A}) \cdot \hat{n} = -y$$

. Integral:

$$\int_0^2 \int_0^2 -ydydz = 2\left[-\frac{y^2}{2}\right]_0^2 = -4.$$

3. **Back (

$$x = 0, \hat{n} = -\hat{i}$$

)**:

$$(\nabla \times \vec{A}) \cdot \hat{n} = y$$

. Integral:

$$\int_0^2 \int_0^2 ydydz = 2\left[\frac{y^2}{2}\right]_0^2 = 4.$$

4. **Right (

$$y = 2, \hat{n} = \hat{j}$$

)**:

$$(\nabla \times \vec{A}) \cdot \hat{n} = z - 1$$

. Integral:

$$\int_0^2 \int_0^2 (z-1) dx dz = 2 \left[\frac{z^2}{2} - z \right]_0^2 = 0.$$

5. **Left (

$$y = 0, \hat{n} = -\hat{j}$$

)**:

$$(\nabla \times \vec{A}) \cdot \hat{n} = -(z - 1)$$

. Integral:

$$\int_0^2 \int_0^2 (1-z) dx dz = 2 \left[z - \frac{z^2}{2} \right]_0^2 = 0.$$

Total Surface Integral $= -4 - 4 + 4 + 0 + 0 = -4$.

Both integrals yield -4. Stokes' theorem is verified.

17.4 Exam Patterns

- Verifying Stokes' theorem over a cube involves 5 surface integrals (if open at the bottom) or 6 surface integrals. Be extremely meticulous with your normal vectors

$$\hat{\imath}, -\hat{\imath}, \hat{\jmath}, -\hat{\jmath}, \hat{k}, -\hat{k}$$

For each face. - When asked to "evaluate using Stokes' theorem", you can choose whichever side of the equation is easier. Usually, converting a complex line integral into a surface integral via curl is the intended path.

---

18 A18: Gauss Divergence Theorem

Exam Weight: Tier 1 | **Appeared in**: 2018, 2021 | **Typical Marks**: 6

The Gauss Divergence Theorem relates the flux of a vector field across a closed surface to the volume integral of the divergence inside that surface. It is extremely powerful for converting difficult surface integrals into simple volume integrals.

18.1 Statement of the Theorem

Let V be a volume bounded by a closed surface S. If $\vec{A}$ is a vector field with continuous first-order partial derivatives, then:

$$\iint_S \vec{A} \cdot \hat{n} dS = \iiint_V (\nabla \cdot \vec{A}) dV$$

where $\hat{n}$ is the unit outward normal vector to the surface S.

(Note: The theorem requires a strictly CLOSED surface. If a surface is open, like a hemisphere without its base, you must add the base to close it before applying the theorem).

18.2 Worked Example: Verification (PYQ 2018, 2021)

Problem: Verify the divergence theorem for

$$\vec{A} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$$

taken over the region bounded by

$$x^2 + y^2 = 4, z = 0, z = 3.$$

Solution:

Step 1: Evaluate the Volume Integral Calculate Divergence:

$$\nabla \cdot \vec{A} = \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) = 4 - 4y + 2z$$

Set up cylindrical coordinates ($x = r\cos\phi, y = r\sin\phi, z = z, dV = r dr d\phi dz$): Limits: $r \in [0, 2]$, $\phi \in [0, 2\pi]$, $z \in [0, 3]$.

$$\iiint_V (\nabla \cdot \vec{A}) dV = \int_0^3 \int_0^{2\pi} \int_0^2 (4 - 4r\sin\phi + 2z) r \, dr \, d\phi \, dz$$

Integrate w.r.t ϕ first (the $\sin\phi$ term evaluates to 0 over 2π):

$$\int_0^{2\pi} (4r - 4r^2\sin\phi + 2rz) d\phi = 2\pi(4r + 2rz)$$

Integrate w.r.t r:

$$2\pi \int_0^2 (4r + 2rz)dr = 2\pi[2r^2 + r^2 z]_0^2 = 2\pi(8 + 4z)$$

Integrate w.r.t z:

$$2\pi \int_0^3 (8 + 4z)dz = 2\pi[8z + 2z^2]_0^3 = 2\pi(24 + 18) = 84\pi$$

Step 2: Evaluate the Surface Integral The closed cylinder has 3 surfaces: Top (S_1), Bottom (S_2), and Curved Wall (S_3).

- **Top Surface S_1 (

$$z = 3, \hat{n} = \hat{k}$$

)**:

$$\vec{A} \cdot \hat{k} = z^2 = 3^2 = 9.$$

$$\iint_{S_1} 9dS = 9 \times (\text{Area of circle}) = 9 \times \pi(2^2) = 36\pi.$$

- **Bottom Surface S_2 (

$$z = 0, \hat{n} = -\hat{k}$$

)**:

$$\vec{A} \cdot (-\hat{k}) = -z^2 = -0^2 = 0.$$

$$\iint_{S_2} 0dS = 0.$$

- **Curved Wall S_3

$$x^2 + y^2 = 4, \hat{n} = \frac{x\hat{i} + y\hat{j}}{2}$$

**:

$$\vec{A} \cdot \hat{n} = (4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}) \cdot \left(\frac{x\hat{i} + y\hat{j}}{2}\right) = 2x^2 - y^3.$$

In cylindrical ($x = 2\cos\phi, y = 2\sin\phi, dS = 2d\phi dz$):

$$\iint_{S_3} (2x^2 - y^3)dS = \int_0^3 \int_0^{2\pi} \left[2(4\cos^2\phi) - 8\sin^3\phi\right] 2d\phi dz$$

We know

$$\int_0^{2\pi} \cos^2 \phi d\phi = \pi$$

and

$$\int_0^{2\pi} \sin^3 \phi d\phi = 0.$$

$$= \int_0^3 2(8\pi - 0)dz = \int_0^3 16\pi dz = 48\pi$$

Total Surface Integral $= 36\pi + 0 + 48\pi = 84\pi$.

Both integrals equal 84π. The divergence theorem is verified.

18.3 The "Big Three" Summary

Vector Calculus revolves around three major integration theorems. Understanding when to use which is critical:

Theorem	Connects...	Common Use Case
Green's	1D Closed Line $\iff$ 2D Plane Area	Evaluating planar work integrals.
Stokes'	1D Closed Line $\iff$ 3D Open Surface	Evaluating work in 3D space by calculating flux of curl.
Gauss's	2D Closed Surface $\iff$ 3D Volume	Evaluating outward flux through a closed solid by integrating divergence.

18.4 Exam Patterns

- Like Green's and Stokes', "verify" means you must compute the volume integral and the surface integral independently.
- The cylinder problem is a classic. Always split the cylinder into top, bottom, and curved surfaces, treating them entirely separately before summing them.
- Cylindrical coordinates (r, ϕ, z) are absolutely essential here.

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19 B01: Matrix Basics and Operations

Exam Weight: Tier 3 | **Typical Marks**: 2–3 (definition questions)

These fundamentals rarely appear as standalone problems. But they are the foundation for everything else. Knowing them helps you avoid careless mistakes.

19.1 What is a Matrix?

A matrix is a rectangular array of numbers arranged in rows and columns. A matrix with m rows and n columns has order $m \times n$.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

The element a_{ij} sits in row i and column j.

19.2 Types of Matrices

Type	Condition	Example
Square	$m = n$	3x3 matrix
Row	$m = 1$	[1 2 3]
Column	$n = 1$	$[1, 2]^T$
Diagonal		

$$a_{ij} = 0$$

for $i \neq j$ | Only diagonal entries non-zero | | **Scalar** | Diagonal with all diagonal entries equal | kI | | **Identity** | Diagonal with all 1s | I_n | | **Null/Zero** | All entries are 0 | O | | **Upper triangular** |

$$a_{ij} = 0$$

for $i > j$ | Zeros below diagonal | | **Lower triangular** |

$$a_{ij} = 0$$

for $i < j$ | Zeros above diagonal |

19.3 Matrix Operations

19.3.1 Addition and Subtraction

Add or subtract matrices of the same order by adding or subtracting corresponding entries:

$$(A + B)_{ij} = a_{ij} + b_{ij}$$

Addition is commutative: $A + B = B + A$.

19.3.2 Scalar Multiplication

Multiply every entry by the scalar:

$$(kA)_{ij} = k \cdot a_{ij}$$

19.3.3 Matrix Multiplication

A $(m \times n)$ times B $(n \times p)$ gives C $(m \times p)$. The (i, j)-th entry of C is:

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Multiplication is NOT commutative: $AB \neq BA$ in general.

Multiplication IS associative: $(AB)C = A(BC)$.

Multiplication IS distributive: $A(B + C) = AB + AC$.

19.3.4 Important: Size Compatibility

You can only multiply $A_{m \times n}$ by $B_{n \times p}$. The inner dimensions must match. The result is $m \times p$.

19.4 Determinant (Quick Reference)

For a 2x2 matrix:

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

For a 3x3 matrix, expand along row 1:

$$|A| = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Properties: - If a row or column is all zeros,

$$|A| = 0.$$

- Swapping two rows changes the sign of the determinant.
-

$$|AB| = |A| \cdot |B|.$$

-

$$|kA| = k^n |A|$$

for an $n \times n$ matrix.

---

20 B02: Transpose, Symmetric, and Skew-Symmetric Matrices

Exam Weight: Tier 2 | **Appeared in**: 2018, 2020, 2021 | **Typical Marks**: 3–6

These definitions are easy marks. The decomposition theorem (symmetric + skew-symmetric) has appeared twice.

20.1 Transpose of a Matrix

20.1.1 Definition

The transpose of matrix A (written A^T or A') is obtained by swapping rows and columns. If A is $m \times n$, then A^T is $n \times m$.

The (i, j)-th element of A^T equals the (j, i)-th element of A:

$$(A^T)_{ij} = a_{ji}$$

20.1.2 Properties of Transpose

1.

$$(A^T)^T = A$$

2.

$$(A + B)^T = A^T + B^T$$

3.

$$(kA)^T = kA^T$$

4.

$$(AB)^T = B^T A^T$$

(order reverses) 5.

$$(A^{-1})^T = (A^T)^{-1}$$

20.2 Symmetric Matrix

20.2.1 Definition

A square matrix A is symmetric if it equals its transpose:

$$A^T = A$$

This means

$$a_{ij} = a_{ji}$$

for all i, j. The matrix is a mirror image across its main diagonal.

20.2.2 Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

Notice: position $(1, 2)$ and $(2, 1)$ both have 2. Position $(1, 3)$ and $(3, 1)$ both have 3.

20.2.3 Key Properties

- Symmetric matrices have only real eigenvalues.
- They can always be diagonalized.
- They are very common in physics and engineering.

20.3 Skew-Symmetric Matrix

20.3.1 Definition

A square matrix A is skew-symmetric if:

$$A^T = -A$$

This means

$$a_{ij} = -a_{ji}$$

for all i, j. The diagonal entries must be zero (since

$$a_{ii} = -a_{ii}$$

forces

$$a_{ii} = 0$$

).

20.3.2 Example

$$A = \begin{bmatrix} 0 & -3 & 5 \\ 3 & 0 & -7 \\ -5 & 7 & 0 \end{bmatrix}$$

All diagonal entries are 0. Each pair of symmetric positions has opposite signs.

20.4 Decomposition Theorem (PYQ 2020, 2021 — Repeated)

20.4.1 Statement

Every square matrix can be uniquely expressed as the sum of a symmetric matrix and a skew-symmetric matrix.

20.4.2 Proof

Let A be any square matrix. Define:

$$P = \frac{1}{2}(A + A^T) \quad \text{and} \quad Q = \frac{1}{2}(A - A^T)$$

Step 1: Show $A = P + Q$.

$$P + Q = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = \frac{1}{2}(2A) = A \quad \checkmark$$

Step 2: Show P is symmetric.

$$P^T = \left[\frac{1}{2}(A + A^T)\right]^T = \frac{1}{2}(A^T + (A^T)^T) = \frac{1}{2}(A^T + A) = P \quad \checkmark$$

Step 3: Show Q is skew-symmetric.

$$Q^T = \left[\frac{1}{2}(A - A^T)\right]^T = \frac{1}{2}(A^T - A) = -\frac{1}{2}(A - A^T) = -Q \quad \checkmark$$

Step 4: Prove uniqueness. Suppose $A = P' + Q'$ where P' is symmetric and Q' is skew-symmetric.

Taking transpose:

$$A^T = P' - Q'.$$

Adding the two equations:

$$A + A^T = 2P'$$

, so

$$P' = \frac{1}{2}(A + A^T) = P.$$

Subtracting:

$$A - A^T = 2Q'$$

, so

$$Q' = \frac{1}{2}(A - A^T) = Q.$$

The decomposition is unique.

20.4.3 Worked Example

Express

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

as symmetric + skew-symmetric.

Symmetric part:

$$P = \frac{1}{2}(A + A^T) = \frac{1}{2}\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix} 2 & 5 \\ 5 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 5/2 \\ 5/2 & 4 \end{bmatrix}$$

Skew-symmetric part:

$$Q = \frac{1}{2}(A - A^T) = \frac{1}{2}\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix}$$

Verify:

$$P + Q = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A.$$

Correct.

20.5 Exam Patterns

- "Define symmetric, skew-symmetric, Hermitian matrices" is a 2-3 mark definition question. Write the formula and one example each.
- The decomposition proof (symmetric + skew-symmetric) appeared in 2020 and 2021. Memorize the formulas

$$P = \frac{1}{2}(A + A^T)$$

and

$$Q = \frac{1}{2}(A - A^T).$$

- The uniqueness part adds 1-2 extra marks if you include it.

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21 B03: Hermitian, Skew-Hermitian, and Special Matrices

Exam Weight: Tier 2 | **Appeared in**: 2018, 2023 | **Typical Marks**: 2–5

Definition questions about matrix types are easy marks. The Hermitian decomposition proof appeared in 2023.

21.1 Hermitian Matrix

21.1.1 Definition

A square matrix A with complex entries is Hermitian if it equals its conjugate transpose:

$$A^\dagger = (\bar{A})^T = A$$

Here $\bar{A}$ is the complex conjugate (replace i with $-i$ in every entry). Then $A^\dagger$ transposes the result.

21.1.2 Properties

- Diagonal entries must be real (since

$$a_{ii} = \overline{a_{ii}}$$

implies $a_{ii} \in \mathbb{R}$). - Off-diagonal entries satisfy

$$a_{ij} = \overline{a_{ji}}.$$

- All eigenvalues of a Hermitian matrix are real.

21.1.3 Example

$$A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$$

Check:

$$\bar{A} = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}, \quad (\bar{A})^T = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix} = A$$

Hermitian.

21.2 Skew-Hermitian Matrix

21.2.1 Definition

A square matrix A is skew-Hermitian if:

$$A^\dagger = -A$$

21.2.2 Properties

- Diagonal entries must be pure imaginary or zero (since

$$\overline{a_{ii}} = -a_{ii}$$

). - Off-diagonal entries satisfy

$$a_{ij} = -\overline{a_{ji}}.$$

21.2.3 Example

$$A = \begin{pmatrix} 2i & 1+i \\ -1+i & 0 \end{pmatrix}$$

21.3 Hermitian Decomposition Theorem (PYQ 2023)

21.3.1 Statement

Every square matrix can be expressed as the sum of a Hermitian matrix and a skew-Hermitian matrix.

21.3.2 Proof

Let A be any square matrix. Define:

$$P = \frac{1}{2}(A + A^\dagger) \quad \text{and} \quad Q = \frac{1}{2}(A - A^\dagger)$$

Show P is Hermitian:

$$P^\dagger = \frac{1}{2}(A^\dagger + (A^\dagger)^\dagger) = \frac{1}{2}(A^\dagger + A) = P$$

Show Q is skew-Hermitian:

$$Q^\dagger = \frac{1}{2}(A^\dagger - (A^\dagger)^\dagger) = \frac{1}{2}(A^\dagger - A) = -Q$$

And $P + Q = A$.

This is the same structure as the symmetric decomposition. Replace transpose with conjugate transpose.

21.4 Special Matrix Types

21.4.1 Nilpotent Matrix

A square matrix A is nilpotent if there exists a positive integer k such that:

$$A^k = O$$

where O is the zero matrix. The smallest such k is the index of nilpotency.

Example:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}. \quad A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O.$$

Nilpotent with index 2.

All eigenvalues of a nilpotent matrix are zero.

21.4.2 Involutory Matrix

A square matrix A is involutory if:

$$A^2 = I$$

This means A is its own inverse:

$$A^{-1} = A.$$

Example:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad A^2 = I.$$

21.4.3 Idempotent Matrix

A square matrix A is idempotent if:

$$A^2 = A$$

Applying the transformation twice gives the same result as applying it once. Projection matrices are idempotent.

Example:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}. \quad A^2 = A.$$

The eigenvalues of an idempotent matrix are only 0 or 1.

21.4.4 Periodic Matrix

A square matrix A is periodic with period k if:

$$A^{k+1} = A$$

An idempotent matrix is periodic with period 1.

21.4.5 Orthogonal Matrix

A square matrix A is orthogonal if:

$$A^T A = AA^T = I$$

This means

$$A^{-1} = A^T$$

. The columns form an orthonormal set. The determinant is ± 1.

Example: Rotation matrices are orthogonal.

21.4.6 Unitary Matrix

A square matrix A is unitary if:

$$A^\dagger A = AA^\dagger = I$$

This is the complex version of orthogonal.

$$A^{-1} = A^\dagger.$$

21.5 Exam Patterns

- "Define commutative, idempotent, involutory matrix" appeared in 2023 for 2 marks. Just write the definition and one example.
- "Define symmetric, skew-symmetric, Hermitian" appeared in 2018 for 3 marks.
- The Hermitian decomposition proof appeared in 2023 for 5 marks.
- These are easy marks. Memorize the one-line definitions.

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22 B04: Matrix Proofs

Exam Weight: Tier 2 | **Appeared in**: 2017, 2018, 2020, 2023 | **Typical Marks**: 3–5

These are standard proofs that appear regularly. They test whether you understand matrix algebra rules. Each proof is short and worth learning by heart.

22.1 Proof 1: Transpose of a Product (PYQ 2020)

22.1.1 Statement

For matrices A (of size $m \times n$) and B (of size $n \times p$):

$$(AB)^T = B^T A^T$$

The order reverses when you transpose a product.

22.1.2 Proof

Let $C = AB$. The (i, j)-th element of C is:

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

The (i, j)-th element of C^T is c_{ji}:

$$(C^T)_{ij} = c_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

Now consider $B^T A^T$. Its (i, j)-th element is:

$$(B^T A^T)_{ij} = \sum_{k=1}^{n} (B^T)_{ik} (A^T)_{kj} = \sum_{k=1}^{n} b_{ki} \cdot a_{jk} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

Since

$$(C^T)_{ij} = (B^T A^T)_{ij}$$

for all i, j:

$$(AB)^T = B^T A^T \qquad \blacksquare$$

22.2 Proof 2: Inverse of a Product (PYQ 2018, 2023 — Repeated)

22.2.1 Statement

If A and B are non-singular matrices of the same order:

$$(AB)^{-1} = B^{-1}A^{-1}$$

22.2.2 Proof

We show that $B^{-1}A^{-1}$ satisfies the definition of the inverse of AB.

Left product:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Right product:

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

Since

$$(AB)(B^{-1}A^{-1}) = I$$

and

$$(B^{-1}A^{-1})(AB) = I :$$

$$(AB)^{-1} = B^{-1}A^{-1} \qquad \blacksquare$$

22.3 Proof 3: Power of Inverse (PYQ 2018)

22.3.1 Statement

For any invertible matrix A and positive integer n:

$$(A^{-1})^n = (A^n)^{-1}$$

22.3.2 Proof (by induction)

Base case ($n = 1$):

$$(A^{-1})^1 = (A^1)^{-1}$$

. True.

Inductive step: Assume

$$(A^{-1})^k = (A^k)^{-1}$$

holds for some k.

For $n = k+1$:

$$(A^{k+1})^{-1} = (A^k \cdot A)^{-1} = A^{-1} \cdot (A^k)^{-1}$$

using Proof 2. By the inductive hypothesis,

$$(A^k)^{-1} = (A^{-1})^k :$$

$$A^{-1} \cdot (A^{-1})^k = (A^{-1})^{k+1}$$

The statement holds for $k+1$. By induction, it holds for all positive integers n.

$$\blacksquare$$

22.4 Proof 4: Rotation Matrix Property (PYQ 2017)

22.4.1 Statement

If

$$A_\alpha = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}$$

then

$$A_\alpha \cdot A_\beta = A_{\alpha+\beta} = A_\beta \cdot A_\alpha.$$

22.4.2 Proof

Multiply $A_\alpha \cdot A_\beta$:

$$A_\alpha \cdot A_\beta = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & \cos\alpha\sin\beta + \sin\alpha\cos\beta \\ -\sin\alpha\cos\beta - \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{pmatrix}$$

Apply the angle addition formulas:

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Substitute:

$$A_\alpha \cdot A_\beta = \begin{pmatrix} \cos(\alpha+\beta) & \sin(\alpha+\beta) \\ -\sin(\alpha+\beta) & \cos(\alpha+\beta) \end{pmatrix} = A_{\alpha+\beta}$$

Since $\alpha + \beta = \beta + \alpha$:

$$A_{\alpha+\beta} = A_{\beta+\alpha} = A_\beta \cdot A_\alpha \qquad \blacksquare$$

22.5 Proof 5: Singular Matrix Definition

22.5.1 Statement

A square matrix A is singular if

$$|A| = 0$$

. It is non-singular if $|A| \neq 0$.

A singular matrix has no inverse. A non-singular matrix is invertible and

$$A^{-1} = \frac{1}{|A|}\text{adj}(A).$$

This definition is often asked as a 1-2 mark opener before a proof question.

22.6 Exam Patterns

- The

$$(AB)^{-1} = B^{-1}A^{-1}$$

proof has appeared in 2018 and 2023. This is the most important proof. - The

$$(AB)^T = B^T A^T$$

proof appeared in 2020. - The rotation matrix property appeared in 2017. - These proofs are short (5-8 lines each). Write them neatly with clear steps. - Always state what you need to prove before starting.

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23 B05: Row Echelon Form and Rank of a Matrix

> **Exam Weight**: Tier 1 | **Appeared in**: 2017, 2018, 2019, 2020, 2021, 2023, 2024 (all 7 papers) + CT | **Typical Marks**: 3–6

This topic has appeared in every single exam paper. It is the most consistent topic in the entire course. You cannot afford to skip it.

23.1 What is the Rank of a Matrix?

23.1.1 Definition

The rank of a matrix A is the maximum number of linearly independent row vectors in it. It also equals the maximum number of linearly independent column vectors. We write it as $\rho(A)$ or $\text{rank}(A)$.

23.1.2 Key Idea

Think of rank as "how many rows actually carry new information." Some rows in a matrix might just be multiples or combinations of other rows. Those rows add no new information. The rank counts only the rows that are truly independent from each other.

A zero matrix has rank 0. A non-zero matrix has rank at least 1.

For an $m \times n$ matrix, the rank can be at most $\min(m, n)$. So a 3×4 matrix has rank at most 3. A 4×3 matrix also has rank at most 3.

23.1.3 Why Rank Matters

Rank shows up everywhere in this course: - It decides if a system of equations has a solution (rank conditions) - It tells you the dimension of the column space and row space - It connects to the number of free variables in a system - It is the order of the largest non-zero minor

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23.2 Elementary Row Operations

Before we find rank, we need row operations. These are tools that change a matrix into a simpler form without changing its rank.

Three types of elementary row operations exist:

1. **Swap two rows**: $R_i \leftrightarrow R_j$
2. **Multiply a row by a non-zero scalar**: $R_i \to kR_i$ where $k \neq 0$
3. **Add a scalar multiple of one row to another**: $R_i \to R_i + kR_j$

Two matrices are called **equivalent** (written $A \sim B$) if one can be reached from the other through row operations. Equivalent matrices always have the same rank.

23.3 Row Echelon Form

23.3.1 Definition

A matrix is in row echelon form if it satisfies these three rules:

1. All zero rows are at the bottom.
2. The first non-zero entry in each row (called the **pivot**) is strictly to the right of the pivot in the row above it.
3. All entries below each pivot are zero.

23.3.2 Example of Echelon Form

This matrix is in echelon form:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Notice how the pivots (the leading 1s) staircase to the right. The zero row sits at the bottom.

This matrix is NOT in echelon form:

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

The zero row is not at the bottom. The last row's pivot is not to the right of the row above it.

23.4 Finding Rank by Echelon Form

23.4.1 The Method

1. Use elementary row operations to convert the matrix to echelon form.
2. Count the number of non-zero rows.
3. That count is the rank.

This is the fastest and most exam-friendly method for finding rank. Every exam question on rank uses this approach.

23.4.2 Worked Example 1: Rank of a 3x4 Matrix

Problem: Find the rank of the matrix (PYQ 2020):

$$A = \begin{bmatrix} 6 & 1 & 8 & 3 \\ 2 & 1 & 0 & 2 \\ 4 & -1 & -8 & -3 \end{bmatrix}$$

Solution:

Swap R_1 and R_2 to get a small leading entry:

$$\begin{bmatrix} 2 & 1 & 0 & 2 \\ 6 & 1 & 8 & 3 \\ 4 & -1 & -8 & -3 \end{bmatrix}$$

Clear the first column. Apply $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 2R_1$:

$$\begin{bmatrix} 2 & 1 & 0 & 2 \\ 0 & -2 & 8 & -3 \\ 0 & -3 & -8 & -7 \end{bmatrix}$$

Clear the second column in row 3. Apply $R_3 \to 2R_3 - 3R_2$:

$$\begin{bmatrix} 2 & 1 & 0 & 2 \\ 0 & -2 & 8 & -3 \\ 0 & 0 & -40 & -5 \end{bmatrix}$$

All three rows are non-zero. So:

$$\rho(A) = 3$$

23.4.3 Worked Example 2: Rank of a 4x4 Matrix

Problem: Find the rank of the matrix (PYQ 2018):

$$A = \begin{pmatrix} 1 & -2 & 1 & -1 \\ 1 & 1 & -2 & 3 \\ 4 & 1 & -5 & 8 \\ 5 & -7 & 2 & -1 \end{pmatrix}$$

Solution:

Clear the first column. Apply $R_2 \to R_2 - R_1$, $R_3 \to R_3 - 4R_1$, $R_4 \to R_4 - 5R_1$:

$$\begin{pmatrix} 1 & -2 & 1 & -1 \\ 0 & 3 & -3 & 4 \\ 0 & 9 & -9 & 12 \\ 0 & 3 & -3 & 4 \end{pmatrix}$$

Clear the second column. Apply $R_3 \to R_3 - 3R_2$ and $R_4 \to R_4 - R_2$:

$$\begin{pmatrix} 1 & -2 & 1 & -1 \\ 0 & 3 & -3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Two non-zero rows remain. So:

$$\rho(A) = 2$$

Notice that rows 3 and 4 were multiples of row 2 after the first set of operations. So they carried no new information.

23.4.4 Worked Example 3: Rank of a 4x5 Matrix

Problem: Find the rank of the matrix (PYQ 2017):

$$A = \begin{bmatrix} 0 & 0 & 1 & -3 & -2 \\ 0 & 1 & 2 & 6 & 0 \\ 0 & 2 & 3 & 9 & 2 \\ 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

Solution:

The first column is all zeros. Swap $R_1 \leftrightarrow R_2$ to put a non-zero entry first:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 2 & 3 & 9 & 2 \\ 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

Clear below the first pivot. Apply $R_3 \to R_3 - 2R_1$ and $R_4 \to R_4 - R_1$:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & -3 & 2 \end{bmatrix}$$

Clear below the second pivot. Apply $R_3 \to R_3 + R_2$ and $R_4 \to R_4 + R_2$:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & -6 & 0 \end{bmatrix}$$

Apply $R_4 \to R_4 - R_3$:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Three non-zero rows remain. So:

$$\rho(A) = 3$$

23.5 Finding Rank by Minor Method

23.5.1 Definition of a Minor

Take any r rows and r columns from an $m \times n$ matrix. The determinant of the resulting $r \times r$ submatrix is a minor of order r.

23.5.2 The Method

1. Start with the highest possible order of a square submatrix.
2. Compute its determinant. If it is non-zero, the rank equals this order.
3. If all minors of that order are zero, drop down one order and check again.
4. The rank is the order of the largest non-zero minor.

23.5.3 When to Use This

The minor method is slower than echelon form for large matrices. But it is useful for small matrices or when the problem specifically asks for "rank by minors." In exams, the echelon method is almost always preferred.

23.5.4 Worked Example: Minor Method

Problem: Find the rank of:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

Solution:

The matrix is 3×3. Start by checking if the full 3×3 determinant is non-zero:

$$|A| = 1(15 - 16) - 2(10 - 12) + 3(8 - 9) = -1 + 4 - 3 = 0$$

The 3×3 minor is zero. So rank is not 3.

Check a 2×2 minor. Take the top-left 2×2 submatrix:

$$\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = 3 - 4 = -1 \neq 0$$

A non-zero 2×2 minor exists. So:

$$\rho(A) = 2$$

23.6 Nullity

23.6.1 Definition

For a square matrix of order n, the nullity is defined as:

$$N(A) = n - \rho(A)$$

Nullity tells you the dimension of the null space (the set of all solutions to $AX = 0$). It equals the number of free variables in the system.

23.6.2 Example

If A is a 4×4 matrix with $\rho(A) = 3$, then:

$$N(A) = 4 - 3 = 1$$

This means the system $AX = 0$ has one free variable and its solution space is one-dimensional.

23.7 Important Properties of Rank

1. $\rho(A) = 0$ if and only if A is the zero matrix.
2. For an $m \times n$ matrix, $\rho(A) \leq \min(m, n)$.
3.

$$\rho(A) = \rho(A^T)$$

. Row rank equals column rank. 4. Row operations do not change the rank. 5. If A is an $n \times n$ matrix, then $\rho(A) = n$ if and only if A is invertible (non-singular). 6. If $\rho(A) < n$ for a square matrix of order n, then

$$|A| = 0.$$

23.8 Exam Patterns

- **Every paper asks rank.** The typical phrasing is: "Define rank of a matrix. Find the rank of…" This gives you 2 marks for the definition and 3–4 marks for the computation.
- Matrices in exams are usually 3×4, 4×4, or 4×5.
- The definition they expect is: "The rank of a matrix is the number of non-zero rows in its row echelon form" or "the maximum number of linearly independent rows."
- Always swap rows first to get a 1 (or the smallest non-zero number) in the top-left corner. This makes arithmetic easier.
- Watch for rows that become identical during reduction. They collapse to zeros and lower the rank.

23.9 Must-Know Problems

23.9.1 Problem 4: Rank of a 4x4 Matrix (PYQ 2024)

Question: Define rank. Find the rank of:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{bmatrix}$$

Solution:

Apply $R_2 \to R_2 - R_1$, $R_3 \to R_3 - 2R_1$, $R_4 \to R_4 - 3R_1$:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & -2 & -5 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

Apply $R_3 \to R_3 + R_2$:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & -8 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

Apply $R_4 \to 4R_4 - 3R_3$:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & -8 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Three non-zero rows. So $\rho(A) = 3$.

23.9.2 Problem 5: Rank of a 3x4 Matrix (PYQ 2023)

Question: Find the rank of:

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

Solution:

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

Apply $R_3 \to R_3 + R_2$:

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Two non-zero rows. So $\rho(A) = 2$.

23.9.3 Problem 6: Echelon Form and Canonical Form (CT)

Question: Find the echelon form, row canonical form, and rank of:

$$A = \begin{bmatrix} 6 & 3 & -4 \\ -4 & 1 & -6 \\ 1 & 2 & -5 \end{bmatrix}$$

Solution:

Step 1: Echelon form.

Swap $R_1 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{bmatrix}$$

Apply $R_2 \to R_2 + 4R_1$ and $R_3 \to R_3 - 6R_1$:

$$\begin{bmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & -9 & 26 \end{bmatrix}$$

Apply $R_3 \to R_3 + R_2$:

$$\begin{bmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & 0 & 0 \end{bmatrix}$$

Scale

$$R_2 \to \frac{1}{9}R_2 :$$

$$\begin{bmatrix} 1 & 2 & -5 \\ 0 & 1 & -\frac{26}{9} \\ 0 & 0 & 0 \end{bmatrix}$$

This is the echelon form.

Step 2: Row canonical form.

Clear above the second pivot. Apply $R_1 \to R_1 - 2R_2$:

$$R_1 = \begin{bmatrix} 1 & 0 & -5 + \frac{52}{9} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{7}{9} \end{bmatrix}$$

The row canonical form is:

$$\begin{bmatrix} 1 & 0 & \frac{7}{9} \\ 0 & 1 & -\frac{26}{9} \\ 0 & 0 & 0 \end{bmatrix}$$

Step 3: Rank.

Two non-zero rows in echelon form. So $\rho(A) = 2$.

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24 B06: Normal (Canonical) Form of a Matrix

Exam Weight: Tier 1-2 | **Appeared in**: 2019, 2021, 2023 | **Typical Marks**: 3–6

The normal form reduces a matrix using both row AND column operations to reach the identity block form. The rank is then read directly from the size of the identity block.

24.1 Definition of Normal Form

A non-zero matrix A of rank r can be reduced to one of these forms using row and column operations:

$$[I_r], \quad [I_r \mid 0], \quad \begin{bmatrix} I_r \\ 0 \end{bmatrix}, \quad \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where I_r is the identity matrix of order r. This is the **normal form** (or **canonical form**).

The rank equals r (the order of I_r in the normal form).

24.2 Difference from Echelon Form

Feature	Echelon Form	Normal Form
Operations used	Row operations only	Row AND column operations
Result	Staircase pattern	Identity block
Off-diagonal entries	May be non-zero above pivots	All zero except I_r
Rank reading	Count non-zero rows	Read order of I_r

24.3 The Method

1. Start with the given matrix.
2. Use row operations to get echelon form (upper triangular with pivots).
3. Scale each pivot to 1.
4. Use column operations to clear all non-zero entries above and beside each pivot.
5. If needed, swap columns to group the 1s together into I_r.

24.4 Worked Example 1: 4x4 Matrix to Normal Form (PYQ 2019)

Problem: Find the rank by reducing to normal form:

$$A = \begin{pmatrix} 3 & -2 & 0 & -7 \\ 0 & 2 & 1 & -5 \\ 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \end{pmatrix}$$

Solution:

Swap $R_1 \leftrightarrow R_3$ for leading 1. Then $R_3 \to R_3 - 3R_1$:

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 2 & 1 & -5 \\ 0 & 4 & 6 & -10 \\ 0 & 1 & 1 & -6 \end{pmatrix}$$

Swap $R_2 \leftrightarrow R_4$. Then $R_3 \to R_3 - 4R_2$ and $R_4 \to R_4 - 2R_2$:

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 2 & 14 \\ 0 & 0 & -1 & 7 \end{pmatrix}$$

Scale

$$R_3 \to \frac{1}{2}R_3$$

. Then $R_4 \to R_4 + R_3$:

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 14 \end{pmatrix}$$

Scale

$$R_4 \to \frac{1}{14}R_4$$

. Now clear upward using column operations.

$C_2 \to C_2 + 2C_1$, $C_3 \to C_3 + 2C_1$, $C_4 \to C_4 - C_1$:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$C_3 \to C_3 - C_2$, $C_4 \to C_4 + 6C_2$:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$C_4 \to C_4 - 7C_3$:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I_4$$

Normal form is I_4. Rank $= 4$.

24.5 Worked Example 2: 4x4 to Normal Form (PYQ 2021)

Problem: Reduce to canonical form and find rank:

$$A = \begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$$

Solution: After row operations, the echelon form has 3 non-zero rows. After column operations to clear off-diagonal entries:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

Normal form is $[I_3 \mid 0]$-type. Rank = 3.

24.6 Exam Patterns

- "Reduce to normal/canonical form and find rank" appears regularly.
- Always show row operations first to reach echelon form. Then show column operations to reach the identity block.
- Write each operation clearly ($R_i \to ...$, $C_j \to ...$).
- State the final rank explicitly.

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25 B07: System of Linear Equations

> **Exam Weight**: Tier 1 | **Appeared in**: 2017, 2018, 2019, 2020, 2021, 2023, 2024 (all 7 papers) + CT | **Typical Marks**: 5–7

This topic has appeared in every single exam paper. Together with rank and eigenvalues, it forms the trio of guaranteed exam topics.

25.1 Matrix Form of a System

25.1.1 Setting Up the System

Any system of linear equations can be written in matrix form as:

$$AX = B$$

Here: - A is the coefficient matrix (contains the coefficients of the variables) - X is the column vector of unknowns - B is the column vector of constants

For example, the system:

$$\begin{aligned} x - 2y - 5z &= 2 \\ -x + y + 3z &= -2 \\ 2x + y + z &= 3 \end{aligned}$$

becomes:

$$\begin{bmatrix} 1 & -2 & -5 \\ -1 & 1 & 3 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$$

25.1.2 The Augmented Matrix

We combine A and B into one matrix called the augmented matrix. We write it as $[A : B]$ or $[A|B]$:

$$[A : B] = \begin{bmatrix} 1 & -2 & -5 & | & 2 \\ -1 & 1 & 3 & | & -2 \\ 2 & 1 & 1 & | & 3 \end{bmatrix}$$

The vertical line separates the coefficients from the constants. We apply row operations to this augmented matrix to solve the system.

25.2 Consistency Conditions

25.2.1 The Rank Test

A system $AX = B$ is either consistent (has at least one solution) or inconsistent (has no solution). The rank decides which case applies.

Let n = number of unknowns, $\rho(A)$ = rank of coefficient matrix, $\rho[A : B]$ = rank of augmented matrix.

Condition	Result
$\rho(A) \neq \rho[A : B]$	**No solution** (inconsistent)
$\rho(A) = \rho[A : B] = n$	**Unique solution**
$\rho(A) = \rho[A : B] < n$	**Infinitely many solutions** (with $n - \rho(A)$ free variables)

25.2.2 Key Idea

When we reduce $[A : B]$ to echelon form, a contradiction row looks like $[0\ 0\ 0 \mid c]$ where $c \neq 0$. This says $0 = c$, which is impossible. Such a row means the system has no solution.

If no contradiction row appears, the system is consistent. Count the non-zero rows in the coefficient part. That is $\rho(A)$. If it equals the number of variables, you get a unique solution. If it is less, you get infinitely many solutions.

25.3 Solving by Row Reduction

25.3.1 The Method

1. Write the augmented matrix $[A : B]$.
2. Use elementary row operations to reach echelon form.
3. Check for contradiction rows.
4. Back-substitute from the bottom row upward.

25.3.2 Worked Example 1: Unique Solution (PYQ 2017)

Problem: Solve by elementary transformation:

$$\begin{aligned} 2x_1 + 2x_2 + x_3 &= 2 \\ 3x_1 + x_2 - 2x_3 &= 1 \\ 4x_1 - 3x_2 - x_3 &= 3 \end{aligned}$$

Solution:

Write the augmented matrix:

$$\begin{bmatrix} 2 & 2 & 1 & | & 2 \\ 3 & 1 & -2 & | & 1 \\ 4 & -3 & -1 & | & 3 \end{bmatrix}$$

Apply $R_2 \to 2R_2 - 3R_1$ and $R_3 \to R_3 - 2R_1$:

$$\begin{bmatrix} 2 & 2 & 1 & | & 2 \\ 0 & -4 & -7 & | & -4 \\ 0 & -7 & -3 & | & -1 \end{bmatrix}$$

Apply $R_3 \to 4R_3 - 7R_2$:

$$\begin{bmatrix} 2 & 2 & 1 & | & 2 \\ 0 & 4 & 7 & | & 4 \\ 0 & 0 & 37 & | & 24 \end{bmatrix}$$

Back-substitute. From row 3:

$$37x_3 = 24 \implies x_3 = \frac{24}{37}$$

From row 2:

$$4x_2 + 7 \cdot \frac{24}{37} = 4 \implies 4x_2 = 4 - \frac{168}{37} = \frac{-20}{37} \implies x_2 = -\frac{5}{37}$$

From row 1:

$$2x_1 + 2 \cdot \left(-\frac{5}{37}\right) + \frac{24}{37} = 2 \implies 2x_1 + \frac{14}{37} = 2 \implies x_1 = \frac{30}{37}$$

The unique solution is:

$$x_1 = \frac{30}{37}, \quad x_2 = -\frac{5}{37}, \quad x_3 = \frac{24}{37}$$

25.3.3 Worked Example 2: Infinite Solutions with Free Variables (PYQ 2019)

Problem: Discuss consistency and find the solution set:

$$\begin{aligned} x + 2y + 2z &= 1 \\ 2x + y + z &= 2 \\ 3x + 2y + 2z &= 3 \\ y + z &= 0 \end{aligned}$$

Solution:

Write the augmented matrix:

$$\begin{bmatrix} 1 & 2 & 2 & | & 1 \\ 2 & 1 & 1 & | & 2 \\ 3 & 2 & 2 & | & 3 \\ 0 & 1 & 1 & | & 0 \end{bmatrix}$$

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & 2 & | & 1 \\ 0 & -3 & -3 & | & 0 \\ 0 & -4 & -4 & | & 0 \\ 0 & 1 & 1 & | & 0 \end{bmatrix}$$

Scale:

$$R_2 \to -\frac{1}{3}R_2$$

and

$$R_3 \to -\frac{1}{4}R_3 :$$

$$\begin{bmatrix} 1 & 2 & 2 & | & 1 \\ 0 & 1 & 1 & | & 0 \\ 0 & 1 & 1 & | & 0 \\ 0 & 1 & 1 & | & 0 \end{bmatrix}$$

Apply $R_3 \to R_3 - R_2$ and $R_4 \to R_4 - R_2$:

$$\begin{bmatrix} 1 & 2 & 2 & | & 1 \\ 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Check consistency: $\rho(A) = 2$ and $\rho[A : B] = 2$. So the system is consistent.

Since there are 3 variables and rank is 2, we have $3 - 2 = 1$ free variable.

From row 2: $y + z = 0 \implies y = -z$.

From row 1: $x + 2(-z) + 2z = 1 \implies x = 1$.

Let $z = k$ where $k \in \mathbb{R}$. The solution set is:

$$x = 1, \quad y = -k, \quad z = k$$

25.3.4 Worked Example 3: Solving a System (PYQ 2023)

Problem: Solve:

$$\begin{aligned} x + 2y - z &= 2 \\ 2x + y + z &= 1 \\ x + 5y - 4z &= 5 \end{aligned}$$

Solution:

Write the augmented matrix:

$$\begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 2 & 1 & 1 & | & 1 \\ 1 & 5 & -4 & | & 5 \end{bmatrix}$$

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 0 & -3 & 3 & | & -3 \\ 0 & 3 & -3 & | & 3 \end{bmatrix}$$

Apply $R_3 \to R_3 + R_2$:

$$\begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 0 & -3 & 3 & | & -3 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Rank is 2 with 3 variables. So there is 1 free variable.

From row 2: $-3y + 3z = -3 \implies y - z = 1 \implies y = z + 1$.

From row 1: $x + 2(z + 1) - z = 2 \implies x + z + 2 = 2 \implies x = -z$.

Let $z = t$. The general solution is:

$$x = -t, \quad y = t + 1, \quad z = t$$

25.4 Finding a System with Parameters (PYQ 2018)

Sometimes the system contains a parameter λ and you must find the values of λ that allow solutions.

25.4.1 Worked Example 4: Parameter in the System

Problem: Find the value of λ so that the system has a solution. Solve completely in each case.

$$\begin{aligned} x + y + z &= 1 \\ x + 2y + 4z &= \lambda \\ x + 4y + 10z &= \lambda^2 \end{aligned}$$

Solution:

Write the augmented matrix:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 1 & 2 & 4 & | & \lambda \\ 1 & 4 & 10 & | & \lambda^2 \end{bmatrix}$$

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 3 & | & \lambda - 1 \\ 0 & 3 & 9 & | & \lambda^2 - 1 \end{bmatrix}$$

Apply $R_3 \to R_3 - 3R_2$:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 3 & | & \lambda - 1 \\ 0 & 0 & 0 & | & \lambda^2 - 3\lambda + 2 \end{bmatrix}$$

For consistency, the last row must not be a contradiction. So we need:

$$\lambda^2 - 3\lambda + 2 = 0 \implies (\lambda - 1)(\lambda - 2) = 0$$

So $\lambda = 1$ or $\lambda = 2$.

Case 1: $\lambda = 1$.

The augmented matrix becomes:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 3 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Let $z = t$. From row 2: $y = -3t$. From row 1: $x = 1 - y - z = 1 + 3t - t = 1 + 2t$.

$$x = 1 + 2t, \quad y = -3t, \quad z = t$$

Case 2: $\lambda = 2$.

The augmented matrix becomes:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 3 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Let $z = t$. From row 2: $y = 1 - 3t$. From row 1: $x = 1 - y - z = 1 - (1 - 3t) - t = 2t$.

$$x = 2t, \quad y = 1 - 3t, \quad z = t$$

25.5 Key Concepts to Remember

25.5.1 Consistent vs Inconsistent

A system is **consistent** if it has at least one solution. It is **inconsistent** if it has no solution at all.

25.5.2 Free Variables

When the rank is less than the number of unknowns, the difference gives the number of free variables. You assign parameters (t, k, etc.) to these free variables and express all other variables in terms of them.

25.5.3 Choosing Free Variables

In echelon form, the columns without pivots correspond to free variables. The columns with pivots correspond to basic (determined) variables.

25.6 Exam Patterns

- **This topic appears every year.** Usually as "Solve the following system by elementary transformation" or "Discuss the consistency and find the solution."
- Marks range from 5 to 7 per question.
- The most common format is 3 equations in 3 unknowns. Sometimes 4 equations appear.
- **Parameter problems** (with λ) appear frequently. See B08 for more on those.
- Always show the augmented matrix and every row operation step. This is where marks are given.
- Write the final answer clearly with a box or line. State the solution as $(x, y, z) = (a, b, c)$ or in parametric form.

25.7 Must-Know Problems

25.7.1 Problem 5: Class Note System

Problem: Solve:

$$\begin{aligned} x - 2y - 5z &= 2 \\ -x + y + 3z &= -2 \\ 2x + y + z &= 3 \end{aligned}$$

Solution:

Augmented matrix:

$$\begin{bmatrix} 1 & -2 & -5 & | & 2 \\ -1 & 1 & 3 & | & -2 \\ 2 & 1 & 1 & | & 3 \end{bmatrix}$$

Apply $R_2 \to R_1 + R_2$ and $R_3 \to 2R_1 - R_3$:

$$\begin{bmatrix} 1 & -2 & -5 & | & 2 \\ 0 & -1 & -2 & | & 0 \\ 0 & -5 & -11 & | & 1 \end{bmatrix}$$

Apply $R_3 \to 5R_2 - R_3$:

$$\begin{bmatrix} 1 & -2 & -5 & | & 2 \\ 0 & -1 & -2 & | & 0 \\ 0 & 0 & 1 & | & -1 \end{bmatrix}$$

Back-substitute. From row 3: $z = -1$.

From row 2: $-y - 2(-1) = 0 \implies y = 2$.

From row 1: $x - 2(2) - 5(-1) = 2 \implies x = 1$.

$$(x, y, z) = (1, 2, -1)$$

25.7.2 Problem 6: Homogeneous System (PYQ 2018)

Problem: Solve:

$$\begin{aligned} 2x_1 - x_2 + x_3 &= 0 \\ 3x_1 + 2x_2 + x_3 &= 0 \\ x_1 - 3x_2 + 5x_3 &= 0 \end{aligned}$$

Solution:

For a homogeneous system $AX = 0$, the trivial solution

$$x_1 = x_2 = x_3 = 0$$

always exists.

Write the coefficient matrix and reduce:

$$\begin{bmatrix} 1 & -3 & 5 \\ 3 & 2 & 1 \\ 2 & -1 & 1 \end{bmatrix}$$

(We swapped $R_1 \leftrightarrow R_3$ to get a leading 1.)

Apply $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 2R_1$:

$$\begin{bmatrix} 1 & -3 & 5 \\ 0 & 11 & -14 \\ 0 & 5 & -9 \end{bmatrix}$$

Apply $R_3 \to 11R_3 - 5R_2$:

$$\begin{bmatrix} 1 & -3 & 5 \\ 0 & 11 & -14 \\ 0 & 0 & -29 \end{bmatrix}$$

Rank = 3 = number of variables. So the system has only the trivial solution:

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0$$

25.7.3 Problem 7: Homogeneous System with Non-Trivial Solution (PYQ 2019)

Problem: Find the complete solution of:

$$\begin{aligned} x + y + z + w &= 0 \\ x + 3y - 2z + 4w &= 0 \\ 2x + y - z - w &= 0 \end{aligned}$$

Solution:

Coefficient matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & 4 \\ 2 & 1 & -1 & -1 \end{bmatrix}$$

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - 2R_1$:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 3 \\ 0 & -1 & -3 & -3 \end{bmatrix}$$

Apply $R_3 \to 2R_3 + R_2$:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 3 \\ 0 & 0 & -9 & -3 \end{bmatrix}$$

Rank $= 3$. Number of variables $= 4$. So there is $4 - 3 = 1$ free variable.

From row 3: $-9z - 3w = 0 \implies w = -3z$.

From row 2: $2y - 3z + 3(-3z) = 0 \implies 2y = 12z \implies y = 6z$.

From row 1: $x + 6z + z + (-3z) = 0 \implies x = -4z$.

Let $z = k$:

$$x = -4k, \quad y = 6k, \quad z = k, \quad w = -3k$$

The trivial solution is $k = 0$. Any $k \neq 0$ gives a non-trivial solution.

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26 B08: Parameter Analysis for Systems of Equations

Exam Weight: Tier 1 | **Appeared in**: 2018, 2020, 2021, 2024 + CT | **Typical Marks**: 5–7

Parameter analysis problems are a favourite of the examiners. The classic question asks: "For what values of λ and μ does the system have no solution, a unique solution, or infinitely many solutions?" One specific problem has appeared word-for-word in multiple years.

26.1 The General Approach

26.1.1 What the Problem Looks Like

You are given a system of equations where one or more coefficients are replaced by parameters (λ, μ, k, etc.). You must find which values of these parameters lead to each of the three cases:

1. **Unique solution**: The system behaves normally. One answer for each variable.
2. **No solution**: The system is contradictory. No set of values can satisfy all equations.
3. **Infinitely many solutions**: The system is underdetermined. Whole families of solutions exist.

26.1.2 The Method

1. Write the augmented matrix $[A : B]$.
2. Row-reduce to echelon form. The parameter will survive in the last row.
3. The last row will look like

$$[0\ 0\ f(\lambda) \mid g(\lambda, \mu)].$$

4. Analyze three cases based on $f(\lambda)$ and $g(\lambda, \mu)$.

26.1.3 The Decision Table

After row reduction, the critical row has the form:

$$[0 \quad 0 \quad f(\lambda) \quad | \quad g(\lambda, \mu)]$$

Condition	Result
$f(\lambda) \neq 0$	Unique solution (for any μ)
$f(\lambda) = 0$ and $g(\lambda, \mu) \neq 0$	No solution (contradiction: $0 = c$)
$f(\lambda) = 0$ and $g(\lambda, \mu) = 0$	Infinitely many solutions

---

26.2 Must-Know Problem 1: The Classic (PYQ 2020, 2024 — Repeated)

Problem: Investigate for what values of λ and μ the system has (i) no solution, (ii) unique solution, (iii) infinitely many solutions:

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 10 \\ x + 2y + \lambda z &= \mu \end{aligned}$$

This exact problem appeared in 2020 (7 marks) and 2024 (5 marks).

Solution:

Write the augmented matrix:

$$\begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 1 & 2 & 3 & | & 10 \\ 1 & 2 & \lambda & | & \mu \end{bmatrix}$$

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 2 & | & 4 \\ 0 & 1 & \lambda - 1 & | & \mu - 6 \end{bmatrix}$$

Apply $R_3 \to R_3 - R_2$:

$$\begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 2 & | & 4 \\ 0 & 0 & \lambda - 3 & | & \mu - 10 \end{bmatrix}$$

Now analyze the last row:

(i) No solution: Need $\lambda - 3 = 0$ and $\mu - 10 \neq 0$:

$$\lambda = 3 \quad \text{and} \quad \mu \neq 10$$

The last row becomes

$$[0\ 0\ 0 \mid \mu - 10]$$

, which says $0 = \mu - 10$. This is a contradiction since $\mu \neq 10$.

(ii) Unique solution: Need $\lambda - 3 \neq 0$:

$$\lambda \neq 3 \quad \text{(for any value of } \mu\text{)}$$

All three rows have pivots. Rank = 3 = number of unknowns.

(iii) Infinitely many solutions: Need $\lambda - 3 = 0$ and $\mu - 10 = 0$:

$$\lambda = 3 \quad \text{and} \quad \mu = 10$$

The last row becomes all zeros. Rank = 2 < 3 unknowns. One free variable.

26.3 Must-Know Problem 2: Single Parameter (PYQ 2021)

Problem: For what values of λ does the system have (i) no solution, (ii) unique solution, (iii) infinitely many solutions?

$$\begin{aligned}\lambda x + y + z &= 1\\ x + \lambda y + z &= \lambda\\ x + y + \lambda z &= \lambda^2\end{aligned}$$

Solution:

This system has λ in the coefficient matrix AND the constant vector. We use the determinant approach.

The coefficient matrix determinant is:

$$D = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix}$$

Add all three rows to R_1:

$$R_1 \to [\lambda + 2, \lambda + 2, \lambda + 2]$$

Factor out $(\lambda + 2)$:

$$D = (\lambda + 2) \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix}$$

Apply $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$:

$$D = (\lambda + 2) \begin{vmatrix} 1 & 0 & 0 \\ 1 & \lambda - 1 & 0 \\ 1 & 0 & \lambda - 1 \end{vmatrix} = (\lambda + 2)(\lambda - 1)^2$$

Analysis:

(ii) Unique solution: $D \neq 0$, which means $\lambda \neq 1$ and $\lambda \neq -2$.

(iii) Infinitely many solutions: When $\lambda = 1$, all three equations become $x + y + z = 1$. They are identical. Infinitely many solutions.

(i) No solution: When $\lambda = -2$, the system becomes:

$$\begin{aligned}-2x + y + z &= 1\\ x - 2y + z &= -2\\ x + y - 2z &= 4\end{aligned}$$

Add all three equations: $0 = 3$. Contradiction. No solution.

26.4 Must-Know Problem 3: CT Problem

Problem: Find the values of k for which the system has (i) unique solution, (ii) no solution, (iii) infinitely many solutions:

$$\begin{aligned} x + y + z &= 1 \\ x + 2y + 3z &= k \\ x + 5y + 9z &= k^2 \end{aligned}$$

Solution:

Augmented matrix:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 1 & 2 & 3 & | & k \\ 1 & 5 & 9 & | & k^2 \end{bmatrix}$$

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & | & k-1 \\ 0 & 4 & 8 & | & k^2-1 \end{bmatrix}$$

Apply $R_3 \to R_3 - 4R_2$:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & | & k-1 \\ 0 & 0 & 0 & | & k^2-4k+3 \end{bmatrix}$$

Factor the last entry:

$$k^2 - 4k + 3 = (k-1)(k-3).$$

(i) Unique solution: The coefficient part has rank 2 (not 3) because the last row of coefficients is all zeros. So this system can NEVER have a unique solution.

Wait. Let me re-examine. The coefficient matrix after reduction is:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of coefficient matrix = 2. Since rank < 3 (number of unknowns), a unique solution is impossible regardless of k.

(ii) No solution: $(k-1)(k-3) \neq 0 \implies k \neq 1$ and $k \neq 3$.

(iii) Infinitely many solutions: $(k-1)(k-3) = 0 \implies k = 1$ or $k = 3$.

When $k = 1$: Let $z = t$. From row 2: $y = -2t$. From row 1: $x = 1 - y - z = 1 + 2t - t = 1 + t$.

$$x = 1 + t, \quad y = -2t, \quad z = t$$

When $k = 3$: Let $z = t$. From row 2: $y = 2 - 2t$. From row 1: $x = 1 - y - z = 1 - (2 - 2t) - t = -1 + t$.

$$x = -1 + t, \quad y = 2 - 2t, \quad z = t$$

26.5 Key Observations

26.5.1 When Two Parameters Appear

If the system has both λ (in the coefficient matrix) and μ (in the constant vector), you get a 2D analysis. The answer typically looks like: - Unique: $\lambda \neq$ some value, $\mu =$ anything - No solution: $\lambda =$ that value, $\mu \neq$ some other value - Infinite: λ and μ both equal specific values

26.5.2 When One Parameter Appears

If λ appears only in the coefficient matrix, the analysis is simpler. You compute the determinant and find where it is zero. At those zero-points, check if the system is consistent or not.

26.5.3 The Determinant Shortcut

For a 3x3 system with parameter λ: - Compute

$$D = |A|$$

as a function of λ. - $D \neq 0$ means unique solution. - $D = 0$ means either no solution or infinite solutions. You must check further.

26.6 Exam Patterns

- **This appears about every other year.** It has shown up in 2018, 2020, 2021, 2024, and CTs.
- The $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ problem is the most repeated. Memorize it.
- Always show the complete row reduction. Marks are given for the steps.
- State all three cases clearly: (i), (ii), (iii) with the exact conditions on λ and μ.
- If the problem asks to "solve completely," you must also give the general solution in the infinite-solutions case.

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27 B09: Homogeneous Systems of Equations

Exam Weight: Tier 2 | **Appeared in**: 2018, 2019 | **Typical Marks**: 5–6

A homogeneous system always has the trivial solution. The real question is whether non-trivial solutions exist.

27.1 What is a Homogeneous System?

A system $AX = 0$ where the constant vector is all zeros:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0 \end{aligned}$$

The trivial solution

$$x_1 = x_2 = \cdots = x_n = 0$$

always exists. It always satisfies every equation.

27.2 When Do Non-Trivial Solutions Exist?

Let $\rho(A)$ be the rank of the coefficient matrix and n be the number of unknowns.

Condition	Result
$\rho(A) = n$	Only the trivial solution (zero solution)
$\rho(A) < n$	Non-trivial solutions exist. The solution space has dimension $n - \rho(A)$

For a square system ($m = n$): - $|A| \neq 0$ means only the trivial solution. -

$$|A| = 0$$

means non-trivial solutions exist.

27.3 The Solution Space

The set of all solutions to $AX = 0$ forms a vector subspace called the **null space** of A. Its dimension is the **nullity** $= n - \rho(A)$.

When non-trivial solutions exist, you express the solution in terms of free variables (parameters). Each free variable gives one dimension.

27.4 Worked Example 1: Only Trivial Solution (PYQ 2018)

Problem: Solve the homogeneous system:

$$\begin{aligned} 2x_1 - x_2 + x_3 &= 0 \\ 3x_1 + 2x_2 + x_3 &= 0 \\ x_1 - 3x_2 + 5x_3 &= 0 \end{aligned}$$

Solution:

Swap $R_1 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & -3 & 5 \\ 3 & 2 & 1 \\ 2 & -1 & 1 \end{bmatrix}$$

Apply $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 2R_1$:

$$\begin{bmatrix} 1 & -3 & 5 \\ 0 & 11 & -14 \\ 0 & 5 & -9 \end{bmatrix}$$

Apply $R_3 \to 11R_3 - 5R_2$:

$$\begin{bmatrix} 1 & -3 & 5 \\ 0 & 11 & -14 \\ 0 & 0 & -29 \end{bmatrix}$$

Rank = 3 = number of unknowns. Only the trivial solution:

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0$$

27.5 Worked Example 2: Non-Trivial Solutions (PYQ 2019)

Problem: Find the complete solution of:

$$\begin{aligned} x + y + z + w &= 0 \\ x + 3y - 2z + 4w &= 0 \\ 2x + y - z - w &= 0 \end{aligned}$$

Solution:

After row reduction:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$

Rank = 3. Variables = 4. So $4 - 3 = 1$ free variable.

From row 3: $3z + w = 0 \implies w = -3z$.

From row 2: $y + 3z + 3(-3z) = 0 \implies y = 6z$.

From row 1: $x + 6z + z - 3z = 0 \implies x = -4z$.

Let $z = k$. The general solution is:

$$x = -4k, \quad y = 6k, \quad z = k, \quad w = -3k$$

In vector form:

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = k \begin{bmatrix} -4 \\ 6 \\ 1 \\ -3 \end{bmatrix}$$

The null space is one-dimensional, spanned by $(-4, 6, 1, -3)^T$.

27.6 Exam Patterns

- Homogeneous systems appear as standalone problems or as part of eigenvalue problems (when solving $(A - \lambda I)X = 0$).
- The key test: compare rank with number of unknowns.
- If the problem says "find the trivial solution," the mathematical intent is usually to find ALL solutions.
- Express solutions in vector form when possible.

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28 B10: Eigenvalues and Eigenvectors

> **Exam Weight**: Tier 1 | **Appeared in**: 2017, 2018, 2019, 2020, 2021, 2023, 2024 (all 7 papers) | **Typical Marks**: 5–12

Eigenvalues and eigenvectors appear in every exam. The same 3x3 matrix has even been reused across years. Master the method and you guarantee yourself 5-12 marks.

28.1 Core Definitions

28.1.1 Eigenvalue and Eigenvector

Let A be a square matrix of order n. If a non-zero vector X and a scalar λ satisfy:

$$AX = \lambda X$$

then λ is an **eigenvalue** of A and X is the corresponding **eigenvector**.

The eigenvector does not change direction under the matrix transformation. It only gets scaled by the factor λ.

28.1.2 Characteristic Matrix

The matrix $A - \lambda I$ is called the characteristic matrix of A. Here I is the identity matrix of the same size as A.

28.1.3 Characteristic Polynomial

The determinant $|A - \lambda I|$ is a polynomial in λ. This polynomial is called the characteristic polynomial.

28.1.4 Characteristic Equation

Setting the characteristic polynomial equal to zero gives the characteristic equation:

$$|A - \lambda I| = 0$$

The roots of this equation are the eigenvalues.

28.2 The Working Rule

Follow these three steps for any eigenvalue/eigenvector problem:

Step 1: Form the characteristic equation. Write down $A - \lambda I$ and compute its determinant. Set it equal to zero.

Step 2: Solve the polynomial. Find all roots

$$\lambda_1, \lambda_2, \ldots, \lambda_n$$

. These are the eigenvalues.

Step 3: Find eigenvectors. For each λ_i, substitute it into

$$(A - \lambda_i I)X = 0$$

. Solve this homogeneous system by row reduction. The non-zero solutions are the eigenvectors.

28.3 Properties of Eigenvalues

These properties help you check your answers and sometimes shortcut the calculation:

1. The sum of all eigenvalues equals the trace (sum of diagonal elements) of A.

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = \mathrm{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$$

2. The product of all eigenvalues equals the determinant of A.

$$\lambda_1 \cdot \lambda_2 \cdots \lambda_n = |A|$$

3. For a triangular matrix (upper or lower), the eigenvalues are the diagonal entries.
4. If λ is an eigenvalue of A, then λ^k is an eigenvalue of A^k.
5. If A is invertible and λ is an eigenvalue, then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1}.
6. An $n \times n$ matrix has exactly n eigenvalues (counting multiplicity). They may be real or complex.
7. Eigenvectors corresponding to distinct eigenvalues are always linearly independent.

28.4 Worked Example 1: 2x2 Matrix (PYQ 2021)

Problem: Find the eigenvalues and eigenvectors of:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

Solution:

Step 1: The characteristic equation is

$$|A - \lambda I| = 0:$$

$$\begin{vmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(4-\lambda) - 3 = 0 \implies \lambda^2 - 6\lambda + 5 = 0$$

Step 2: Factor: $(\lambda - 1)(\lambda - 5) = 0$. So

$$\lambda_1 = 1$$

and

$$\lambda_2 = 5.$$

Check: Trace $= 2 + 4 = 6 = 1 + 5$. Determinant $= 8 - 3 = 5 = 1 \times 5$. Correct.

Step 3a: For $\lambda = 1$, solve $(A - I)X = 0$:

$$\begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $x + 3y = 0 \implies x = -3y$. Let $y = 1$:

$$X_1 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

Step 3b: For $\lambda = 5$, solve $(A - 5I)X = 0$:

$$\begin{bmatrix} -3 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $x - y = 0 \implies x = y$. Let $y = 1$:

$$X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

28.5 Worked Example 2: 3x3 Matrix (PYQ 2017, 2018 — Repeated)

Problem: Find the eigenvalues and eigenvectors of:

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

This exact matrix appeared in both 2017 (12 marks) and 2018 (6 marks).

Solution:

Step 1: Compute

$$|A - \lambda I| = 0 :$$

$$\begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = 0$$

Expand the determinant:

$$(2-\lambda)[(3-\lambda)(2-\lambda) - 2] - 2[(2-\lambda) - 1] + 1[2 - (3-\lambda)] = 0$$

$$(2-\lambda)[\lambda^2 - 5\lambda + 4] - 2[1-\lambda] + [\lambda - 1] = 0$$

After simplification:

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

Step 2: Test $\lambda = 1$: $1 - 7 + 11 - 5 = 0$. Yes, it is a root.

Divide by $(\lambda - 1)$:

$$(\lambda - 1)(\lambda^2 - 6\lambda + 5) = 0 \implies (\lambda - 1)(\lambda - 1)(\lambda - 5) = 0.$$

Eigenvalues: $\lambda = 1, 1, 5$.

Check: Trace $= 2 + 3 + 2 = 7 = 1 + 1 + 5$. Correct.

Step 3a: For $\lambda = 5$, solve $(A - 5I)X = 0$:

$$\begin{bmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 2 & -3 \end{bmatrix}$$

Swap $R_1 \leftrightarrow R_2$, then $R_2 \to R_2 + 3R_1$, $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & -4 & 4 \\ 0 & 4 & -4 \end{bmatrix}$$

Apply $R_3 \to R_3 + R_2$: row 3 becomes all zeros. From row 2: $y = z$. From row 1: $x = 2y - z = z$.

So $x = y = z$. Let $z = 1$:

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Step 3b: For $\lambda = 1$ (repeated), solve $(A - I)X = 0$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

All three rows are identical. This gives one equation: $x + 2y + z = 0 \implies x = -2y - z$.

Two free variables (y and z). Let $y = s$, $z = t$:

$$X = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Two independent eigenvectors:

$$X_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

28.6 Worked Example 3: 3x3 Matrix (PYQ 2020)

Problem: Find eigenvalues and eigenvectors of:

$$A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

Solution:

Step 1-2: The characteristic equation simplifies to:

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0 \implies (\lambda - 1)(\lambda + 1)(\lambda - 2) = 0$$

Eigenvalues: $\lambda = 1, -1, 2$.

Step 3a: For $\lambda = 1$:

$$\begin{bmatrix} 0 & 1 & -2 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

From row 1: $y = 2z$. From row 2: $x = y + z = 3z$. Let $z = 1$:

$$X_1 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Step 3b: For $\lambda = -1$:

$$\begin{bmatrix} 2 & 1 & -2 \\ -1 & 3 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

From row 3: $y = 0$. From row 1: $2x - 2z = 0 \implies x = z$. Let $z = 1$:

$$X_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Step 3c: For $\lambda = 2$:

$$\begin{bmatrix} -1 & 1 & -2 \\ -1 & 0 & 1 \\ 0 & 1 & -3 \end{bmatrix}$$

From row 2: $x = z$. From row 3: $y = 3z$. Let $z = 1$:

$$X_3 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

28.7 Worked Example 4: 3x3 Matrix (PYQ 2023)

Problem: Find eigenvalues and eigenvectors of:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Solution:

The characteristic equation gives:

$$(2-\lambda)(\lambda-1)(\lambda-3) = 0$$

Eigenvalues: $\lambda = 1, 2, 3$.

For $\lambda = 1$: Solving $(A - I)X = 0$ gives $y = 0$, $x = -z$. Eigenvector:

$$X_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

For $\lambda = 2$: Solving $(A - 2I)X = 0$ gives $x = -z$, $y = -z$. Eigenvector:

$$X_2 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

For $\lambda = 3$: Solving $(A - 3I)X = 0$ gives $x = 0$, $y = -z$. Eigenvector:

$$X_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

28.8 Special Cases

28.8.1 Triangular Matrices

For any upper or lower triangular matrix, the eigenvalues are the diagonal entries. No computation needed.

Example (PYQ 2017): The eigenvalues of:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{pmatrix}$$

are simply $\lambda = 1, -4, 7$.

28.8.2 Repeated Eigenvalues

When an eigenvalue has multiplicity m (it appears m times as a root), the number of independent eigenvectors can range from 1 to m. If you get fewer than m independent eigenvectors, the matrix cannot be diagonalized. This is important for the diagonalization topic (see B13).

28.8.3 Complex Eigenvalues

If the characteristic equation has complex roots, the eigenvectors will also have complex entries. Complex eigenvalues always come in conjugate pairs for real matrices. See the class notes for the worked example with

$$\lambda = \frac{3 \pm i\sqrt{3}}{2}.$$

28.9 Exam Patterns

- **Every paper asks eigenvalues.** The phrasing is always "Find the eigenvalues and eigenvectors of…"
- Usually a 3x3 matrix. Marks range from 5 to 12.
- The same matrix has appeared multiple years (the $[2, 2, 1; 1, 3, 1; 1, 2, 2]$ matrix appeared in 2017 and 2018).
- **Always verify** using trace and determinant. The trace must equal the sum of eigenvalues. The determinant must equal their product.
- For 3x3 matrices, the characteristic polynomial is a cubic. Try integer values (0, 1, 2, -1) first. One root is usually an integer, and you can factor out.
- When solving $(A - \lambda I)X = 0$, you solve a homogeneous system. Use echelon form and express variables in terms of free variables.
- Write eigenvectors with integer entries. Choose the free variable value to avoid fractions.

28.10 Eigenvalue Quick-Check Table

For a 3x3 matrix with eigenvalues

$$\lambda_1, \lambda_2, \lambda_3 :$$

Property	Formula
Sum	

$$\lambda_1 + \lambda_2 + \lambda_3 = a_{11} + a_{22} + a_{33}$$

Product |

$$\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = |A|$$

Sum of products of pairs |

$$\lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 =$$

sum of 2×2 cofactors |

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29 B11: Cayley-Hamilton Theorem

Exam Weight: Tier 2 | **Appeared in**: 2018, 2019, 2024 | **Typical Marks**: 5–6

The Cayley-Hamilton theorem connects a matrix to its own characteristic equation. The exam typically asks you to state the theorem, verify it for a given matrix, and then use it to find the inverse.

29.1 Statement

Every square matrix satisfies its own characteristic equation.

If the characteristic equation of matrix A of order n is:

$$\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0 = 0$$

then replacing λ with A and the constant term with $c_0 I$ gives:

$$A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I = O$$

where O is the zero matrix.

29.1.1 Key Idea

The characteristic equation is a polynomial equation in the scalar λ. The Cayley-Hamilton theorem says the same polynomial, when you plug in the matrix A instead of λ, gives the zero matrix. Constants become scalar multiples of the identity matrix I.

29.2 Full Proof

Let A be an $n \times n$ square matrix. Its characteristic polynomial is:

$$p(\lambda) = |A - \lambda I| = c_0 + c_1\lambda + c_2\lambda^2 + \cdots + (-1)^n\lambda^n$$

The adjoint of the characteristic matrix $(A - \lambda I)$ can be written as a polynomial in λ:

$$\text{adj}(A - \lambda I) = B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \cdots + B_0$$

where $B_0, B_1, \ldots, B_{n-1}$ are constant $n \times n$ matrices. This works because each cofactor of $(A - \lambda I)$ is a polynomial in λ of degree at most $n - 1$.

By the matrix identity

$$D \cdot \text{adj}(D) = |D| \cdot I$$

applied to $D = A - \lambda I$:

$$(A - \lambda I) \cdot \text{adj}(A - \lambda I) = |A - \lambda I| \cdot I$$

Substitute both polynomial expressions:

$$(A - \lambda I)(B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \cdots + B_0) = (c_0 + c_1\lambda + \cdots + (-1)^n\lambda^n)I$$

Expand the left side and equate coefficients of each power of λ:

For λ^0:

$$AB_0 = c_0 I$$

For λ^1:

$$AB_1 - B_0 = c_1 I$$

For λ^2:

$$AB_2 - B_1 = c_2 I$$

$\vdots$

For λ^{n-1}:

$$AB_{n-1} - B_{n-2} = c_{n-1} I$$

For λ^n:

$$-B_{n-1} = (-1)^n I$$

Now pre-multiply these equations by $I, A, A^2, \ldots, A^n$ respectively:

$$AB_0 = c_0 I$$

$$A^2B_1 - AB_0 = c_1 A$$

$$A^3B_2 - A^2B_1 = c_2 A^2$$

$$\vdots$$

$$A^nB_{n-1} - A^{n-1}B_{n-2} = c_{n-1}A^{n-1}$$

$$-A^nB_{n-1} = (-1)^n A^n$$

Add all these equations. The left side telescopes (each A^kB_{k-1} appears with $+$ in one equation and $-$ in the next). Everything cancels:

$$0 = c_0 I + c_1 A + c_2 A^2 + \cdots + (-1)^n A^n = p(A)$$

So $p(A) = O$. The matrix satisfies its own characteristic equation.

29.3 How to Verify the Theorem

The verification problem is a standard exam question. Follow these steps:

1. Find the characteristic equation

$$|A - \lambda I| = 0.$$

2. Write the matrix version: replace λ^k with A^k and constants with cI.
3. Compute A^2 by multiplying $A \cdot A$.
4. Compute A^3 by multiplying $A^2 \cdot A$ (for 3x3 matrices).
5. Substitute into the equation and verify every entry is zero.

29.4 Using Cayley-Hamilton to Find the Inverse

This is the most important application. If the characteristic equation is:

$$A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I = O$$

Multiply both sides by A^{-1}:

$$A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I + c_0 A^{-1} = O$$

Rearrange:

$$A^{-1} = -\frac{1}{c_0}\left(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I\right)$$

This requires $c_0 \neq 0$ (which means $|A| \neq 0$, so A must be invertible).

For a 3x3 matrix with equation

$$A^3 + aA^2 + bA + cI = O:$$

$$A^{-1} = -\frac{1}{c}(A^2 + aA + bI)$$

For a 2x2 matrix with equation

$$A^2 + aA + bI = O:$$

$$A^{-1} = -\frac{1}{b}(A + aI)$$

29.5 Worked Example 1: 2x2 Matrix (Lecture)

Problem: Verify Cayley-Hamilton and find A^{-1} for:

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

Solution:

Step 1: Characteristic equation.

$$|A - \lambda I| = (1-\lambda)(3-\lambda) + 2 = \lambda^2 - 4\lambda + 5 = 0$$

Step 2: Matrix equation:

$$A^2 - 4A + 5I = O.$$

Step 3: Compute A^2:

$$A^2 = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix}$$

Step 4: Verify:

$$A^2 - 4A + 5I = \begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ -4 & 12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \quad \checkmark$$

Step 5: Find A^{-1}. Multiply by A^{-1}:

$$A - 4I + 5A^{-1} = O \implies A^{-1} = \frac{1}{5}(4I - A) = \frac{1}{5}\begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$$

29.6 Worked Example 2: 3x3 Matrix (PYQ 2019)

Problem: Verify Cayley-Hamilton for the following matrix and find A^{-1}:

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

Solution:

Step 1: Characteristic equation.

$$|A - \lambda I| = 0 \implies \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

Step 2: Matrix equation:

$$A^3 - 6A^2 + 9A - 4I = O.$$

Step 3: Compute A^2:

$$A^2 = \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix}$$

Compute

$$A^3 = A^2 \cdot A :$$

$$A^3 = \begin{pmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{pmatrix}$$

Step 4: Verify. Check diagonal entry $(1,1)$: $22 - 6(6) + 9(2) - 4 = 22 - 36 + 18 - 4 = 0$. Check off-diagonal entry $(1,2)$: $-21 - 6(-5) + 9(-1) - 0 = -21 + 30 - 9 = 0$. All entries verify to zero.

Step 5: Find A^{-1}. Multiply by A^{-1}:

$$A^2 - 6A + 9I = 4A^{-1}$$

$$4A^{-1} = \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix} - \begin{pmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{pmatrix} + \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{pmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{pmatrix}$$

29.7 Worked Example 3: 3x3 Matrix (PYQ 2024)

Problem: Verify Cayley-Hamilton for A and compute A^{-1}:

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Solution:

Step 1: Characteristic equation:

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0.$$

Step 2: Matrix equation:

$$A^3 - 7A^2 + 11A - 5I = O.$$

Step 3: Compute:

$$A^2 = \begin{bmatrix} 7 & 12 & 6 \\ 6 & 13 & 6 \\ 6 & 12 & 7 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 32 & 62 & 31 \\ 31 & 63 & 31 \\ 31 & 62 & 32 \end{bmatrix}$$

Step 4: Verify. Entry $(1, 1)$: $32 - 49 + 22 - 5 = 0$. Entry $(1, 2)$: $62 - 84 + 22 + 0 = 0$. All entries zero.

Step 5: Find A^{-1}:

$$5A^{-1} = A^2 - 7A + 11I = \begin{bmatrix} 7 & 12 & 6 \\ 6 & 13 & 6 \\ 6 & 12 & 7 \end{bmatrix} - \begin{bmatrix} 14 & 14 & 7 \\ 7 & 21 & 7 \\ 7 & 14 & 14 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & -1 \\ -1 & -2 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & -1 \\ -1 & -2 & 4 \end{bmatrix}$$

29.8 Exam Patterns

- The typical question is “State and prove Cayley-Hamilton theorem” (6 marks) or “Verify Cayley-Hamilton for matrix A and find A^{-1}” (5-6 marks).
- The proof was asked in 2018. Verification was asked in 2019 and 2024.
- When verifying, show A^2 computation fully. You can abbreviate A^3 by showing a few entries if space is short.
- The inverse step is almost always attached. Do not skip it.
- The matrix from PYQ 2024 ($[2, 2, 1; 1, 3, 1; 1, 2, 2]$) is the same matrix used in the eigenvalue problem. Examiners pair these topics.

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30 B12: Adjoint and Inverse of a Matrix

Exam Weight: Tier 1-2 | **Appeared in**: 2017, 2019, 2020, 2021, 2023 | **Typical Marks**: 5–6

Finding the inverse of a matrix is one of the most common exam questions. Two methods exist: the adjoint method and the row operations method. You must know both.

30.1 Minors and Cofactors

30.1.1 Minor

The minor M_{ij} of element a_{ij} in a matrix A is the determinant of the submatrix obtained by deleting row i and column j from A.

30.1.2 Cofactor

The cofactor C_{ij} of element a_{ij} is the signed minor:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The sign follows a checkerboard pattern:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

30.1.3 Example

For matrix:

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}$$

The minor M_{11} is obtained by deleting row 1 and column 1:

$$M_{11} = \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 2 - 3 = -1$$

The cofactor

$$C_{11} = (-1)^{1+1}(-1) = -1.$$

30.2 Adjoint Matrix

30.2.1 Definition

The adjoint of a square matrix A (written $\text{adj}(A)$) is the transpose of the cofactor matrix.

Step 1: Find all cofactors C_{ij}.

Step 2: Form the cofactor matrix C.

Step 3: Transpose it:

$$\text{adj}(A) = C^T.$$

30.2.2 Key Property

$$A \cdot \text{adj}(A) = |A| \cdot I$$

This is the foundation of the adjoint method for finding the inverse.

30.3 Inverse by Adjoint Method

30.3.1 Definition

A square matrix A is invertible if there exists a matrix B such that $AB = BA = I$. This B is called A^{-1}.

A matrix is invertible if and only if $|A| \neq 0$.

30.3.2 Formula

$$A^{-1} = \frac{1}{|A|}\text{adj}(A)$$

30.3.3 Worked Example 1: Adjoint Method (PYQ 2017)

Problem: Find the adjoint and inverse of:

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}$$

Solution:

Step 1: Find the determinant:

$$|A| = 0(2-3) - 1(1-9) + 2(1-6) = 0 + 8 - 10 = -2$$

Since $|A| \neq 0$, the inverse exists.

Step 2: Find all cofactors:

$$C_{11} = +(2-3) = -1, \quad C_{12} = -(1-9) = 8, \quad C_{13} = +(1-6) = -5$$

$$C_{21} = -(1-2) = 1, \quad C_{22} = +(0-6) = -6, \quad C_{23} = -(0-3) = 3$$

$$C_{31} = +(3-4) = -1, \quad C_{32} = -(0-2) = 2, \quad C_{33} = +(0-1) = -1$$

Step 3: Adjoint = transpose of cofactor matrix:

$$\text{adj}(A) = \begin{pmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{pmatrix}$$

Step 4: Inverse:

$$A^{-1} = \frac{1}{-2}\begin{pmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & -1 \\ 5/2 & -3/2 & 1/2 \end{pmatrix}$$

30.3.4 Worked Example 2: Rotation Matrix Inverse (PYQ 2020)

Problem: Find the adjoint and inverse of:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution:

Determinant:

$$|A| = \cos^2\theta + \sin^2\theta = 1.$$

Cofactors:

$$C_{11} = \cos\theta, \quad C_{12} = -\sin\theta, \quad C_{13} = 0$$

$$C_{21} = \sin\theta, \quad C_{22} = \cos\theta, \quad C_{23} = 0$$

$$C_{31} = 0, \quad C_{32} = 0, \quad C_{33} = 1$$

Adjoint (transpose of cofactor matrix):

$$\text{adj}(A) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since

$$|A| = 1:$$

$$A^{-1} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Notice:

$$A^{-1} = A^T$$

. This is because A is an orthogonal matrix (rotation matrix).

30.4 Inverse by Row Operations

30.4.1 The Method

1. Write the augmented matrix $[A \mid I]$.
2. Apply elementary row operations to transform A into I.
3. The same operations transform I into A^{-1}.
4. When the left side becomes I, the right side is A^{-1}.

30.4.2 Worked Example 3: Row Operations (PYQ 2019)

Problem: Find the inverse of:

$$A = \begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$$

Solution:

Set up $[A|I]$:

$$\begin{bmatrix} 1 & 3 & 3 & | & 1 & 0 & 0 \\ 1 & 4 & 3 & | & 0 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{bmatrix}$$

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 3 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1 \end{bmatrix}$$

Apply $R_1 \to R_1 - 3R_2$:

$$\begin{bmatrix} 1 & 0 & 3 & | & 4 & -3 & 0 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1 \end{bmatrix}$$

Apply $R_1 \to R_1 - 3R_3$:

$$\begin{bmatrix} 1 & 0 & 0 & | & 7 & -3 & -3 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1 \end{bmatrix}$$

The inverse is:

$$A^{-1} = \begin{pmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

30.4.3 Worked Example 4: Row Operations (PYQ 2021)

Problem: Find the inverse by row operations:

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

Solution:

Set up $[A|I]$ and swap $R_1 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 2 & 2 & 1 & | & 0 & 1 & 0 \\ 2 & 1 & 2 & | & 1 & 0 & 0 \end{bmatrix}$$

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 2R_1$:

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 0 & -2 & -3 & | & 0 & 1 & -2 \\ 0 & -3 & -2 & | & 1 & 0 & -2 \end{bmatrix}$$

Scale

$$R_2 \to -\frac{1}{2}R_2 :$$

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 0 & 1 & 3/2 & | & 0 & -1/2 & 1 \\ 0 & -3 & -2 & | & 1 & 0 & -2 \end{bmatrix}$$

Apply $R_3 \to R_3 + 3R_2$:

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 0 & 1 & 3/2 & | & 0 & -1/2 & 1 \\ 0 & 0 & 5/2 & | & 1 & -3/2 & 1 \end{bmatrix}$$

Scale

$$R_3 \to \frac{2}{5}R_3$$

, then back-substitute upward. After all operations:

$$A^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 2 & -3 \\ -3 & 2 & 2 \\ 2 & -3 & 2 \end{pmatrix}$$

30.4.4 Worked Example 5: Row Operations (PYQ 2023)

Problem: Find the inverse by row transformation:

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 5 \\ 3 & 7 & 6 \end{bmatrix}$$

Solution:

Set up $[A|I]$. Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 3 & 5 & | & 1 & 0 & 0 \\ 0 & -2 & -5 & | & -2 & 1 & 0 \\ 0 & -2 & -9 & | & -3 & 0 & 1 \end{bmatrix}$$

Apply $R_3 \to R_3 - R_2$:

$$\begin{bmatrix} 1 & 3 & 5 & | & 1 & 0 & 0 \\ 0 & -2 & -5 & | & -2 & 1 & 0 \\ 0 & 0 & -4 & | & -1 & -1 & 1 \end{bmatrix}$$

Scale

$$R_2 \to -\frac{1}{2}R_2$$

and

$$R_3 \to -\frac{1}{4}R_3$$

. Then back-substitute:

$$A^{-1} = \frac{1}{8}\begin{bmatrix} -11 & 17 & -5 \\ 3 & -9 & 5 \\ 2 & 2 & -2 \end{bmatrix}$$

30.5 Which Method to Use?

Method	Best For	Pros	Cons
Adjoint	Small matrices, when adjoint is also asked	Gives both adjoint and inverse	Tedious for large matrices (9 cofactors for 3x3)
Row operations	Any size matrix, when only inverse is asked	Systematic, fewer calculations	Does not give the adjoint
Cayley-Hamilton	When eigenvalues or characteristic equation is already known	Elegant, uses known information	Requires A^2 computation

If the problem says "find adjoint and inverse," use the adjoint method. If the problem says "find inverse by row operations," use row operations. If the problem says "verify Cayley-Hamilton and find inverse," use Cayley-Hamilton (see B11).

30.6 Important Proofs

30.6.1 Proof: Inverse of a Product

Statement:

$$(AB)^{-1} = B^{-1}A^{-1}$$

(the order reverses).

Proof: We need to show that $B^{-1}A^{-1}$ satisfies the definition of the inverse of AB.

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

Since

$$(AB)(B^{-1}A^{-1}) = I$$

and

$$(B^{-1}A^{-1})(AB) = I$$

, we have

$$(AB)^{-1} = B^{-1}A^{-1}.$$

This proof has appeared in 2018 and 2023.

30.7 Exam Patterns

- **Inverse appears in 5 out of 7 papers.** Usually "Find the inverse by row operations" or "Find the adjoint and inverse."
- Row operations method is more common in recent papers (2019, 2021, 2023).
- The adjoint method was popular in older papers (2017, 2020).
- The proof

$$(AB)^{-1} = B^{-1}A^{-1}$$

has appeared twice. Memorize it. - Always check: $|A| \neq 0$ before attempting inverse. State this explicitly. - Verify your answer by multiplying AA^{-1} (at least check one row) if time permits.

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31 B13: Diagonalization, Matrix Exponential, and Matrix ODE

Exam Weight: Tier 2-3 | **Appeared in**: 2017, 2020, 2021 | **Typical Marks**: 6–12

This topic combines eigenvalues with differential equations. The matrix ODE problem appeared identically in 2020 and 2021.

31.1 Diagonalization

31.1.1 What is Diagonalization?

A square matrix A is diagonalizable if there exists an invertible matrix P such that:

$$P^{-1}AP = D$$

where D is a diagonal matrix. The diagonal entries of D are the eigenvalues of A. The columns of P are the corresponding eigenvectors.

31.1.2 When is Diagonalization Possible?

An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors.

- If all n eigenvalues are distinct, the matrix is always diagonalizable.
- If eigenvalues are repeated, diagonalization depends on whether enough independent eigenvectors exist. If the geometric multiplicity (number of independent eigenvectors) equals the algebraic multiplicity (repeat count) for each eigenvalue, the matrix is diagonalizable.

31.1.3 The Modal Matrix

The matrix P whose columns are the eigenvectors is called the **modal matrix**:

$$P = [X_1 \ X_2 \ \dots \ X_n]$$

Then

$$A = PDP^{-1}.$$

31.2 When Diagonalization Fails (PYQ 2017)

The matrix

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

has eigenvalue $\lambda = 1$ with algebraic multiplicity 2.

Solving $(A - I)X = 0$:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} X = 0$$

This gives only one eigenvector:

$$X = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Geometric multiplicity $= 1 <$ algebraic multiplicity $= 2$. So A is NOT diagonalizable.

31.3 Matrix Exponential

31.3.1 Definition

The matrix exponential e^A is defined by the power series:

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

31.3.2 Computation Using Diagonalization

If

$$A = PDP^{-1}:$$

$$e^A = Pe^D P^{-1}$$

where e^D is simply:

$$e^D = \begin{bmatrix} e^{\lambda_1} & 0 & \dots \\ 0 & e^{\lambda_2} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

31.3.3 Computation Using Decomposition (PYQ 2017)

When diagonalization fails, decompose $A = I + B$ where B is nilpotent.

For

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}:$$

Set

$$B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then

$$B^2 = O.$$

$$e^A = e^{I+B} = e^I \cdot e^B = e(I + B) = e \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e & e \\ 0 & e \end{bmatrix}$$

For time-dependent version:

$$e^{At} = e^t(I + Bt) = e^t \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix}$$

31.4 Solving Matrix ODE

31.4.1 The System

Given a system of first-order linear ODEs:

$$\frac{dX}{dt} = MX$$

where M is a constant matrix and X is the vector of unknowns.

31.4.2 The Method

1. Find eigenvalues of M.
2. Find eigenvectors of M.
3. The general solution is:

$$X(t) = C_1 e^{\lambda_1 t} v_1 + C_2 e^{\lambda_2 t} v_2 + \dots$$

where λ_i are eigenvalues and v_i are corresponding eigenvectors.

31.5 Must-Know Problem: Matrix ODE (PYQ 2020, 2021 — Repeated)

Problem: Solve by matrix method:

$$\frac{dx}{dt} = 6x - 3y, \quad \frac{dy}{dt} = 2x + y$$

Solution:

Write in matrix form

$$\frac{dX}{dt} = MX$$

where:

$$M = \begin{bmatrix} 6 & -3 \\ 2 & 1 \end{bmatrix}$$

Step 1: Eigenvalues.

$$|M - \lambda I| = 0:$$

$$(6 - \lambda)(1 - \lambda) + 6 = \lambda^2 - 7\lambda + 12 = (\lambda - 3)(\lambda - 4) = 0$$

So

$$\lambda_1 = 3$$

and

$$\lambda_2 = 4.$$

Step 2: Eigenvectors.

For $\lambda = 3$: $3x - 3y = 0 \implies x = y$. Eigenvector:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For $\lambda = 4$:

$$2x - 3y = 0 \implies x = \frac{3}{2}y$$

. Eigenvector:

$$v_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Step 3: General solution:

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = C_1 e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C_2 e^{4t} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Component form:

$$x(t) = C_1 e^{3t} + 3C_2 e^{4t}$$

$$y(t) = C_1 e^{3t} + 2C_2 e^{4t}$$

31.6 Exam Patterns

- Matrix ODE ($dx/dt = 6x - 3y$, $dy/dt = 2x + y$) appeared identically in 2020 and 2021. Memorize it.
- The 2017 paper had a 12-mark diagonalization + e^A + ODE problem.
- The method is always: eigenvalues then eigenvectors then write the general solution.
- If the matrix is not diagonalizable, mention it and use the decomposition method for e^A.

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32 C01: Vector Space Definition and Properties

Exam Weight: Tier 1-2 | **Appeared in**: 2017, 2018, 2024 | **Typical Marks**: 2–8

A vector space is the fundamental structure of linear algebra. It is a set of elements (vectors) that can be scaled and added together according to specific mathematical rules.

32.1 Formal Definition

Let V be a non-empty set of objects called vectors. Let K be a field of scalars (usually real numbers $\mathbb{R}$ or complex numbers $\mathbb{C}$).

The set V is a **vector space** over the field K if it satisfies two closure conditions and eight axioms under vector addition and scalar multiplication:

Closure Properties: 1. **Closure under Addition:** If $u, v \in V$, then $u + v \in V$. 2. **Closure under Scalar Multiplication:** If $u \in V$ and $k \in K$, then $ku \in V$.

Addition Axioms: 3. **Commutativity:** $u + v = v + u$ 4. **Associativity:** $(u + v) + w = u + (v + w)$ 5. **Additive Identity:** There exists a zero vector $0 \in V$ such that $u + 0 = u$. 6. **Additive Inverse:** For every $u \in V$, there exists $-u \in V$ such that $u + (-u) = 0$.

Scalar Multiplication Axioms: 7. **Distributivity (Scalar over Vector Addition):** $k(u+v) = ku+kv$ 8. **Distributivity (Vector over Scalar Addition):** $(a+b)u = au+bu$ 9. **Associativity of Multiplication:** $a(bu) = (ab)u$ 10. **Unitary Identity:** $1u = u$, where 1 is the multiplicative identity in K.

32.2 Must-Know Proofs: Basic Properties (PYQ 2017)

You must be able to prove four basic properties of vector spaces using *only* the axioms above. This is an 8-mark question.

32.2.1 Proof 1

We know $0 + 0 = 0$. Multiply by scalar k: $k(0 + 0) = k0$. By distributivity: $k0 + k0 = k0$. Add the additive inverse $-(k0)$ to both sides: $(k0 + k0) + (-(k0)) = k0 + (-(k0))$ $k0 + (k0 - k0) = 0$ $k0 + 0 = 0 \implies k0 = 0$. ■

32.2.2 Proof 2

In the scalar field, $0 + 0 = 0$. Multiply by vector u: $(0 + 0)u = 0u$. By distributivity: $0u + 0u = 0u$. Add the additive inverse $-(0u)$ to both sides: $(0u + 0u) + (-(0u)) = 0u + (-(0u))$ $0u + (0u - 0u) = 0$ $0u + 0 = 0 \implies 0u = 0$. ■

32.2.3 Proof 3: If, then or

Assume $ku = 0$. If $k = 0$, the statement holds. If $k \neq 0$, then its multiplicative inverse k^{-1} exists in the field K. Multiply both sides by k^{-1}:

$$k^{-1}(ku) = k^{-1}(0)$$

By associativity and Proof 1:

$$(k^{-1}k)u = 0$$

$1u = 0$ By the unitary axiom: $u = 0$. Thus, either $k = 0$ or $u = 0$. ■

32.2.4 Proof 4

The vector $-ku$ is the unique additive inverse of ku, meaning $ku + (-ku) = 0$. Show that $(-k)u$ is the additive inverse: $ku + (-k)u = (k + (-k))u = 0u = 0$ (by Proof 2). Thus, $(-k)u = -ku$.

Show that $k(-u)$ is the additive inverse: $ku + k(-u) = k(u + (-u)) = k(0) = 0$ (by Proof 1). Thus, $k(-u) = -ku$. Therefore, $(-k)u = k(-u) = -ku$. ■

32.3 Exam Patterns

- When asked to "define a vector space", you must list the properties clearly. Bullet points are fine, but be sure to mention the field K, the closure properties, and the 8 axioms.
- The 4-part proof from 2017 is a classic test of abstract algebra logic. Always explicitly state which axiom you are using at each step (e.g., "by distributivity").

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33 C02: Vector Subspaces

Exam Weight: Tier 2 | **Appeared in**: 2017, 2018, 2024 | **Typical Marks**: 3–4

A subspace is a smaller vector space entirely contained within a larger one.

33.1 Definition of a Subspace

Let W be a non-empty subset of a vector space V over a field K. W is called a **subspace** of V if W is itself a vector space under the same vector addition and scalar multiplication defined on V.

33.1.1 The Subspace Test

To prove that a subset W is a subspace, you do *not* need to check all 10 vector space axioms. You only need to verify three things:

1. **Non-empty / Zero Vector**: The zero vector of V must be in W ($0 \in W$).
2. **Closed under Addition**: For any vectors $u, v \in W$, their sum must also be in W ($u + v \in W$).
3. **Closed under Scalar Multiplication**: For any vector $u \in W$ and any scalar $k \in K$, the scaled vector must also be in W ($ku \in W$).

Note: Conditions 2 and 3 are sometimes combined into a single condition: For any scalars a, b and vectors u, v, $au + bv \in W$.

33.2 Worked Example: Subspace Verification (PYQ 2024)

Problem: Let U consist of all vectors in $\mathbb{R}^3$ whose entries are equal, that is $U = \{(a, b, c) \mid a = b = c\}$. Prove that U is a vector space over $\mathbb{R}$.

Solution: Since U is a subset of the known vector space $\mathbb{R}^3$, we only need to show it is a subspace. We apply the subspace test.

Step 1: Contains the Zero Vector The zero vector of $\mathbb{R}^3$ is $(0, 0, 0)$. Since $0 = 0 = 0$ satisfies the condition $a = b = c$, the zero vector is in U.

Step 2: Closed Under Addition Let

$$u_1 = (a_1, a_1, a_1) \in U$$

and

$$u_2 = (a_2, a_2, a_2) \in U.$$

We add these vectors:

$$u_1 + u_2 = (a_1 + a_2, a_1 + a_2, a_1 + a_2)$$

Since all three components of the sum vector are identical (equal to $a_1 + a_2$), the sum vector is in U.

Step 3: Closed Under Scalar Multiplication Let $u = (a, a, a) \in U$ and $k \in \mathbb{R}$ be a scalar. We multiply the vector by the scalar:

$$ku = (ka, ka, ka)$$

Since all three components are identical (equal to ka), this vector is in U.

Since U satisfies all three conditions, it is a subspace of $\mathbb{R}^3$. Because a subspace is itself a vector space, U is a vector space over $\mathbb{R}$. ■

33.3 Common Examples of Subspaces

- **Trivial Subspaces**: For any vector space V, the set containing only the zero vector $\{0\}$ and the entire space V itself are both subspaces.
- **Lines and Planes**: In $\mathbb{R}^3$, any line that passes through the origin is a subspace. Any flat plane that passes through the origin (e.g., $z = 0$) is a subspace. *Warning: If a line or plane does NOT pass through the origin (e.g., $z = 5$), it is NOT a subspace because it fails the zero vector test.*

33.4 Exam Patterns

- When asked to prove something is a vector space, usually it is a subset of a known space (like $\mathbb{R}^n$ or the space of polynomials). Do not list all 10 axioms; just use the 3-step subspace test.
- Always check the zero vector first. It is the easiest way to prove a set is *not* a subspace. If plugging in zeros for the variables breaks the given condition, you're done.

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34 C03: Sum and Direct Sum of Subspaces

Exam Weight: Tier 1-2 | **Appeared in**: 2019, 2023, 2024 | **Typical Marks**: 2–6

When you have multiple subspaces within a vector space, you can combine them. The most important way to combine them is via a "Direct Sum", which guarantees a unique decomposition of vectors.

34.1 Definitions and Examples

Sum of Subspaces ($U + W$): Let U and W be subspaces of a vector space V. The sum $U + W$ is the set of all possible vectors $u + w$ where $u \in U$ and $w \in W$. - *Example*: In $\mathbb{R}^2$, let U be the x-axis and W be the line $y = x$. Their sum $U + W$ covers the entire $\mathbb{R}^2$ plane.

Direct Sum ($U \oplus W$): The sum $U + W$ is called a **direct sum** (written $U \oplus W$) if every element in the sum can be written *uniquely* as $u + w$. - This occurs if and only if the only vector shared by both subspaces is the zero vector: $U \cap W = \{0\}$. - *Example*: In $\mathbb{R}^2$, let U be the x-axis (vectors $(x, 0)$) and W be the y-axis (vectors $(0, y)$). Their intersection is only the origin $(0, 0)$. Thus,

$$\mathbb{R}^2 = U \oplus W$$

, because any vector (x, y) has exactly one decomposition: $(x, 0) + (0, y)$.

34.2 Must-Know Proof (PYQ 2019)

Problem: Prove that the vector space V is the direct sum of its subspaces U and W if and only if (i) $V = U + W$, and (ii) $U \cap W = \{0\}$.

Proof:

Part 1: Forward Direction ($\implies$) Assume V is the direct sum of U and W ($V = U \oplus W$). By definition, every vector $v \in V$ can be *uniquely* written as $v = u + w$ (where $u \in U, w \in W$). - Since every v can be written as $u + w$, we have $V = U + W$. - To prove $U \cap W = \{0\}$, let $v \in U \cap W$. Since $v \in U$, we can write it as $v = v + 0$ (where $v \in U$ and $0 \in W$). Since $v \in W$, we can write it as $v = 0 + v$ (where $0 \in U$ and $v \in W$). Because the decomposition must be unique, these two expressions must be identical. Therefore, $v = 0$. Thus, $U \cap W = \{0\}$.

Part 2: Reverse Direction ($\impliedby$) Assume (i) $V = U + W$ and (ii) $U \cap W = \{0\}$. Since $V = U + W$, every $v \in V$ can be written as $v = u + w$. We must show this representation is unique. Suppose v can be represented in two ways:

$$v = u_1 + w_1 \quad \text{and} \quad v = u_2 + w_2$$

Equating them:

$$u_1 + w_1 = u_2 + w_2 \implies u_1 - u_2 = w_2 - w_1$$

Let this new vector be x. Because U is a subspace,

$$x = u_1 - u_2 \in U.$$

Because W is a subspace,

$$x = w_2 - w_1 \in W.$$

Therefore, $x \in U \cap W$. But we are given $U \cap W = \{0\}$. Thus, $x = 0$.

$$u_1 - u_2 = 0 \implies u_1 = u_2$$

$$w_2 - w_1 = 0 \implies w_1 = w_2$$

Since the components are identical, the representation is unique. Therefore, $V = U \oplus W$. ■

34.3 Exam Patterns

- The proof above is a highly standard 6-mark linear algebra theorem. Memorize the trick for the forward direction ($v = v + 0$ vs $v = 0 + v$) and the reverse direction (

$$u_1 - u_2 = w_2 - w_1$$

). - When asked to "define" direct sum for 2 marks, always include the mathematical condition $U \cap W = \{0\}$ and provide the x-axis / y-axis example in $\mathbb{R}^2$ for full credit.

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35 C04: Linear Combination and Span

Exam Weight: Tier 2 | **Appeared in**: 2020, 2021, 2023, 2024 | **Typical Marks**: 3–5

A vector is a **linear combination** of a set of vectors if it can be created by scaling those vectors and adding them together.

35.1 The Core Concept

To express a vector v as a linear combination of vectors $e_1, e_2, \ldots, e_n$, we set up the equation:

$$v = c_1 e_1 + c_2 e_2 + \cdots + c_n e_n$$

This generates a system of linear equations where the unknowns are the scalars $c_1, c_2, \ldots, c_n$. Solving this system (usually via an augmented matrix and row reduction) gives the required coefficients.

35.2 Worked Example 1: Coordinate Vectors (PYQ 2020, 2021)

Problem: Write the vector $v = (1, -2, 5)$ as a linear combination of the vectors

$$e_1 = (1, 1, 1),$$

$$e_2 = (1, 2, 3)$$

, and

$$e_3 = (2, -1, 1).$$

Solution: Set up the relation:

$$(1, -2, 5) = c_1(1, 1, 1) + c_2(1, 2, 3) + c_3(2, -1, 1)$$

This gives the system of equations (grouping by components):

$$c_1 + c_2 + 2c_3 = 1$$

$$c_1 + 2c_2 - c_3 = -2$$

$$c_1 + 3c_2 + c_3 = 5$$

Write the augmented matrix and apply row operations to reach row echelon form:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 1 & 2 & -1 & | & -2 \\ 1 & 3 & 1 & | & 5 \end{bmatrix}$$

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & -3 & | & -3 \\ 0 & 2 & -1 & | & 4 \end{bmatrix}$$

Apply $R_3 \to R_3 - 2R_2$:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & -3 & | & -3 \\ 0 & 0 & 5 & | & 10 \end{bmatrix}$$

Solve by back substitution: - From R_3:

$$5c_3 = 10 \implies c_3 = 2$$

- From R_2:

$$c_2 - 3(2) = -3 \implies c_2 = 3$$

- From R_1:

$$c_1 + 3 + 2(2) = 1 \implies c_1 = -6$$

The linear combination is:

$$v = -6e_1 + 3e_2 + 2e_3$$

35.3 Worked Example 2: Polynomial Vectors (PYQ 2024)

Problem: Express the polynomial

$$v = 3t^2 + 5t - 5$$

as a linear combination of the polynomials

$$P_1 = t^2 + 2t + 1,$$

$$P_2 = 2t^2 + 5t + 4$$

, and

$$P_3 = t^2 + 3t + 6.$$

Solution: Set up the relation:

$$v = c_1 P_1 + c_2 P_2 + c_3 P_3$$

$$3t^2 + 5t - 5 = c_1(t^2 + 2t + 1) + c_2(2t^2 + 5t + 4) + c_3(t^2 + 3t + 6)$$

Expand and group terms by powers of t:

$$3t^2 + 5t - 5 = (c_1 + 2c_2 + c_3)t^2 + (2c_1 + 5c_2 + 3c_3)t + (c_1 + 4c_2 + 6c_3)$$

Match the coefficients to form a system of equations: 1.

$$c_1 + 2c_2 + c_3 = 3$$

2.

$$2c_1 + 5c_2 + 3c_3 = 5$$

3.

$$c_1 + 4c_2 + 6c_3 = -5$$

You can solve this using substitution or matrix row reduction. Let's use substitution from Eq 1 (

$$c_1 = 3 - 2c_2 - c_3$$

): Substitute into Eq 2:

$$2(3 - 2c_2 - c_3) + 5c_2 + 3c_3 = 5 \implies 6 - 4c_2 - 2c_3 + 5c_2 + 3c_3 = 5 \implies c_2 + c_3 = -1$$

Substitute into Eq 3:

$$(3 - 2c_2 - c_3) + 4c_2 + 6c_3 = -5 \implies 3 + 2c_2 + 5c_3 = -5 \implies 2c_2 + 5c_3 = -8$$

Now solve the 2×2 system:

$$c_2 = -1 - c_3.$$

$$2(-1 - c_3) + 5c_3 = -8 \implies -2 - 2c_3 + 5c_3 = -8 \implies 3c_3 = -6 \implies c_3 = -2$$

Find c_2:

$$c_2 = -1 - (-2) = 1.$$

Find c_1:

$$c_1 = 3 - 2(1) - (-2) = 3.$$

The linear combination is:

$$v = 3P_1 + P_2 - 2P_3$$

35.4 Exam Patterns

- When given polynomials, simply treat their coefficients like a standard vector (e.g., $3t^2 + 5t - 5$ becomes $(3, 5, -5)$). The math is identical to coordinate vector problems.
- If the system of equations is inconsistent (e.g., you get $0 = 5$ during row reduction), then it is *not possible* to write the vector as a linear combination of the given set.

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36 C05: Linear Dependence and Independence

Exam Weight: Tier 1 | **Appeared in**: 2018, 2020, 2021, 2024 | **Typical Marks**: 4–6

A set of vectors is **linearly independent** if no vector in the set can be written as a linear combination of the others. If at least one vector can be, the set is **linearly dependent**.

36.1 Definitions and The Determinant Test

Linearly Independent: A set of vectors

$$\{v_1, v_2, \ldots, v_n\}$$

is linearly independent if the equation

$$c_1 v_1 + c_2 v_2 + \cdots + c_n v_n = 0$$

has *only the trivial solution* (

$$c_1 = c_2 = \cdots = c_n = 0$$

). **Linearly Dependent**: The set is dependent if the equation has a solution where at least one coefficient is not zero.

36.1.1 The Determinant Test

If you have exactly n vectors in $\mathbb{R}^n$, you can form an $n \times n$ matrix with the vectors as its rows (or columns). - If Determinant $\neq 0 \implies$ Linearly **Independent**. - If Determinant $= 0 \implies$ Linearly **Dependent**.

*(Note: If you are given a theoretical proof question (PYQ 2024) asking you to show why this determinant rule is true, set

$$a\vec{A} + b\vec{B} + c\vec{C} = 0$$

, convert it to a homogeneous system, and state that a homogeneous system only has the trivial solution if the determinant of the coefficient matrix is non-zero).*

36.2 Worked Example 1: Finding the Dependence Relation (PYQ 2018)

Problem: Show that

$$\vec{A} = 2\hat{\imath} + \hat{\jmath} - 3\hat{k},$$

$$\vec{B} = \hat{\imath} - 4\hat{k},$$

$$\vec{C} = 4\hat{\imath} + 3\hat{\jmath} - \hat{k}$$

are linearly dependent, and determine a relation between them.

Solution: **Step 1: Check Determinant**

$$D = \begin{vmatrix} 2 & 1 & -3 \\ 1 & 0 & -4 \\ 4 & 3 & -1 \end{vmatrix} = 2(0-(-12)) - 1(-1-(-16)) - 3(3-0) = 24 - 15 - 9 = 0$$

Since $D = 0$, the vectors are linearly dependent.

Step 2: Find the Relation Set

$$x\vec{A} + y\vec{B} + z\vec{C} = 0 :$$

$$x(2,1,-3) + y(1,0,-4) + z(4,3,-1) = (0,0,0)$$

This yields the system: 1. $2x + y + 4z = 0$ 2. $x + 3z = 0 \implies x = -3z$ 3. $-3x - 4y - z = 0$

Substitute $x = -3z$ into Equation 1:

$$2(-3z) + y + 4z = 0 \implies -6z + y + 4z = 0 \implies y = 2z$$

Let $z = 1$ (we can pick any non-zero value). This gives $x = -3$ and $y = 2$. Substitute the coefficients back into the original relation:

$$-3\vec{A} + 2\vec{B} + 1\vec{C} = 0 \implies \vec{C} = 3\vec{A} - 2\vec{B}$$

36.3 Worked Example 2: Abstract Vectors (PYQ 2020, 2021)

Problem: Suppose vectors u, v, w are linearly independent. Show that $u + v, u - v, u - 2v + w$ are also linearly independent.

Solution: Set a linear combination equal to the zero vector:

$$c_1(u + v) + c_2(u - v) + c_3(u - 2v + w) = 0$$

Group the terms by the original vectors u, v, w:

$$(c_1 + c_2 + c_3)u + (c_1 - c_2 - 2c_3)v + c_3 w = 0$$

Since we are *given* that u, v, w are linearly independent, the only way their linear combination equals 0 is if their coefficients are exactly 0. Thus: 1.

$$c_1 + c_2 + c_3 = 0$$

2.

$$c_1 - c_2 - 2c_3 = 0$$

3.

$$c_3 = 0$$

Substitute

$$c_3 = 0$$

into the first two equations:

$$c_1 + c_2 = 0$$

$$c_1 - c_2 = 0$$

Adding them gives

$$2c_1 = 0 \implies c_1 = 0$$

. This means

$$c_2 = 0.$$

Since the only solution is

$$c_1 = c_2 = c_3 = 0$$

, the new vectors are linearly independent. ■

36.4 Exam Patterns

- If you are asked to test 3 vectors in $\mathbb{R}^3$, immediately calculate the determinant.
- If asked to "find a relation", you must set up the x, y, z system, express two variables in terms of the third, arbitrarily set the third to 1, and write the final formula.
- Abstract linear independence proofs (like Example 2) are extremely common and very easy to score full marks on if you group coefficients properly.

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37 C06: Basis and Dimension

> **Exam Weight**: Tier 1 | **Appeared in**: 2018, 2020, 2021, 2023, 2024 (twice) | **Typical Marks**: 4–5

A basis is the most efficient set of vectors needed to define a space. It contains enough vectors to reach everywhere in the space (span), but no unnecessary vectors (linear independence).

37.1 Definitions

- **Basis**: A set of vectors

$$B = \{v_1, v_2, \dots, v_n\}$$

is a basis of a vector space V if: 1. The set B is linearly independent. 2. The set B spans V (any vector in V can be written as a linear combination of vectors in B). - **Dimension**: The dimension of a vector space is exactly equal to the number of vectors in its basis.

37.2 Finding a Basis and Dimension (The Row Reduction Method)

A very common exam question gives you a set of vectors and asks you to find a basis and the dimension of the subspace they span.

The Algorithm: 1. Write the given vectors as the **rows** of a matrix. 2. Apply elementary row operations to reduce the matrix to **Row Echelon Form** (zeros below the leading ones). 3. The **non-zero rows** of the resulting matrix form the basis. 4. The **number** of non-zero rows is the dimension.

(Note: If a row becomes entirely zeros, it means that original vector was redundant and was a linear combination of the others).

37.3 Worked Example 1: Vectors in (PYQ 2018)

Problem: Let W be the subspace of $\mathbb{R}^4$ generated by $(1, -2, 5, -3)$, $(2, 3, 1, -4)$, and $(3, 8, -3, -5)$. Find a basis and the dimension of W.

Solution: Write vectors as rows:

$$\begin{bmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{bmatrix}$$

Apply row operations to eliminate the first column below row 1 ($R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$):

$$\begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{bmatrix}$$

Eliminate the second column below row 2 ($R_3 \to R_3 - 2R_2$):

$$\begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The non-zero rows form the basis:

$$\text{Basis} = \{(1, -2, 5, -3), (0, 7, -9, 2)\}$$

Since there are 2 vectors in the basis:

$$\text{Dimension} = 2$$

37.4 Worked Example 2: Polynomial Spaces (PYQ 2020, 2023)

Problem: Find a basis and dimension of the space generated by the polynomials:

$$v_1 = t^3 - 2t^2 + 4t + 1$$

$$v_2 = 2t^3 - 3t^2 + 9t - 1$$

$$v_3 = t^3 + 6t - 5$$

$$v_4 = 2t^3 - 5t^2 + 7t + 5$$

Solution: Represent each polynomial as a coordinate vector in the standard basis $\{t^3, t^2, t, 1\}$ and form a matrix:

$$\begin{bmatrix} 1 & -2 & 4 & 1 \\ 2 & -3 & 9 & -1 \\ 1 & 0 & 6 & -5 \\ 2 & -5 & 7 & 5 \end{bmatrix}$$

Apply $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$, $R_4 \to R_4 - 2R_1$:

$$\begin{bmatrix} 1 & -2 & 4 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 2 & 2 & -6 \\ 0 & -1 & -1 & 3 \end{bmatrix}$$

Apply $R_3 \to R_3 - 2R_2$ and $R_4 \to R_4 + R_2$:

$$\begin{bmatrix} 1 & -2 & 4 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Convert the non-zero rows back into polynomials to form the basis:

$$\text{Basis} = \{t^3 - 2t^2 + 4t + 1, \quad t^2 + t - 3\}$$

Since there are 2 polynomials in the basis:

$$\text{Dimension} = 2$$

37.5 Exam Patterns

- This exact row-reduction algorithm appeared 5 times in the past 6 years. It is highly reliable.
- If asked "Determine whether these 4 vectors form a basis of $\mathbb{R}^4$", perform the row reduction. If you end up with 4 non-zero rows, yes they do. If any row zeros out (meaning dimension < 4), they do not form a basis. (See PYQ 2021/2024).

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38 C07: Linear Transformations

Exam Weight: Tier 3 | **Appeared in**: 2023 | **Typical Marks**: 4

A linear transformation (or mapping) is a function between two vector spaces that preserves the operations of vector addition and scalar multiplication.

38.1 The Linearity Conditions

Let V and U be vector spaces over the same field K. A function $F : V \to U$ is a **linear mapping** if for all vectors $u, v \in V$ and all scalars $k \in K$:

1. **Preservation of Addition**: $F(u + v) = F(u) + F(v)$
2. **Preservation of Scalar Multiplication**: $F(ku) = kF(u)$

Shortcut: These two conditions can be combined into one: $F(au + bv) = aF(u) + bF(v)$.

Crucial Property: Any linear transformation MUST map the zero vector of the domain to the zero vector of the codomain. If $F(0_V) \neq 0_U$, the mapping is definitely not linear.

38.2 Worked Example: Verifying Linearity (PYQ 2023)

Problem: Identify if the following mappings F are linear or not linear: (i)

$$F : \mathbb{R}^3 \to \mathbb{R}$$

defined by $F(x, y, z) = 2x - 3y + 4z$ (ii)

$$F : \mathbb{R}^2 \to \mathbb{R}^3$$

defined by $F(x, y) = (x + 1, 2y, x + y)$

Solution:

(i) Mapping $F(x, y, z) = 2x - 3y + 4z$ Let

$$u = (x_1, y_1, z_1)$$

and

$$v = (x_2, y_2, z_2).$$

Check Addition:

$$F(u + v) = F(x_1 + x_2, y_1 + y_2, z_1 + z_2) = 2(x_1 + x_2) - 3(y_1 + y_2) + 4(z_1 + z_2)$$

$$F(u + v) = (2x_1 - 3y_1 + 4z_1) + (2x_2 - 3y_2 + 4z_2) = F(u) + F(v)$$

Check Scalar Multiplication:

$$F(ku) = F(kx_1, ky_1, kz_1) = 2(kx_1) - 3(ky_1) + 4(kz_1) = k(2x_1 - 3y_1 + 4z_1) = kF(u)$$

Both conditions hold. The mapping is **linear**.

(ii) Mapping $F(x, y) = (x + 1, 2y, x + y)$ We use the zero-vector shortcut. A linear mapping must map the zero vector $(0, 0)$ to the zero vector $(0, 0, 0)$.

$$F(0, 0) = (0 + 1, 2(0), 0 + 0) = (1, 0, 0) \neq (0, 0, 0)$$

Since $F(0_V) \neq 0_U$, the mapping is **not linear**.

38.3 Exam Patterns

- When given a function with a constant addition term (like $x + 1$), it is automatically non-linear because it is an "affine" shift, not a linear one. Use the $F(0) \neq 0$ trick to prove it in one line.
- For functions without constants (like $2x - 3y$), you must grind out the full $F(u + v)$ and $F(ku)$ proofs.

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39 C08: Kernel, Image, Rank, and Nullity

Exam Weight: Tier 1-2 | **Appeared in**: 2019 | **Typical Marks**: 6

Every linear mapping $F : V \to U$ is associated with two crucial subspaces: the Kernel (inside the domain V) and the Image (inside the codomain U).

39.1 Definitions

- **Kernel (Null Space)**: The set of all vectors in V that map exactly to the zero vector in U.

$$\text{Ker}(F) = \{v \in V \mid F(v) = 0_U\}$$

- **Image (Range Space)**: The set of all vectors in U that are "hit" by the mapping (i.e., they have at least one pre-image in V).

$$\text{Im}(F) = \{u \in U \mid \exists v \in V \text{ such that } F(v) = u\}$$

39.2 Must-Know Proof (PYQ 2019)

Problem: Let $F : V \to U$ be a linear mapping. Show that the Kernel of F is a subspace of V and the Image of F is a subspace of U.

Part 1: Prove Ker(F) is a subspace of V We must verify the three subspace properties for Ker(F): 1. **Zero Vector**: Since F is linear,

$$F(0_V) = 0_U$$

. Therefore, $0_V \in \text{Ker}(F)$. 2. **Closure under Addition**: Let $v_1, v_2 \in \text{Ker}(F)$. By definition,

$$F(v_1) = 0_U$$

and

$$F(v_2) = 0_U.$$

$$F(v_1 + v_2) = F(v_1) + F(v_2) = 0_U + 0_U = 0_U$$

. Thus, $v_1 + v_2 \in \text{Ker}(F)$. 3. **Closure under Scalar Multiplication**: Let $v \in \text{Ker}(F)$ and let k be a scalar.

$$F(kv) = kF(v) = k(0_U) = 0_U$$

. Thus, $kv \in \text{Ker}(F)$. Since all three hold, Ker(F) is a subspace. ■

Part 2: Prove Im(F) is a subspace of U We must verify the three subspace properties for Im(F): 1. **Zero Vector**: Since

$$F(0_V) = 0_U$$

, the zero vector 0_U has a pre-image. Thus, $0_U \in \text{Im}(F)$. 2. **Closure under Addition**: Let $u_1, u_2 \in \text{Im}(F)$. Then there exist $v_1, v_2 \in V$ such that

$$F(v_1) = u_1$$

and

$$F(v_2) = u_2.$$

$$u_1 + u_2 = F(v_1) + F(v_2) = F(v_1 + v_2).$$

Since $v_1 + v_2 \in V$, the sum $u_1 + u_2$ is the image of a vector in V. Thus, $u_1 + u_2 \in \text{Im}(F)$. 3. **Closure under Scalar Multiplication**: Let $u \in \text{Im}(F)$ and let k be a scalar. There exists $v \in V$ such that $F(v) = u$. $ku = kF(v) = F(kv)$. Since $kv \in V$, ku is the image of a vector in V. Thus, $ku \in \text{Im}(F)$. Since all three hold, $\text{Im}(F)$ is a subspace. ■

39.3 Rank-Nullity Theorem

The dimensions of these two subspaces are given special names: - **Nullity**: The dimension of the Kernel. - **Rank**: The dimension of the Image.

The Rank-Nullity Theorem states that for any linear mapping $F : V \to U$, the dimensions perfectly balance out to equal the dimension of the starting domain V:

$$\dim(\text{Ker}(F)) + \dim(\text{Im}(F)) = \dim(V)$$

$$\text{Nullity} + \text{Rank} = \dim(V)$$

39.4 Exam Patterns

- The proof that the Kernel and Image are subspaces is a classic 6-mark theoretical question. Memorize the 3-step subspace check for both.
- If asked to find the Kernel of a matrix mapping, set the matrix times a vector $\vec{x}$ to the zero vector:

$$A\vec{x} = 0$$

Then solve for $\vec{x}$ using row reduction.

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40 C09: Matrix Representation and Change of Basis

Exam Weight: Tier 3 | **Appeared in**: Curriculum Reference | **Typical Marks**: 3–4

Every linear transformation between finite-dimensional vector spaces can be entirely represented by a matrix. Changing the basis of the vector space changes the matrix representation in a predictable way.

40.1 Matrix Representation of a Linear Operator

Let $T : V \to V$ be a linear operator on a vector space V with dimension n. Let

$$B = \{v_1, v_2, \ldots, v_n\}$$

be a basis for V.

When you apply the transformation T to the first basis vector v_1, the result $T(v_1)$ is still a vector in V. Therefore, it can be written as a linear combination of the basis vectors:

$$T(v_1) = a_{11}v_1 + a_{21}v_2 + \cdots + a_{n1}v_n$$

The coefficients

$$(a_{11}, a_{21}, \ldots, a_{n1})$$

form the **first column** of the matrix representation $[T]_B$. You repeat this for every basis vector. The resulting $n \times n$ matrix is the **Matrix Representation** of the linear operator.

40.2 Change of Basis Matrix

Suppose you have two different bases for the same vector space V: an old basis B and a new basis B'. You can write each vector of the *new* basis B' as a linear combination of the *old* basis B.

Taking the column coefficients of these combinations creates the **Change of Basis Matrix** (or Transition Matrix), denoted as P. - The matrix P converts coordinates from the new basis to the old basis:

$$[x]_B = P[x]_{B'}.$$

- The inverse matrix P^{-1} converts coordinates from the old basis to the new basis:

$$[x]_{B'} = P^{-1}[x]_B.$$

40.3 Similarity and Matrix Equivalency

When you change the basis of the vector space from B to B', the matrix representation of the linear operator T also changes.

If

$$[T]_B = A$$

(the matrix in the old basis) and

$$[T]_{B'} = C$$

(the matrix in the new basis), they are related by the formula:

$$C = P^{-1}AP$$

Two matrices A and C that satisfy this relationship for some invertible matrix P are called **Similar Matrices**. Similar matrices represent the *exact same linear transformation*, just viewed from the perspective of two different coordinate systems. Because they represent the same transformation, similar matrices share identical: - Determinants - Traces - Eigenvalues

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40.4 Exam Patterns

- While deep calculations involving Change of Basis rarely appear in short exams, understanding the formula

$$C = P^{-1}AP$$

is critical for diagonalization questions (which appear in the Matrix Theory sections).

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41 C10: Inner Product Spaces

Exam Weight: Tier 2 | **Appeared in**: Assignment / Final | **Typical Marks**: 4–5

An inner product space is a vector space equipped with a way to multiply vectors together, resulting in a scalar. This generalizes the concept of the dot product, allowing us to define angles and lengths in abstract spaces.

41.1 Inner Product Axioms

Let V be a vector space over the complex numbers $\mathbb{C}$ (or real numbers $\mathbb{R}$). An inner product is a function $\langle x, y\rangle$ that maps two vectors to a scalar, satisfying the following axioms for all $x, y, z \in V$ and scalar α:

1. **Conjugate Symmetry**: $\langle x, y\rangle = \overline{\langle y, x\rangle}$ *(If the field is real, this is simply commutativity: $\langle x, y\rangle = \langle y, x\rangle$)*
2. **Linearity in the First Argument**: $\langle x + y, z\rangle = \langle x, z\rangle + \langle y, z\rangle$ $\langle \alpha x, y\rangle = \alpha\langle x, y\rangle$
3. **Positive-Definiteness**: $\langle x, x\rangle \geq 0$, and $\langle x, x\rangle = 0$ if and only if $x = 0$.

41.2 Must-Know Proof 1: The Induced Norm

Problem: Prove that every inner product space is a normed space.

Proof: Let V be an inner product space. We define the induced norm as

$$\|x\| = \sqrt{\langle x, x\rangle}$$

. We must prove that this satisfies the three axioms of a norm.

1. **Positivity**: By the positive-definite axiom of inner products, $\langle x, x\rangle \geq 0$. Thus,

$$\|x\| = \sqrt{\langle x, x\rangle} \geq 0$$

. Additionally, $\langle x, x\rangle = 0 \iff x = 0$, so

$$\|x\| = 0 \iff x = 0.$$

2. **Absolute Homogeneity**:

$$\|\alpha x\| = \sqrt{\langle \alpha x, \alpha x\rangle} = \sqrt{\alpha\overline{\alpha}\langle x, x\rangle} = \sqrt{|\alpha|^2\langle x, x\rangle} = |\alpha|\sqrt{\langle x, x\rangle} = |\alpha|\|x\|$$

3. **Triangle Inequality**:

$$\|x + y\|^2 = \langle x + y, x + y\rangle = \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle$$

$$\|x + y\|^2 = \|x\|^2 + \langle x, y\rangle + \overline{\langle x, y\rangle} + \|y\|^2 = \|x\|^2 + 2\text{Re}(\langle x, y\rangle) + \|y\|^2$$

By the Cauchy-Schwarz inequality (

$$|\langle x, y\rangle| \leq \|x\|\|y\|$$

) and knowing

$$\mathrm{Re}(z) \leq |z| :$$

$$2\mathrm{Re}(\langle x, y\rangle) \leq 2|\langle x, y\rangle| \leq 2\|x\|\|y\|$$

`Substituting this back:`

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2$$

`Taking the square root gives $\|x + y\| \leq \|x\| + \|y\|$.`

Since all three axioms hold, the induced norm makes the inner product space a normed space. ■

41.3 Must-Know Proof 2: The Parallelogram Law

Problem: Prove the parallelogram law in an inner product space.

Proof: The parallelogram law states that the sum of the squares of the lengths of the four sides of a parallelogram equals the sum of the squares of the lengths of the two diagonals:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

Expand the left-hand side using the inner product:

$$\|x + y\|^2 = \langle x + y, x + y\rangle = \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle$$

$$= \|x\|^2 + \langle x, y\rangle + \langle y, x\rangle + \|y\|^2$$

Expand the second term:

$$\|x - y\|^2 = \langle x - y, x - y\rangle = \langle x, x\rangle - \langle x, y\rangle - \langle y, x\rangle + \langle y, y\rangle$$

$$= \|x\|^2 - \langle x, y\rangle - \langle y, x\rangle + \|y\|^2$$

Add the two equations together:

$$\|x + y\|^2 + \|x - y\|^2 = (\|x\|^2 + \langle x, y\rangle + \langle y, x\rangle + \|y\|^2) + (\|x\|^2 - \langle x, y\rangle - \langle y, x\rangle + \|y\|^2)$$

The cross terms $\langle x, y\rangle$ and $-\langle x, y\rangle$ cancel out, as do $\langle y, x\rangle$ and $-\langle y, x\rangle$.

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

■

41.4 Exam Patterns

- If asked to prove something is an inner product space, you just need to grind through the 3 axioms (linearity, symmetry, positive-definite).
- The Parallelogram Law proof is incredibly straightforward and a very easy 5 marks if it appears.

42 C11: Normed Linear Spaces & Banach Spaces

Exam Weight: Tier 1-2 | **Appeared in**: Assignment / Final | **Typical Marks**: 3–5

While an inner product gives a vector space "angles", a norm gives a vector space "length". When a normed space is mathematically "complete", it is called a Banach Space.

42.1 Normed Linear Space Definitions

A normed linear space is a vector space V equipped with a norm $\|\cdot\| : V \to \mathbb{R}$, which assigns a non-negative real length to each vector. It must satisfy three properties for all $x, y \in V$ and scalars α: 1. **Positivity**: $\|x\| \geq 0$, and

$$\|x\| = 0 \iff x = 0.$$

2. **Absolute Homogeneity**:

$$\|\alpha x\| = |\alpha|\|x\|.$$

3. **Triangle Inequality**: $\|x + y\| \leq \|x\| + \|y\|$.

Example Verification: Verify whether

$$\|x\| = |x_1| + |x_2|$$

defines a norm on $\mathbb{R}^2$. 1. **Positivity**: $|x_1| \geq 0$ and $|x_2| \geq 0$, so

$$|x_1| + |x_2| \geq 0$$

. If

$$\|x\| = 0$$

, then

$$|x_1| = 0$$

and

$$|x_2| = 0$$

, so $x = (0, 0)$. 2. **Homogeneity**:

$$\|\alpha x\| = |\alpha x_1| + |\alpha x_2| = |\alpha|(|x_1| + |x_2|) = |\alpha|\|x\|.$$

3. **Triangle Inequality**:

$$\|x + y\| = |x_1 + y_1| + |x_2 + y_2| \leq (|x_1| + |y_1|) + (|x_2| + |y_2|) = \|x\| + \|y\|.$$

Since all three hold, it is a valid norm (the L^1 norm).

42.2 Banach Space Definitions

A **Banach space** is defined as a **complete** normed linear space. “Complete” means that every Cauchy sequence of vectors in the space successfully converges to a limit that is *also contained within the space.* There are no “missing holes” in the space.

42.2.1 Example of a Banach Space

The space ℓ^∞, which consists of all bounded infinite sequences of complex numbers

$$x = (x_1, x_2, \dots)$$

, is equipped with the supremum norm:

$$\|x\|_\infty = \sup_i |x_i|$$

Because $\mathbb{C}$ is complete, Cauchy sequences of coordinates converge to bounded limits within ℓ^∞.

42.2.2 Example of an INCOMPLETE Normed Space

The space $P[a, b]$ of all polynomial functions on $[a, b]$, equipped with the supremum norm. It is not complete because a Cauchy sequence of polynomials (like the Taylor series expansion for e^t) will converge to a limit function (e^t) that is continuous but *not* a polynomial. Since the limit escapes the space of polynomials, the space is not complete.

42.3 Must-Know Proof: Finite-Dimensional Spaces

Problem: Prove that every finite-dimensional normed linear space is complete.

Proof: Let V be an n-dimensional normed linear space. Let $\{e_1, \dots, e_n\}$ be a basis. Any vector x can be uniquely written as

$$x = \sum_{i=1}^{n} \alpha_i e_i.$$

Define a reference norm:

$$\|x\|_1 = \sum_{i=1}^{n} |\alpha_i|$$

. In a finite-dimensional space, all norms are equivalent, meaning there exist constants $c, C > 0$ such that

$$c\|x\|_1 \leq \|x\| \leq C\|x\|_1.$$

Let $(x^{(m)})$ be a Cauchy sequence in V with respect to $\|\cdot\|$. Because the norms are equivalent, it is also Cauchy with respect to $\|\cdot\|_1$. Let

$$x^{(m)} = \sum_{i=1}^{n} \alpha_i^{(m)} e_i$$

. Since $(x^{(m)})$ is Cauchy in $\|\cdot\|_1$, for a given $\epsilon > 0$, for large p, q:

$$\|x^{(p)} - x^{(q)}\|_1 = \sum_{i=1}^{n} |\alpha_i^{(p)} - \alpha_i^{(q)}| < \epsilon$$

This implies that for each coordinate index i, the sequence

$$(\alpha_i^{(m)})_{m=1}^{\infty}$$

is a Cauchy sequence of scalars. Since $\mathbb{R}$ (and $\mathbb{C}$) is complete, each coordinate sequence converges to a limit α_i.

Let

$$x = \sum_{i=1}^{n} \alpha_i e_i$$

. Clearly, $x \in V$. We check convergence:

$$\|x^{(m)} - x\|_1 = \sum_{i=1}^{n} |\alpha_i^{(m)} - \alpha_i| \to 0$$

as $m \to \infty$. Because norms are equivalent,

$$\|x^{(m)} - x\| \le C\|x^{(m)} - x\|_1 \to 0$$

, meaning the sequence converges to x in the original norm. Thus, every Cauchy sequence converges within V, making V complete. ■

42.4 Exam Patterns

- The definitions are highly testable. Remember: Banach = Complete + Normed.
- You may be asked for a 3-mark definition and example of a Banach space. Provide the ℓ^∞ or $C[a, b]$ example.

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43 C12: Hilbert Spaces

Exam Weight: Tier 1-2 | **Appeared in**: Assignment / Final | **Typical Marks**: 4–5

If a Banach space is a complete space with a norm (length), a Hilbert space goes one step further: it is a complete space with an inner product (length and angle).

43.1 Definition

A **Hilbert Space** is a complete inner product space. - It has an inner product $\langle x, y\rangle$. - This inner product induces a norm

$$\|x\| = \sqrt{\langle x, y\rangle}.$$

- The space is complete with respect to the metric defined by this induced norm (every Cauchy sequence converges to a limit within the space).

43.2 The Hierarchy of Spaces

The spaces you have learned form a strict nested hierarchy: 1. **Vector Space**: Has addition and scalar multiplication. 2. **Normed Space**: A Vector Space that also has a concept of length ($\|x\|$). 3. **Banach Space**: A Normed Space that is "complete" (no missing limits). 4. **Inner Product Space**: A Vector Space that has a dot-product-like operation (angles). *Note: All inner product spaces are automatically Normed Spaces via the induced norm, but they are not necessarily complete.* 5. **Hilbert Space**: An Inner Product Space that is complete.

43.2.1 Must-Know Proof: Every Hilbert is a Banach (PYQ / Assignment)

Problem: Prove that every Hilbert space is a Banach space.

Proof: By definition, a Hilbert space is an inner product space that is complete with respect to the distance metric induced by its inner product. The induced metric is based on the induced norm, defined as

$$\|x\| = \sqrt{\langle x, x\rangle}.$$

As proven previously (see C10), the induced norm

$$\|x\| = \sqrt{\langle x, x\rangle}$$

satisfies all the axioms of a norm (positivity, absolute homogeneity, and the triangle inequality via the Cauchy-Schwarz inequality). Therefore, every inner product space is automatically a **normed linear space** when equipped with this induced norm.

Because a Hilbert space is, by definition, **complete** with respect to this specific induced norm, it fulfills both conditions of a Banach space: 1. It is a normed linear space. 2. It is complete. Thus, every Hilbert space is a Banach space. ■ *(Note: The converse is not true. Not all norms are induced by an inner product. If a norm does not satisfy the Parallelogram Law, it cannot come from an inner product, so a space with that norm could be Banach but not Hilbert).*

43.3 Worked Example: The Space

Problem: Show that the space ℓ^2 is a Hilbert space.

Solution Overview: The space ℓ^2 consists of all infinite sequences of complex numbers

$$x = (x_1, x_2, \dots)$$

such that the sum of the squares of their absolute values converges:

$$\sum_{i=1}^{\infty} |x_i|^2 < \infty.$$

1. Define the Inner Product The inner product is defined as

$$\langle x, y \rangle = \sum_{i=1}^{\infty} x_i \overline{y_i}.$$

By the Cauchy-Schwarz inequality for sequences, this series converges absolutely, so the inner product is mathematically well-defined. It trivially satisfies linearity, conjugate symmetry, and positive-definiteness.

2. Prove Completeness Let $(x^{(n)})$ be a Cauchy sequence in ℓ^2. For $\epsilon > 0$, there is an N such that for $n, m > N$, we have

$$\|x^{(n)} - x^{(m)}\|_2 < \epsilon.$$

This implies that for each coordinate i, the sequence

$$(x_i^{(n)})_{n=1}^{\infty}$$

is a Cauchy sequence in $\mathbb{C}$. Since $\mathbb{C}$ is complete, each coordinate converges to a limit $x_i \in \mathbb{C}$. Let

$$x = (x_1, x_2, \dots)$$

. By taking the limit as $m \to \infty$ in the finite partial sums of the Cauchy condition, and extending to the infinite sum, we find $x^{(n)} - x \in \ell^2$. Since $x^{(n)} \in \ell^2$ and ℓ^2 is a vector space,

$$x = x^{(n)} - (x^{(n)} - x) \in \ell^2.$$

Finally, this shows $\|x^{(n)} - x\|_2 \to 0$ as $n \to \infty$, meaning the sequence converges to x in the ℓ^2 norm. Because every Cauchy sequence converges to an element within ℓ^2, the space is complete, making it a Hilbert space.

43.4 Exam Patterns

- "Define a Hilbert space and give an example" is a highly probable short question. Provide the definition and the ℓ^2 sequence space example.
- The 4-mark theoretical proof "Every Hilbert is Banach" is very easy if you clearly state definitions.
